Geometric Class Field Theory

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Abstract

We calculate the ramification of the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ of a local system $\mathcal{F}$ on a curve $C \setminus \mathfrak{m}$. Assuming the ramification of $\mathcal{F}$ is bounded by $\mathfrak{m}$, we find that the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ is at most tamely ramified at the generic point of the exceptional divisor $E_{\mathfrak{m}}$ of the blowup of $C^{(\deg \mathfrak{m})}$ at $\mathfrak{m}$. As a primary application, we utilize this result to prove Geometric Class Field Theory. Our approach builds upon the geometric framework for the unramified case originally established by Deligne for the rank-one Langlands correspondence, following the subsequent extensions to the ramified case developed by Takeuchi and Guignard.

Contents

Introduction

In this thesis, we give an elementary proof of a certain imporant geometric theorem occuring in Deligne's approach to geometric class field theory. We (usually) work over a perfect field k, C is a projective smooth geometrically connected curve over k, with genus g. One of the main geometric ingredients in the approach, is showing why a local system $\mathcal{F}$ with ramification bounded by a modulus $\mathfrak{m} = \sum n_i P_i$ on $U = C \setminus \mathfrak{m}$ descends via the Abel-Jacobi $\Phi : U \to \mathrm{Pic}_{C,\mathfrak{m}}$ to $\mathrm{Pic}_{C,\mathfrak{m}}$. The approach, innovated by Deligne, relies on analyzing the symmetric powers $\mathcal{F}^{[d]}$ of $\mathcal{F}$ on the symmetric powers $U^{(d)}$ of U, and showing that for sufficiently large d, $\mathcal{F}^{[d]}$ descends to $\mathrm{Pic}^d_{C,\mathfrak{m}}$ via the degree d Abel-Jacobi map $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$. The geometric-fibers of Φ_d (for $d \geq \deg \mathfrak{m} + 2g - 1$) over any point are isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg \mathfrak{m} - g+1}_{k^{sep}} & \text{if } \mathfrak{m} > 0 \\ \mathbb{P}^{d-g}_{k^{sep}} & \text{if } \mathfrak{m} = 0 \end{cases}$$

Where g is the genus of the curve C. The unramified case ($\mathfrak{m} = 0$) is relatively simple, as the Abel-Jacobi map is proper, surjective with geometrically connected fibers, which follows from the fact that it is a fibration in projective spaces. Thus, by using the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $U^{(d)} (= C^{(d)})$ and that of $\mathrm{Pic}^d_{C,\mathfrak{m}} (= \mathrm{Pic}^d_C)$.

The ramified case ($\mathfrak{m} > 0$) is more subtle, as the Abel-Jacobi map is not proper anymore, and one needs to analyze the ramification of $\mathcal{F}^{[d]}$ "along the boundary" of $U^{(d)}$ in $C^{(d)}$.

Previous work has generalized Deligne's approach to the ramified case, most notably by Guignard [Gui19] and Takeuchi [Tak19]. Their approaches differ. To descend, Guignard proves that the restriction of $\mathcal{F}^{[d]}$ to any line in the fiber of the degree d Abel-Jacobi map is a constant étale sheaf. He achieves this by demonstrating that the restriction is at most tamely ramified and invoking the triviality of the tame fundamental group of $\mathbb{A}^1_k$. His analysis relies on local geometric class field theory. It is also worth noting that Guignard's method generalizes to relative curves over arbitrary base schemes. Takeuchi, on the other hand, constructs a compactification of $U^{(d)}$ by blowing up $C^{(d)}$ along certain well-chosen centers. This compactification, denoted by $\tilde{C}^{(d)}_{\mathfrak{m}}$, has $U^{(d)}$ as an open subscheme with a codimension 1 closed subscheme H as complement. He then shows that the Abel-Jacobi map extends to a proper morphism from $\tilde{C}^{(d)}_{\mathfrak{m}}$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$, which is a fibration in projective spaces. Thus, by the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $\tilde{C}^{(d)}_{\mathfrak{m}}$ and that of $\mathrm{Pic}^d_{C,\mathfrak{m}}$. To conclude the descent, Takeuchi analyzes the ramification of $\mathcal{F}^{[d]}$ along the boundary H of $\tilde{C}^{(d)}_{\mathfrak{m}}$, showing that it is tamely ramified there, which suffices. His methods relies on the theory of refined Swan conductors.

For an account of these approaches, see [Gui19] and [Tak19]. For a full approach following Deligne's method in the unramified case, and the tamely ramified case see [Ten15], and [Tót11].

In this thesis, we calculate the ramification of $\mathcal{F}^{[d]}$ directly to show that it is at most tame at the generic point of H, avoiding the use of Swan conductors. To prove Geometric Class Field Theory, we utilize Takeuchi's construction of the compactification $\tilde{C}_{\mathfrak{m}}^{(d)}$ of $U^{(d)}$ via the blowing up of $C^{(d)}$.

In the rest of the introduction, we state the main theorem of geometric class field theory Theorem 1, and its reductions to Theorem 2 and Theorem 3, which we prove in this thesis.

Let k be a perfect field, and let C be a projective smooth geometrically connected curve over k, with genus g. Geometric class field theory gives a geometric description of abelian coverings of C by relating it to isogenies of the generalized Picard schemes.

Fix a modulus $\mathfrak{m}$, i.e. an effective Cartier divisor of C and let U be its complement in C. The pairs $(\mathcal{L}, \alpha)$, where $\mathcal{L}$ is an invertible $\mathcal{O}_C$-module and α is a rigidification of $\mathcal{L}$ along $\mathfrak{m}$, are parametrized by a k-group scheme $\mathrm{Pic}_{C,\mathfrak{m}}$, called the rigidified Picard scheme. The Abel-Jacobi morphism

$$\Phi : U \to \mathrm{Pic}_{C,\mathfrak{m}}$$

is the morphism which sends a section x of U to the pair $(\mathcal{O}(x), 1)$. The fundamental result of GCFT can be formulated as:

Theorem 1 (Geometric Class Field Theory)**.** *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on U, with ramification bounded by $\mathfrak{m}$. Then, there exists a unique (up to isomorphism) multiplicative*[1] *étale sheaf of Λ-modules $\mathcal{G}$ on $\mathrm{Pic}_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}$ by Φ is isomorphic to $\mathcal{F}$.*

Let d be a positive integer. We denote by $U^{(d)}$ the d-th symmetric power of U over k. For an étale sheaf $\mathcal{F}$ on $U_{\text{ét}}$, we denote by $\mathcal{F}^{[d]}$ the d-th symmetric power of $\mathcal{F}$ on $U^{(d)}$ The degree d Abel-Jacobi morphism is defined as the map

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$$

which sends a section $x_1 + \cdots + x_d$ of $U^{(d)}$ to the pair $(\mathcal{O}(x_1 + \cdots + x_d), 1)$.

A method of descent shows that to prove Theorem 1, it suffices to prove the following reduced version[2]:

Theorem 2. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{\text{ét}}$, with ramification bounded by $\mathfrak{m}$. Then, for sufficiently large integer d, there exists a unique (up to isomorphism) étale sheaf of Λ-modules $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}_d$ by Φ_d is isomorphic to $\mathcal{F}^{[d]}$.*

To prove Theorem 2 we follow the work of [Tak19] (and similiary done in [Tót11]), analyzing the ramification of $\mathcal{F}^{[d]}$ after blowing up $C^{(d)}$. We analyze this ramification using elementary methods. We prove the following Theorem:

Theorem 3. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{\text{ét}}$, with ramification bounded by $\mathfrak{m} = \sum n_i P_i$. Let $\mathfrak{n} = n_i P_i \subset \mathfrak{m}$ be a sub modulus of $\mathfrak{m}$. Then $\mathcal{F}^{[\deg \mathfrak{n}]}$ is tamely ramified at the generic point $\eta_{\mathfrak{n}}$ of the exceptional divisor $E_{\mathfrak{n}}$ of the blowup $X_{\mathfrak{n}}$ of $C^{(\deg \mathfrak{n})}$ at the closed point $\mathfrak{n}$.*

The thesis is organized as follows:

[1] The notion of a multiplicative locally free Λ-module of rank 1 is due to [Gui19] and corresponds to isogenies $G \to \mathrm{Pic}_{C,\mathfrak{m}}$ with constant kernel $\Lambda^\times$. This concept corresponds to multiplicative characters of $H^1(\mathrm{Pic}_{C,\mathfrak{m}}, \mathbb{Q}/\mathbb{Z})$ in the formulation of [Tak19], and generalizes Hecke eigensheaves in the context of [Ten15].

[2] See the last page of [Gui19], Section 8.3 of [Ten15], or the proof of Theorem 1.2 in [Tak19] for details on this reduction.

Chapter 1 - General Preliminaries introduces torsors, their relationship with rank-one local systems, the symmetric-power constructions used throughout the thesis, and the basic geometric facts about symmetric powers and blowups needed later. In particular, it explains why the main theorems are stated for local systems while their ramification calculations are carried out using torsors.

Chapter 2 - Ramification After Blowup Is Tame is devoted to the proof of Theorem 3, along with several corollaries used in the proof of Theorem 2. The local ramification theory required only for this calculation is recalled at the beginning of the chapter.

Chapter 3 - Geometric Class Field Theory presents the proof of Theorem 2.

Appendix A - A Wrong Path documents an earlier attempted approach to Theorem 3, and the difficulties that led us to abandon it.

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Chapter 1

General Preliminaries

This chapter collects the constructions and conventions used in both main parts of the thesis. Rank-one étale local systems are the natural objects in the statement of geometric class field theory. For the geometric and ramification-theoretic arguments, however, it is more convenient to work with torsors: they are represented by finite étale covers, admit concrete contracted products and quotients, and make ramification accessible through extensions of local fields.

If Λ is a finite commutative ring and $G = \Lambda^{\times}$, the sheaf of frames of a rank-one Λ-local system is a G-torsor, and conversely a G-torsor determines an associated rank-one local system. We establish this dictionary below and record its compatibility with pullback, tensor products, symmetric powers, and ramification. Thus, the main results will be stated in the language of local systems, while the constructions and ramification calculations will generally be carried out in the language of torsors.

1.1 Torsors and Rank-One Local Systems

In what follows, we largely adhere to the treatment of torsors and group objects found in published notes of Alex Youcis [You20]. Let $\mathscr{C} = (\mathcal{C}, J)$ be a site and let $\mathcal{E} = Sh(\mathscr{C})$ be the associated topos. Let $\mathcal{G}$ be a group object in $\mathcal{E}$. We denote by $\mathcal{G}\mathcal{E}$ the category of objects in $\mathcal{E}$ endowed with a left $\mathcal{G}$-action.

Definition 4. A **$\mathcal{G}$-torsor** in $\mathcal{E}$ is an object $\mathcal{P}$ of $\mathcal{G}\mathcal{E}$ satisfying the following conditions:

1. The structural morphism $\mathcal{P} \to 1$ is an epimorphism in $\mathcal{E}$ (i.e., $\mathcal{P}$ is locally non-empty).
2. The map $\mathcal{G} \times \mathcal{P} \to \mathcal{P} \times \mathcal{P}$ defined by $(g, p) \mapsto (g \cdot p, p)$ is an isomorphism in $\mathcal{E}$ (i.e., $\mathcal{G}$ acts simply transitively on $\mathcal{P}$).

Since $\mathcal{E}$ is the topos of sheaves on a site $\mathscr{C}$, the definition can be reformulated in terms of covers. A $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if:

1. For every object $X \in \mathcal{C}$, there exists a covering $\{U_i \to X\} \in J$ such that $\mathcal{P}(U_i) \neq \emptyset$ for all i.
2. For any $X \in \mathcal{C}$ where $\mathcal{P}(X)$ is non-empty, the action of $\mathcal{G}(X)$ on $\mathcal{P}(X)$ is simply transitively.

A fundamental property of torsors is their local triviality: a $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if and only if it is locally isomorphic to the trivial torsor. Specifically, for every $X \in \mathcal{C}$, there must exist a cover

$\{U_i \to X\}$ such that the restriction $\mathcal{P}|_{U_i}$ is isomorphic, as a $\mathcal{G}|_{U_i}$-sheaf, to $\mathcal{G}|_{U_i}$ acting on itself by left multiplication.

A **morphism of $\mathcal{G}$-torsors** $f : \mathcal{P}_1 \to \mathcal{P}_2$ is a morphism of sheaves that is equivariant with respect to the $\mathcal{G}$-action.

Proposition 5. *Every morphism of $\mathcal{G}$-torsors is an isomorphism. Consequently, the category of $\mathcal{G}$-torsors in $\mathcal{E}$ forms a groupoid.*

Proof. Locally on a cover trivializing both torsors, such a morphism is a $\mathcal{G}$-equivariant endomorphism of the trivial torsor $\mathcal{G}$, i.e. right translation by a section of $\mathcal{G}$, hence an isomorphism; and being an isomorphism may be checked on a cover. □

Definition 6. We denote the groupoid of $\mathcal{G}$-torsors in $\mathcal{E}$ by $\mathbf{Tors}(\mathcal{E}, \mathcal{G})$. The set of isomorphism classes of $\mathcal{G}$-torsors is denoted by $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$.

Torsors exhibit functoriality with respect to the group object:

Definition 7. Let $\varphi : \mathcal{G}_1 \to \mathcal{G}_2$ be a morphism of group sheaves on $\mathscr{C}$, and let $\mathcal{P}$ be a $\mathcal{G}_1$-torsor. We define the **contracted product** $\mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$ as the quotient sheaf $\mathcal{G}_1 \backslash (\mathcal{G}_2 \times \mathcal{P})$, where $\mathcal{G}_1$ acts on the product by:

$$g_1 \cdot (g_2, p) = (g_2 \varphi(g_1)^{-1}, g_1 \cdot p)$$

The contracted product inherits a natural left $\mathcal{G}_2$-action given on local sections by $h \cdot [g_2, p] = [hg_2, p]$, which endows it with the structure of a $\mathcal{G}_2$-torsor.

This construction yields a functor:

$$\varphi_* : \mathbf{Tors}(\mathcal{E}, \mathcal{G}_1) \to \mathbf{Tors}(\mathcal{E}, \mathcal{G}_2), \quad \mathcal{P} \mapsto \mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$$

On the level of isomorphism classes, φ_* induces a map of pointed sets $\mathrm{Tors}(\mathcal{E}, \mathcal{G}_1) \to \mathrm{Tors}(\mathcal{E}, \mathcal{G}_2)$, sending the class of the trivial $\mathcal{G}_1$-torsor to the class of the trivial $\mathcal{G}_2$-torsor.

When $\mathcal{G}$ is a **sheaf of abelian groups** (an *abelian sheaf*), the pointed set $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$ inherits the structure of an abelian group:

Definition 8. Assume $\mathcal{G}$ is abelian sheaf. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be objects of $\mathbf{Tors}(\mathcal{E}, \mathcal{G})$. We define the **product** $\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2$ to be the quotient sheaf $(\mathcal{P}_1 \times \mathcal{P}_2)/\mathcal{G}$, where $\mathcal{G}$ acts on T-points by having:

$$g \cdot (f_1, f_2) := (gf_1, g^{-1} f_2)$$

The group object $\mathcal{G}$ then acts on the resulting quotient via its action on the presheaf quotient, which is given on classes by:

$$g \cdot [(f_1, f_2)] = [(gf_1, f_2)] = [(f_1, gf_2)]$$

where the square brackets denote the class in the quotient set.[1]

Let $[\mathcal{P}_1]$ and $[\mathcal{P}_2]$ be the isomorphism classes of $\mathcal{P}_1$ and $\mathcal{P}_2$ in $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$. We define the sum $[\mathcal{P}_1] + [\mathcal{P}_2]$ to be the class $[\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2]$. This structure turns $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$ into an abelian group, where the identity is the class of the trivial torsor and the inverse is obtained by the opposite action.

[1] Equivantly, the sum is obtained as the contracted product of the $\mathcal{G} \times \mathcal{G}$-torsor $\mathcal{P}_1 \times \mathcal{P}_2$ along the multiplication map $m : \mathcal{G} \times \mathcal{G} \to \mathcal{G}$.

We now record the dictionary that explains the role of torsors in this thesis. Let Λ be a commutative ring object of $\mathcal{E}$ and let $G = \Lambda^\times$ be its group of units. We write $\mathbf{Pic}(\mathcal{E}, \Lambda)$ for the groupoid of locally free Λ-modules of rank one. For $\mathcal{L}$ in this groupoid, its **frame torsor** is

$$\mathrm{Fr}(\mathcal{L}) := \underline{\mathrm{Isom}}_\Lambda(\Lambda, \mathcal{L}).$$

The group G acts on frames by precomposition with multiplication on Λ. Conversely, if $\mathcal{P}$ is a G-torsor, its **associated Λ-module** is the contracted product

$$\mathcal{P} \times^G \Lambda := (\mathcal{P} \times \Lambda)/G, \qquad g \cdot (p, \lambda) = (gp, g^{-1}\lambda),$$

where G acts on Λ by multiplication.

Proposition 9 (The torsor–local-system dictionary)**.** *The functors*

$$\mathcal{P} \longmapsto \mathcal{P} \times^G \Lambda, \qquad \mathcal{L} \longmapsto \mathrm{Fr}(\mathcal{L})$$

are mutually quasi-inverse symmetric monoidal equivalences

$$(\mathbf{Tors}(\mathcal{E}, G), \otimes^G, G) \xrightarrow{\sim} (\mathbf{Pic}(\mathcal{E}, \Lambda), \otimes_\Lambda, \Lambda).$$

These equivalences commute with pullback.

Proof. Both assertions may be checked locally. On a cover on which $\mathcal{P}$ is the trivial torsor G, the associated module $\mathcal{P} \times^G \Lambda$ identifies with Λ. Conversely, on a cover on which $\mathcal{L}$ is free of rank one, $\mathrm{Fr}(\mathcal{L})$ identifies with G acting on itself by translation. These local identifications are compatible with the descent data and give the two quasi-inverse equivalences. Applying the same argument to products of local trivializations identifies contracted products with tensor products. The construction is defined by fiber products and sheaf quotients, and therefore commutes with pullback. $\square$

When $\mathcal{E} = \mathrm{Sh}(X_{\text{ét}})$ and Λ is a finite constant ring, the objects of $\mathbf{Pic}(\mathcal{E}, \Lambda)$ are precisely the rank-one Λ-local systems on X. We retain local-system language in the statements of the main theorems and use the equivalent torsor language for the geometric constructions and ramification calculations. We will pass between the two languages using Proposition 9.

Lastly, for any object $\mathcal{X} \in \mathcal{E}$, there is a canonical identification between the slice category $(\mathcal{G}\mathcal{E})/\mathcal{X}$ and the category of group objects $\mathcal{G}(\mathcal{E}/\mathcal{X})$, where $\mathcal{X}$ is viewed as having the trivial $\mathcal{G}$-action.

We denote by $\mathbf{Tors}(\mathcal{X}, \mathcal{G})$ the category of G-torsors over $\mathcal{X}$ in $\mathcal{G}\mathcal{E}_{/\mathcal{X}}$.

1.2 Torsors on the Étale Sites

Fix a scheme X, and let G be a smooth affine X-group scheme. Denote by $X_{\text{Ét}}$ and $X_{\text{ét}}$ the big and small étale sites on X, respectively. The functor of points of G, which we denote by $G(\cdot) = \mathrm{Hom}_X(-, G)$, defines a group sheaf on $X_{\text{ét}}$.

Definition 10. A *principal G-bundle* (or *principal homogeneous space*) for G is a scheme $f : Y \to X$ equipped with a left G-action $\rho : G \times_X Y \to Y$ satisfying:

1. The morphism $G \times_X Y \to Y \times_X Y$ sending $(g, y) \mapsto (gy, y)$ is an isomorphism of X-schemes.

2. There exists an étale covering $\{U_i \to X\}$ such that $Y_{U_i} \cong G_{U_i}$ as G-schemes, where G acts on itself by left translation.

Remark. Since $G \to X$ is smooth and Y is étale-locally isomorphic to G, the morphism $f : Y \to X$ is smooth and affine. Note that Y is generally not étale over X unless G is an étale group scheme (e.g., a finite constant group). Similarly, if G is finite then $Y \to X$ is finite.

Similarly to the case of G-torsors of a topos, we define a morphism of principal G-bundles to be a morphism of X-schemes commuting with the G-action. We now have:

Theorem 11. *The morphism sending $Y \mapsto \mathrm{Hom}_X(-, Y)$ is an equivalence of categories from the category of principal G-bundles to the category of G-torsors on $X_{\acute{E}t}$ Similarly, the morphism sending Y to $\mathrm{Hom}_X(-, Y)$ to the category of G-torsors on $X_{\acute{e}t}$ is an equivalence.*

Corollary 12. *There is a natural equivalence $\mathbf{Tors}(X_{\acute{e}t}, G) \cong \mathbf{Tors}(X_{\acute{E}t}, G)$ inducing a bijection of pointed sets $\mathrm{Tors}(X_{\acute{e}t}, G) \xrightarrow{\cong} \mathrm{Tors}(X_{\acute{E}t}, G)$ which is an isomorphism of abelian groups if G is abelian. And every G-torsor, in each of the above sites can be realized as a scheme, which is a prinicipal G-bundle.*

Constant Finite Group Torsors

Let G be a finite group, within this section, we denote the associated constant group scheme over X by $\underline{G}$ to maintain a clear distinction between the group as a set and the group as a scheme. In subsequent sections, we shall follow the standard practice of identifying G with its constant group scheme and omit the underline for brevity.

$\underline{G}$ is given by $\coprod_{g\in G} X$ with the action shuffeling the X's according to multiplication.

By following the definitions, one sees that if X be a connected scheme and $f : Y \to X$ is a finite Galois cover with Galois group G. Then, $f : Y \to X$ is a principal $\underline{G}$-bundle. (Recall that a *finite Galois cover* is a finite étale surjection $Y \to X$ with Y connected and such that $G = \mathrm{Aut}(Y/X)$ acts transitively on the geometric points of Y lying over any geometric point of X).

On the otherhand if $f : Y \to X$ is a principal $\underline{G}$-bundle with Y connected, then Y is a finite Galois cover with automorphism group G.

However, not all $\underline{G}$-torsors are connected. If $H \subset G$ is a proper subgroup then any connected finite étale cover $f : Y \to X$ with Galois group H gives rise to a non-connected $\underline{G}$-torsor by looking at the induced $\underline{G}$-torsor $\varphi_*(Y)$ under the inclusion $\varphi : \underline{H} \to \underline{G}$. On the otherhand, if we fix a geometric point $\overline{x} \to X$, then to give an homomorphism $\rho \in \mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X, x), G)$ is equivilant to give a connected pointed Galois cover $(Y, \overline{y}) \to (X, \overline{x})$ with Galois group $H = \rho(\pi_1^{\mathrm{et}}(X, x)) \subset G$. Thus, pushing forward to $\underline{G}$ we get a prinicipal $\underline{G}$-bundle. The choice of a different geometric point $\overline{x'} \to X$ differ the homomorphism by an inner automorphism, thus we have:

Theorem 13. *Let X be a connected scheme and $\overline{x}$ a geometric point of X. Suppose in addition that G is a finite abstract group. Define a map*

$$\mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X, \overline{x}), G)/\mathrm{Inn}(G) \to \mathrm{Tors}(X_{\acute{E}t}, \underline{G}) \tag{1.1}$$

by sending a homomorphism $\rho : \pi_1^{\mathrm{et}}(X, \overline{x}) \to G$ to the principal $\underline{G}$-bundle $\varphi_*(Y)$ where Y is the principal $\underline{\rho(\pi_1^{\mathrm{et}}(X, \overline{x}))}$-bundle obtained above and φ is the inclusion $\rho(\pi_1^{\mathrm{et}}(X, \overline{x})) \hookrightarrow G$. Then, the map is a bijection of pointed sets where the trivial homomorphisms (which is the only element of its $\mathrm{Inn}(G)$-orbit) is the distinguished element of the left hand side.

If G is abelian then $\mathrm{Inn}(G)$ is trivial, and we obtain:

Corollary 14. *Let G be a finite abelian group, X a connected scheme, and $\overline{x}$ a geometric point of X. Then, the map from Theorem 13 induces an isomorphism of abelian groups*

$$\mathrm{Hom}_{\mathrm{cont.}}(\pi_1^{\acute{e}t}(X,\overline{x}),G) \xrightarrow{\cong} \mathrm{Tors}(X_{\acute{E}t},\underline{G}) \tag{1.2}$$

For $G = \Lambda^\times$, the character in Corollary 14 is also the monodromy character of the associated rank-one Λ-local system. In particular, the torsor and the local system have the same restrictions to inertia and to the upper ramification groups. Thus, one has ramification bounded by a given modulus, or is tamely ramified along a given divisor, if and only if the other does.

Quotients and Decomposition of Torsors

Let $\mathcal{H}$ be a sheaf of groups acting on a sheaf $\mathcal{P}$ on a site $\mathscr{C}$. The **quotient sheaf** $\mathcal{P}/\mathcal{H}$ is the coequalizer in $\mathrm{Sh}(\mathscr{C})$

$$\mathcal{H}\times\mathcal{P} \rightrightarrows \mathcal{P} \longrightarrow \mathcal{P}/\mathcal{H},$$

where the parallel arrows are the action and the projection. Equivalently, it is the sheaf associated with the presheaf of orbits $U \mapsto \mathcal{H}(U)\backslash\mathcal{P}(U)$. We use the notation $\mathcal{P}/\mathcal{H}$ even though our torsor actions are written on the left. In particular, the quotient always exists as a sheaf. It is characterized by the following universal property: for every sheaf $\mathcal{F}$, composition with $\mathcal{P}\to\mathcal{P}/\mathcal{H}$ identifies morphisms $\mathcal{P}/\mathcal{H}\to\mathcal{F}$ with the $\mathcal{H}$-invariant morphisms $\mathcal{P}\to\mathcal{F}$. A quotient sheaf need not in general be representable by a scheme; representability in the finite-torsor situation used here is part of the next proposition.

Let G be a finite abelian group, let $H\subseteq G$ be a subgroup, and let $\pi : G\to G/H$ be the quotient homomorphism.

Proposition 15. *Let $\mathcal{P}\to X$ be a G-torsor in $X_{\acute{e}t}$. The quotient sheaf $\mathcal{P}/H$ is represented by a finite étale X-scheme, and there is a canonical isomorphism of G/H-torsors*

$$\mathcal{P}/H \xrightarrow{\sim} \pi_*\mathcal{P} = (G/H)\times^G\mathcal{P}.$$

Consequently, $\mathcal{P}\to X$ admits a canonical factorization

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}/H \xrightarrow{\psi} X,$$

where ϕ is an H-torsor and ψ is a G/H-torsor. This factorization is unique up to a unique isomorphism commuting with the projections.

Proof. The canonical morphism

$$\mathcal{P}\longrightarrow\pi_*\mathcal{P},\qquad p\longmapsto[1,p],$$

is H-invariant because $\pi(H)=1$. The universal property of the quotient sheaf therefore gives a canonical morphism

$$\mathcal{P}/H\longrightarrow\pi_*\mathcal{P}.$$

Choose an étale cover $U\to X$ that trivializes $\mathcal{P}$. Over U this morphism identifies with the identity of $(G/H)\times U$, since

$$(G\times U)/H\cong(G/H)\times U \qquad \text{and} \qquad \pi_*\mathcal{P}|_U\cong(G/H)\times U.$$

It is therefore an isomorphism of sheaves. The contracted product $\pi_*\mathcal{P}$ is a G/H-torsor and, by Theorem 11, is represented by a finite étale X-scheme. Hence the same is true of $\mathcal{P}/H$.

Under the same local trivialization, the morphism $\mathcal{P} \to \mathcal{P}/H$ is the product of the quotient map with the identity,

$$G \times U \longrightarrow (G/H) \times U,$$

and is thus an H-torsor. Finally, suppose $\mathcal{P} \to \mathcal{Q} \to X$ is another such factorization. Since an H-torsor is locally split, every H-invariant morphism out of $\mathcal{P}$ descends uniquely through $\mathcal{Q}$. Thus $\mathcal{Q}$ satisfies the same universal property as $\mathcal{P}/H$, which gives the unique isomorphism compatible with the projections. □

1.3 Symmetric Powers of Schemes and Torsors

This section reviews the construction of quotients for schemes and torsors under finite group actions, specifically focusing on symmetric powers. To ensure these quotients exist as schemes, we utilize the framework of admissible actions from [SGA1]. Our treatment here closely follows the exposition in [Gui19] The definitions and results presented below are adapted from their work, the difference being that we ask Γ to act on the left, to make the later construction of symmetric powers more natural. This foundation provides the necessary criteria for admissibility and base change required to define the symmetric powers of a scheme X and a G-torsor $\mathcal{P}$ over X.

Let S be a scheme.

Definition 16 (([SGA1], V.1.7).)**.**

- Let T be an object of a category $\mathcal{C}$ endowed with a right action of a group Γ. We say that **the quotient T/Γ exists** in $\mathcal{C}$ if the covariant functor

$$\begin{aligned} \mathcal{C} &\to \mathrm{Sets} \\ U &\mapsto \mathrm{Hom}_{\mathcal{C}}(T, U)^{\Gamma} \end{aligned}$$

  is representable by an object of $\mathcal{C}$.

- Let T be an S-scheme. An action of a finite group Γ on T is **admissible** if there exists an affine Γ-invariant morphism $f : T \to T'$ such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^{\Gamma}$.

Proposition 17. *The following holds:*

1. ***([SGA1] V.1.3).*** *Let T be an S-scheme endowed with an admissible right action of a finite group Γ. If $f : T \to T'$ is an affine Γ-invariant morphism such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^{\Gamma}$, then the quotient T/Γ exists and is isomorphic to T'.*

2. ***([SGA1], V.1.8).*** *Let T be an S-scheme endowed with a right action of a finite group Γ. Then, the action of Γ on T is admissible if and only if T is covered by Γ-invariant affine open subsets.*

3. *(*[*SGA1*]*, V.1.9). Let T be an S-scheme endowed with an admissible right action of a finite group Γ, and let S' be a flat S-scheme. Then, the action of Γ on the S'-scheme $T \times_S S'$ is admissible, and the canonical morphism*

$$(T \times_S S')/\Gamma \to (T/\Gamma) \times_S S'$$

is an isomorphism.

Now let X be an S-scheme and let $d \geq 0$ be an integer. The group S_d of permutations of $[\![1, d]\!]$ acts on the right on the S-scheme $X^{\times_S d} = X \times_S \cdots \times_S X$ by the formula

$$(x_i)_{i \in [\![1,d]\!]} \cdot \sigma = (x_{\sigma(i)})_{i \in [\![1,d]\!]}.$$

Proposition 18 ([Gui19] Proposition 2.27)**.** *If X is a scheme, Zariski locally quasi-projective over S, then the right action of the symmetric group S_d on the d-fold fiber product $X^{\times_S d}$ is admissible. Consequently, the quotient $\mathrm{Sym}^d_S(X) = X^{\times_S d}/S_d$ exists as a scheme over S.*

Definition 19. Under the hypotheses of the proposition above, we define the **relative symmetric product** of X over S of degree d as the quotient

$$\mathrm{Sym}^d_S(X) := X^{\times_S d}/S_d.$$

When the base scheme S is clear from the context, we shall denote this quotient by $X^{(d)}$.

Guingard shows that when $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ then $\mathrm{Sym}^d_S(X)$ is representable by an affine S-scheme (See [Gui19] Remark 2.28).

Proposition 20 ([Gui19] Proposition 2.28)**.** *If X is flat and Zariski-locally quasi-projective over S, then $Sym^d_S(X)$ is flat over S. Moreover, for any S-scheme S', the canonical morphism*

$$Sym^d_{S'}(X \times_S S') \to Sym^d_S(X) \times_S S'$$

is an isomorphism.

Geometric Properties of Symmetric Powers

We record the geometric properties of symmetric powers that will be used in both main parts of the thesis.

Theorem 21. *Let X and Y be integral schemes over a field k. If X is geometrically integral, then $X \times_k Y$ is integral. If both X and Y are geometrically integral, then $X \times_k Y$ is geometrically integral.*

Proposition 22. *Let $f : X \to S$ be a smooth, Zariski-locally quasi-projective morphism of relative dimension one. Then the relative symmetric power $X^{(d)}_S = \mathrm{Sym}^d_S(X)$ is smooth over S of relative dimension d.*

Proof. Since $f : X \to S$ is smooth, the d-fold product X^d_S is smooth over S. By Proposition 20, the quotient $X^{(d)}_S$ is flat over S. Smoothness is a fiberwise property for flat morphisms of finite presentation (cf. [Stacks, Tag 01V8]); thus, it suffices to verify the smoothness of the fibers. For every $s \in S$, the fiber of the symmetric power is given by:

$$(\mathrm{Sym}^d_S(X))_s \cong \mathrm{Sym}^d_{\kappa(s)}(X_s)$$

Consequently, we may reduce to the case where $S = \mathrm{Spec}(k)$ for a field k. Since smoothness is preserved under base change to the algebraic closure, we may further assume $k = \bar{k}$. Let $z = \sum_{i=1}^{r} d_i \cdot x_i \in X^{(d)}$ be a point represented by an effective cycle of degree d, where the points $x_i \in X(k)$ are distinct and $\sum d_i = d$. Choose pairwise disjoint Zariski-open subsets $U_i \subset X$ such that $x_i \in U_i$. Let $W \subset X^{(d)}$ be the open subset consisting of cycles whose support is contained in $\bigcup U_i$ and which meet each U_i with degree exactly d_i. There is a canonical isomorphism:

$$W \cong \prod_{i=1}^{r} U_i^{(d_i)}$$

Thus, it suffices to show that each $U_i^{(d_i)}$ is smooth of dimension d_i at the point $d_i \cdot x_i$.

We may therefore restrict our attention to the "worst" case: the diagonal point $z = d \cdot x$. Let $R = \mathcal{O}_{X,x}$ be the local ring of the curve at x. Since X is a smooth curve over k, R is a regular local ring of dimension 1, i.e., a discrete valuation ring. Its completion is $\hat{R} \cong k[[t]]$. The completed local ring of the product X^d at the point $(x, \ldots, x)$ is:

$$\hat{\mathcal{O}}_{X^d,(x,\ldots,x)} \cong \hat{R} \hat{\otimes}_k \ldots \hat{\otimes}_k \hat{R} \cong k[[t_1, \ldots, t_d]]$$

where t_i is the uniformizer for the i-th copy of X. The symmetric group S_d acts on this power series ring by permuting the variables t_i. By the *Fundamental Theorem of Symmetric Polynomials*, the ring of invariants is:

$$\big(k[[t_1, \ldots, t_d]]\big)^{S^d} = k[[e_1, \ldots, e_d]]$$

where e_j is the j-th elementary symmetric power series in the variables t_i.

The ring $k[[e_1, \ldots, e_d]]$ is a formal power series ring in d variables. A fundamental theorem in commutative algebra states that a local ring is regular if and only if its completion is a formal power series ring over a field. Since $\hat{\mathcal{O}}_{X^{(d)},z} \cong k[[e_1, \ldots, e_d]]$, the local ring $\mathcal{O}_{X^{(d)},z}$ is a regular local ring of dimension d. This implies that $X^{(d)}$ is smooth over k of dimension d, completing the proof. □

Corollary 23. *Let C be a smooth, projective, geometrically connected curve over a field k. For every $d \geq 0$, the scheme $C^{(d)}$ is smooth and geometrically integral. Moreover, every finite product of symmetric powers of C is geometrically integral.*

Proof. The smoothness follows from Proposition 22. The curve C is geometrically integral by [Stacks, Tag 0366]. After any field extension, C^d is integral and its image under the quotient map $C^d \to C^{(d)}$ is irreducible. Since $C^{(d)}$ is smooth, it is geometrically reduced, and hence geometrically integral. The assertion about finite products now follows inductively from Theorem 21. □

Symmetric Powers of Torsors

Let G be a finite abelian group, let $\mathcal{P}$ be a G-torsor over an S-scheme X in $S_{\text{Ét}}$.

Proposition 24 ([[SGA1]], IX.5.8)**.** *Assume that $\mathcal{P}$ and X are endowed with right actions from a finite group Γ such that the morphism $\mathcal{P} \to X$ is Γ-equivariant, and that the following properties hold:*

(a) The right Γ-action on $\mathcal{P}$ commutes with the left G-action.

(b) The right Γ-action on X is admissible, and the quotient morphism $X \to X/\Gamma$ is finite.

(c) For any geometric point $\bar{x}$ of X, the action of the stabilizer $\Gamma_{\bar{x}}$ of $\bar{x}$ in Γ on the fiber $\mathcal{P}_{\bar{x}}$ of $\mathcal{P}$ at $\bar{x}$ is trivial.

Then the action of Γ on $\mathcal{P}$ is admissible, and $\mathcal{P}/\Gamma$ is a G-torsor over X/Γ in $S_{\acute{E}t}$.

By Theorem 11, $\mathcal{P}$ is representable by a finite étale X-scheme. For each $i \in [\![1,d]\!]$ let $p_i : X^{\times_S d} \to X$ be the projection on i-th factor, and let us consider the G-torsor

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} = K_G \backslash \mathcal{P}^{\times_S d}$$

over $X^{\times_S d}$, where $K_G \subseteq G^d$ is the kernel of the multiplication morphism $G^d \to G$ (the dependence on d is suppressed in the notation). [2] The object $K_G \backslash \mathcal{P}^{\times_S d}$ of $S_{\acute{E}t}$ is too representable by an S-scheme which is finite étale over $X^{\times_S d}$. The group S_d acts on the right on $K_G \backslash \mathcal{P}^{\times_S d}$ by the formula

$$(p_i)_{i\in[\![1,d]\!]} \cdot \sigma = (p_{\sigma(i)})_{i\in[\![1,d]\!]}.$$

This action of S_d commutes with the left action of G on $K_G \backslash \mathcal{P}^{\times_S d}$.

Proposition 25 ([Gui19] Proposition 2.32.)**.** *If X is Zariski-locally quasi-projective on S, then the right action of S_d on $K_G \backslash \mathcal{P}^{\times_S d}$ is admissible, so that the quotient $\mathcal{P}^{[d]}$ of $K_G \backslash \mathcal{P}^{\times_S d}$ by S_d exists as a scheme over S. Moreover, the canonical morphism $\mathcal{P}^{[d]} \to \mathrm{Sym}^d_S(X)$ is a G-torsor, and the morphism*

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} \to r^{-1}\mathcal{P}^{[d]}$$

where $r : X^{\times_S d} \to \mathrm{Sym}^d_S(X)$ is the canonical projection, is an isomorphism of G-torsors over $X^{\times_S d}$.

Definition 26 (Symmetric powers of rank-one local systems)**.** Let Λ be a finite commutative ring, let $G = \Lambda^\times$, and let $\mathcal{F}$ be a rank-one Λ-local system on X. If $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$, we define

$$\mathcal{F}^{[d]} := \mathcal{P}^{[d]} \times^G \Lambda$$

on $X^{(d)}$. Equivalently, $\mathcal{P}^{[d]}$ is the frame torsor of $\mathcal{F}^{[d]}$. By Proposition 25 and the monoidal compatibility in Proposition 9, its pullback along $r : X^{\times_S d} \to X^{(d)}$ satisfies

$$r^{-1}\mathcal{F}^{[d]} \cong p_1^{-1}\mathcal{F} \otimes_\Lambda \cdots \otimes_\Lambda p_d^{-1}\mathcal{F}.$$

Thus this definition agrees with the usual descent construction of the symmetric power of a rank-one local system.

When G is abelian, and letting the S_d-action be a left action (via the inverse), then the condition $\prod_{i=1}^d g_i = 1$ defining K_G is invariant under permuting the coordinates. Therefore, the left action of S_d normalizes the action of K_G. The group generated by these two actions is the semidirect product $\Xi = K_G \rtimes S_d$, which is a finite group acting on the left on $\mathcal{P}^d$. By Proposition 25, the induced action of S_d on $K_G \backslash \mathcal{P}^{\times_S d}$ is admissible. Hence the iterated quotient

$$(K_G \backslash \mathcal{P}^{\times_S d})/S_d$$

exists as an S-scheme. Moreover, since K_G is normal in $\Xi = K_G \rtimes S_d$, a morphism from $\mathcal{P}^{\times_S d}$ is Ξ-invariant if and only if it is K_G-invariant and the induced morphism from $K_G \backslash \mathcal{P}^{\times_S d}$ is S_d-invariant. Consequently, the universal properties of the quotients give a canonical isomorphism

$$\Xi \backslash \mathcal{P}^{\times_S d} \cong (K_G \backslash \mathcal{P}^{\times_S d})/S_d.$$

[2] Equivantly, the sum is obtained as the contracted product of the G^d-torsor $p_1^{-1}\mathcal{P}_1 \times \cdots \times p_d^{-1}\mathcal{P}_d$ along the multiplication map $m : G^d \to G$.

Here Ξ acts on $\mathcal{P}^{\times_S d}$ by the formula

$$((g_1,\ldots,g_d),\sigma)\star(p_1,\ldots,p_d)=(g_1p_{\sigma^{-1}(1)},g_2p_{\sigma^{-1}(2)},\ldots,g_dp_{\sigma^{-1}(d)})$$

Corollary 27. *Under the assumptions of the above discussion, there is a natural canonical morphism*

$$q:S_d\backslash\mathcal{P}^{\times_S d}\to\Xi\backslash\mathcal{P}^{\times_S d}$$

or equivlantly

$$q:S_d\backslash\mathcal{P}^{\times_S d}\to(K_G\backslash\mathcal{P}^{\times_S d})/S_d$$

between the symmetric power of $\mathcal{P}$ as a scheme and the symmetric power of $\mathcal{P}$ as a G-torsor. We make it a convention to denote the symmetric power of $\mathcal{P}$ as a scheme by $\mathcal{P}^{(d)}$, and the symmetric power of $\mathcal{P}$ as a G-torsor by $\mathcal{P}^{[d]}$. So that the above qoutient map is also written as: $q:\mathcal{P}^{(d)}\to\mathcal{P}^{[d]}$.

Proof. S_d is a subgroup of Ξ, hence any map that is Ξ-invariant is automatically S_d-invariant. □

Remark. Note that the above discussion does not require that $\mathcal{P}$ is a G-torsor over X, only that it is a scheme with a left G-action.

Compatibility with Addition

The construction of symmetric powers is compatible with a natural addition law. For any integers $d_1,d_2\geq 0$, there exists a canonical **addition morphism**

$$+_{d_1,d_2}:X^{(d_1)}\times_S X^{(d_2)}\to X^{(d_1+d_2)}$$

induced by the equivariant isomorphism $X^{\times_S d_1}\times_S X^{\times_S d_2}\cong X^{\times_S(d_1+d_2)}$ with respect to the inclusion of the product of symmetric groups $S_{d_1}\times S_{d_2}\subseteq S_{d_1+d_2}$.

Proposition 28. *Let $\mathcal{P}$ be a G-torsor over X. There is a canonical isomorphism of G-torsors over $X^{(d_1)}\times_S X^{(d_2)}$:*

$$+_{d_1,d_2}^{-1}\left(\mathcal{P}^{[d_1+d_2]}\right)\cong r_1^{-1}\mathcal{P}^{[d_1]}\otimes^G r_2^{-1}\mathcal{P}^{[d_2]}.$$

Proof. Let $\pi=r_{d_1}\times r_{d_2}$. By the commutativity of the following diagram

$$\begin{tikzcd}
X^{\times_S d_1}\times_S X^{\times_S d_2} \arrow[r, "\sim"] \arrow[d, "\pi"'] & X^{\times_S(d_1+d_2)} \arrow[d, "r_{d_1+d_2}"] \\
X^{(d_1)}\times_S X^{(d_2)} \arrow[r, "+_{d_1,d_2}"] & X^{(d_1+d_2)}
\end{tikzcd}$$

and applying Proposition 25, we obtain canonical isomorphisms:

$$\pi^{-1}\left(+_{d_1,d_2}^{-1}\mathcal{P}^{[d_1+d_2]}\right)\cong r_{d_1+d_2}^{-1}\mathcal{P}^{[d_1+d_2]}\cong\bigotimes_{i=1}^{d_1+d_2}{}^{G}\,p_i^{-1}\mathcal{P}$$

and

$$\pi^{-1}(r_1^{-1}\mathcal{P}^{[d_1]}\otimes^G r_2^{-1}\mathcal{P}^{[d_2]})\cong\left(\bigotimes_{i=1}^{d_1}{}^{G}\,p_i^{-1}\mathcal{P}\right)\otimes^G\left(\bigotimes_{j=d_1+1}^{d_1+d_2}{}^{G}\,p_j^{-1}\mathcal{P}\right).$$

Since both pullbacks identify with

$$\bigotimes_{k=1}^{d_1+d_2}{}^{G}\, p_k^{-1}\mathcal{P}$$

in an $S_{d_1} \times S_{d_2}$-equivariant manner, the isomorphism descends to the quotient $X^{(d_1)} \times_S X^{(d_2)}$ by Proposition 20. □

1.4 Basic Geometric Lemmas

We conclude with two elementary facts about codimension-one points and blowups. They are placed here because they enter the geometric arguments in both later chapters.

Proposition 29. *Let $f : X \to Y$ be a finite flat morphism between integral schemes of finite type over a field k. If $Z \subset X$ is a prime divisor with generic point η_Z, then $f(Z) \subset Y$ is a prime divisor, and its generic point is $f(\eta_Z)$.*

Proof. f is finite hence proper hence closed so $f(Z)$ is closed subset of Y, it is irreducible as the image of an irreducible. Since $Z = \overline{\{\eta_Z\}}$ we get:

$$\{f(\eta_Z)\} \subset f(Z) = f(\overline{\{\eta_Z\}}) \subseteq \overline{f(\{\eta_Z\})} = \overline{\{f(\eta_Z)\}}$$

And since $f(Z)$ is closed we get $f(Z) = \overline{\{f(\eta_Z)\}}$.

For flat map of integral schemes we have for every $x \in X$, $y = f(x)$ the dimension formula:

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X_y,x})$$

And since $\dim(\mathcal{O}_{X_y,x}) = 0$ we get $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y})$ concluding that $f(Z)$ is a prime divisor as well. □

Blowups

Theorem 30 ([Stacks, Tag 0805]). *Let $X_1 \to X_2$ be a flat morphism of schemes, let $Z_2 \subset X_2$ be a closed subscheme, and let Z_1 be its inverse image in X_1. If X_i' is the blowup of X_i along Z_i, then the diagram*

$$\begin{tikzcd} X_1' \arrow[r] \arrow[d] & X_2' \arrow[d] \\ X_1 \arrow[r] & X_2 \end{tikzcd}$$

is cartesian.

Proposition 31. *Let X be a smooth k-scheme of pure dimension $d \geq 1$, let $z \in X(k)$, and let*

$$\pi : \widetilde{X} = \mathrm{Bl}_z(X) \longrightarrow X$$

be the blowup at z. Then the exceptional divisor $E = \pi^{-1}(z)$ is isomorphic to $\mathbb{P}_k^{d-1}$. In particular, E is irreducible and has codimension 1 in $\widetilde{X}$. Consequently, it has a unique generic point η_E, and $\overline{\{\eta_E\}} = E$.

Proof. Because X is smooth of dimension d at the k-rational point z, after replacing X by a sufficiently small open neighborhood of z there is an étale morphism

$$f : X \longrightarrow \mathbb{A}^d_k$$

that sends z to the origin 0. We may shrink this neighborhood further so that z is the scheme-theoretic inverse image of 0. Since an étale morphism is flat, Theorem 30 reduces the assertion to the blowup of $\mathbb{A}^d_k$ at the origin.

Write $x_1, \ldots, x_d$ for the affine coordinates on $\mathbb{A}^d_k$ and $[u_1 : \cdots : u_d]$ for the homogeneous coordinates on $\mathbb{P}^{d-1}_k$. The blowup of the origin is the closed subscheme

$$\mathrm{Bl}_0(\mathbb{A}^d_k) = \big\{((x_1, \ldots, x_d), [u_1 : \cdots : u_d]) \bigm| x_i u_j = x_j u_i \text{ for all } i, j\big\} \subset \mathbb{A}^d_k \times_k \mathbb{P}^{d-1}_k.$$

Over the origin all the x_i vanish, so the displayed equations impose no condition on $[u_1 : \cdots : u_d]$. Hence the exceptional fiber is

$$\{0\} \times_k \mathbb{P}^{d-1}_k \cong \mathbb{P}^{d-1}_k.$$

It is therefore irreducible of dimension $d - 1$. The blowup has dimension d (for example, each standard chart $u_i \neq 0$ has d affine coordinates), so the exceptional divisor has codimension 1. $\square$

Theorem 32 ([Stacks, Tag 01OF], Lemma 31.33.9)**.** *If X is integral and $Z \subsetneq X$ is a proper closed subscheme, then the blowup $\mathrm{Bl}_Z(X)$ is integral.*

Chapter 2

Ramification After Blowup Is Tame

Throughout this chapter, let C be a smooth, projective, geometrically connected curve over k. Fix a modulus

$$\mathfrak{m} = \sum_i n_i P_i$$

on C, and set $U = C \setminus \mathfrak{m}$.

For each single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, set $d_{\mathfrak{n}} = \deg(\mathfrak{n})$. The effective divisor $\mathfrak{n}$ determines a k-rational closed point

$$Z_{\mathfrak{n}} \hookrightarrow C^{(d_{\mathfrak{n}})} = \mathrm{Sym}_k^{d_{\mathfrak{n}}}(C).$$

Let

$$\pi_{\mathfrak{n}} \colon X_{\mathfrak{n}} := \mathrm{Bl}_{Z_{\mathfrak{n}}}\big(C^{(d_{\mathfrak{n}})}\big) \longrightarrow C^{(d_{\mathfrak{n}})}$$

be the blowup at this point. We denote its exceptional divisor by

$$E_{\mathfrak{n}} := Z_{\mathfrak{n}} \times_{C^{(d_{\mathfrak{n}})}} X_{\mathfrak{n}}$$

By Propositions 22 and 31, $E_{\mathfrak{n}} \cong \mathbb{P}_k^{d_{\mathfrak{n}}-1}$; in particular, it is irreducible and has codimension 1 in $X_{\mathfrak{n}}$. We denote its unique generic point by $\eta_{\mathfrak{n}}$, so that $\overline{\{\eta_{\mathfrak{n}}\}} = E_{\mathfrak{n}}$. Thus we have the Cartesian diagram

$$\begin{tikzcd}
E_{\mathfrak{n}} = \overline{\{\eta_{\mathfrak{n}}\}} \arrow[r, hook] \arrow[d] & X_{\mathfrak{n}} \arrow[d, "\pi_{\mathfrak{n}}"] \\
Z_{\mathfrak{n}} \arrow[r, hook] & C^{(d_{\mathfrak{n}})}.
\end{tikzcd} \tag{2.1}$$

The purpose of this chapter is to prove Theorem 3. Let $\mathcal{F}$ be the rank-one Λ-local system appearing in that theorem, and let

$$\mathcal{P} = \mathrm{Fr}(\mathcal{F})$$

be its $G = \Lambda^\times$-torsor of frames. By Proposition 9 and definition 26, there is a canonical identification

$$\mathrm{Fr}\big(\mathcal{F}^{[d_\mathfrak{n}]}\big) \cong \mathcal{P}^{[d_\mathfrak{n}]}.$$

In particular, the two objects have the same ramification. It therefore suffices to prove the following torsor formulation.

Theorem 33. *Let G be a finite abelian group, and let $\mathcal{P} \to U$ be a G-torsor whose ramification is bounded by $\mathfrak{m}$. For every single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, the symmetric-power torsor $\mathcal{P}^{[d_\mathfrak{n}]}$ is tamely ramified at $\eta_\mathfrak{n}$.*

After recalling the necessary local ramification theory, we make two reductions. First, the general finite abelian case is reduced to cyclic groups. Second, formal–local invariance reduces the geometric calculation on C to a calculation on $\mathbb{G}_m \subset \mathbb{P}^1$. These reductions motivate the Kummer, Artin–Schreier, and Artin–Schreier–Witt theories used in the cyclic calculations that follow.

2.1 Ramification of Discrete Valuations

This section reviews the ramification of discrete valuations, beginning with the general theory of *discrete valuation rings* and their integral closures in finite separable extensions. We then specialize to complete rings in Galois extensions to describe the ramification filtration via lower and upper numbering, following the treatments in [Stacks, Tag 0EXQ] and [Ser79].

Definition 34 ([Stacks, Tag 09E4])**.** We say that $A \to B$ or $A \subset B$ is an extension of discrete valuation rings if A and B are discrete valuation rings and $A \to B$ is injective and local. In particular, if π_A and π_B are uniformizers of A and B, then $\pi_A = u\pi_B^e$ for some $e \geq 1$ and unit u of B. The integer e does not depend on the choice of the uniformizers as it is also the unique integer ≥ 1 such that

$$\mathfrak{m}_A B = \mathfrak{m}_B^e$$

The integer e is called the *ramification index* of B over A. We say that B is *weakly unramified* over A if $e = 1$. If the extension of residue fields $\kappa_A = A/\mathfrak{m}_A \subset \kappa_B = B/\mathfrak{m}_B$ is finite, then we set $f = [\kappa_B : \kappa_A]$ and we call it the *residual degree* or residue degree of the extension $A \subset B$.

Note that we do not require the extension of fraction fields to be finite.

Now let A be a discrete valuation ring with fraction field K, let L/K be a finite separable field extension and let $B \subset L$ be the integral closure of A in L. Then the ring extension $A \subset B$ is finite, hence B is Noetherian. The dimension of B is 1, hence B is a Dedekind domain. Let $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$ be the maximal ideals of B (i.e., the primes lying over $\mathfrak{m}_A$). We obtain extensions of discrete valuation rings

$$A \subset B_{\mathfrak{m}_i}$$

and hence ramification indices e_i and residue degrees f_i. We have

$$[L:K]=\sum_{i=1,\dots,n} e_i f_i$$

Note that if A is henselian (e.g. A is complete) then $n=1$.

Definition 35 ([Stacks, Tag 09E9])**.** Let A be a discrete valuation ring with fraction field K. Let L/K be a finite separable extension. With B and $\mathfrak{m}_i$, $i=1,\dots,n$ as above, we say the extension L/K is

1. unramified with respect to A if $e_i=1$ and the extension $\kappa(\mathfrak{m}_i)/\kappa_A$ is separable for all i,
2. tamely ramified with respect to A if either the characteristic of κ_A is 0 or the characteristic of κ_A is $p>0$, the field extensions $\kappa(\mathfrak{m}_i)/\kappa_A$ are separable, and the ramification indices e_i are prime to p, and
3. totally ramified with respect to A if $n=1$ and the residue field extension $\kappa(\mathfrak{m}_1)/\kappa_A$ is trivial.

If the discrete valuation ring A is clear from context, then we sometimes say L/K is unramified, totally ramified, or tamely ramified for short.

For the remainder of this section, assume A is a *complete discrete valuation ring* with uniformizer π and a perfect residue field κ. In the equal characteristic case, where $\mathrm{char}(A)=\mathrm{char}(\kappa)=p>0$, the Cohen Structure Theorem implies that A contains a coefficient field $k\cong\kappa$, such that $A\cong k[[\pi]]\cong k[[t]]$. Consequently, the fraction field is given by $K=k((\pi))$. Let L/K be a finite Galois extension with Galois group G. The integral closure B of A in L is itself a complete discrete valuation ring. Since the residue field κ is perfect, the extension of residue fields is separable, and there exists a uniformizer $\Pi\in B$ such that $B=A[\Pi]$.

The *lower numbering ramification filtration* of G, denoted by $(G_i)_{i\geq -1}$, is defined as:

$$G_i=\{\sigma\in G \mid v_B(\sigma(x)-x)\geq i+1 \text{ for all } x\in B\}$$

where v_B is the valuation on L associated with B. In this indexing, $G_{-1}=G$ and G_0 corresponds to the *inertia group* of the extension L/K. Note that L/K is unramified if and only if $G_0=\{1\}$, and it is tamely ramified if and only if $G_1=\{1\}$.

A standard result in the theory of local fields (complete local fields with a discrete valuation and perfect residue field) ensures that the condition in the definition of G_i need only be checked for the uniformizer Π of B. Specifically, if we define the function

$$i_K^L(\sigma)=v_B(\sigma(\Pi)-\Pi)$$

for $\sigma\in G$, then $G_i=\{\sigma\in G \mid i_K^L(\sigma)\geq i+1\}$. The subgroups G_i are normal in G and become trivial for sufficiently large i.

Consider a tower of fields $K\subset E\subset L$, and let $H=\mathrm{Gal}(L/E)\subset G$. The lower numbering is compatible with subgroups:

$$G_i\cap H=H_i \quad \text{for all } i\geq -1$$

which follows from the identity $i_E^L=i_K^L|_H$. However, the lower numbering does not behave as simply with respect to quotients. If H is normal in G, the image of G_i in G/H is generally not $(G/H)_i$. Instead, we have $G_iH/H=(G/H)_j$, where the index j is given by the formula:

$$j=\frac{1}{e_{L/E}}\sum_{\tau\in H}\min(i_K^L(\tau),i+1)-1$$

In the literature, one reindexes the ramification groups by defining the Herbrand function $\phi_{L/K} : [-1,\infty) \to [-1,\infty)$:

$$\phi_{L/K}(i) = \frac{1}{e_{L/K}} \sum_{\sigma \in G} \min(i_K^L(\sigma), i+1) - 1 = \int_0^i \frac{1}{[G_0 : G_t]} dt$$

This function is continuous, strictly increasing, and piecewise linear, making it a bijection. It satisfies the composition law $\phi_{L/K} = \phi_{E/K} \circ \phi_{L/E}$ for a tower of extensions $K \subset E \subset L$. Using this function, we define the *upper numbering ramification groups* as:

$$G^i = G_{\phi_{L/K}^{-1}(i)}$$

The primary advantage of this reindexing is its compatibility with quotients: for any normal subgroup $H \subset G$, we have:

$$G^i H/H = (G/H)^i$$

for all $i \geq -1$.

We record, for orientation, the structure of the inertia group and of the graded pieces of the filtration; these facts are not used in the sequel, but they explain the dichotomy — prime-to-p versus p-power — that shapes the whole chapter.

Proposition 36 ([Ser79], IV Corollary 3)**.** *The quotients G_i/G_{i+1}, $i \geq 1$, are abelian groups, and are direct products of cyclic groups of order p. The group G_1 is a p-group.*

Proposition 37 ([Ser79], IV Corollary 4)**.** *The inertia group G_0 has the following property: It is the semi-direct product of a cyclic group of order prime to p with a normal subgroup whose order is a power of p.*

2.2 Ramification of G-Torsors

Generally speaking, assume we are given a locally Noetherian scheme X, a dense open $U \subset X$, a finite étale morphism $f : Y \to U$ and a prime divisor $Z \subset X$ such that $Z \cap U = \emptyset$ and the local ring $\mathcal{O}_{X,\xi}$ of X at the generic point ξ of Z is a discrete valuation ring. Setting $K_\xi = \mathrm{Frac}(\mathcal{O}_{X,\xi})$ we obtain a cartesian square

$$\begin{tikzcd} \mathrm{Spec}(K_\xi) \arrow[r] \arrow[d] & U \arrow[d] \\ \mathrm{Spec}(\mathcal{O}_{X,\xi}) \arrow[r] & X \end{tikzcd}$$

of schemes. In particular, we see that $Y \times_U \mathrm{Spec}(K_\xi)$ is the spectrum of a finite separable algebra L_ξ/K_ξ.

Definition 38. We say:

1. Y is *unramified* over X at (Z,ξ) resp. Y is *tamely ramified* over X at (Z,ξ) if L_ξ/K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ (Definition 35)

2. Y is unramified over X in codimension 1, resp. Y is tamely ramified over X in codimension 1 if L_ξ/K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ for every (Z,ξ) as above.

More precisely, we decompose L_ξ into a product of finite separable field extensions of K_ξ and we require each of these to be unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$.

Now back to our original assumptions, C is a projective smooth geometrically connected curve over a field k and $U = C \setminus \mathfrak{m}$. Every (Z, ξ) is now a closed point P and $\mathcal{O}_{C,P}$ is a discrete valuation ring. Assume $f : Y \to U$ is actually a G-torsor $\mathcal{P} \to U$ in $U_{\text{Ét}}$ for G a finite abelian group.

By passing to the completion $\widehat{\mathcal{O}}_{C,P}$, we obtain the complete local field $\widehat{K}_P$. Restricting the torsor $\mathcal{P}$ to the punctured formal neighborhood $\operatorname{Spec}(\widehat{K}_P)$ allows us to study the ramification in a strictly local context. This restriction yields a G-torsor over a field, which necessarily decomposes into a disjoint union of spectra of finite separable field extensions

$$\mathcal{P}|_{\operatorname{Spec}(\widehat{K}_P)} \cong \bigsqcup \operatorname{Spec}(F)$$

such that:

1. These extensions $F/\widehat{K}_P$ are all isomorphic and Galois.
2. The Galois group $H = \operatorname{Gal}(F/\widehat{K}_P)$ identifies with the stabilizer of any connected component of the pullback $\mathcal{P}|_{\operatorname{Spec}(\widehat{K}_P)}$, which is a subgroup of G.
3. The number of such components is then given by the index $[G : H]$.

Definition 39. We say that the extension $F/\widehat{K}_P$ *has ramification bounded by* r if the ramification group H^r is trivial. We say $\mathcal{P}$ *has ramification bounded by* r *at* P if the corresponding local field extensions $F/\widehat{K}_P$ satisfy this condition.

Definition 40. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. We say the G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if, for each point P_i, the local restriction $\mathcal{P}|_{\operatorname{Spec}(\widehat{K}_{P_i})}$ has ramification bounded by the coefficient n_i.

This local property remains invariant under base change to the separable closure. Let $\bar{s} = \operatorname{Spec}(\bar{k}) \to \operatorname{Spec}(k)$ be a geometric point. The higher ramification groups for $\mathcal{P}|_{\operatorname{Spec}(\widehat{K}_P)}$ are isomorphic to those of the geometric restriction $\mathcal{P}_{\bar{k}}$ at any point $\bar{x}$ lying over P. Thus an equivalent definition:

Definition 41. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. The G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if for every geometric point $\bar{x}$ of $\mathfrak{m}$ (with image $\bar{s}$ in $\operatorname{Spec}(k)$), the restriction of $\mathcal{P}$ to

$$\operatorname{Spec}(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}) \times_{C_{\bar{k}}} U_{\bar{k}}$$

has ramification bounded by the multiplicity of $\mathfrak{m}_{\bar{s}}$ at $\bar{x}$ in the sense of Definition 39.

These definitions are equivalent because $\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}} \cong \widehat{\mathcal{O}}_{C,P} \otimes_k \bar{k}$. Furthermore, the notions of tame ramification and unramifiedness derived here coincide with the codimension-1 criteria in Definition 38.

There is another equivalent formulation through the language of characters. The group G, being a finite abelian group over k, is étale. Consequently, the correspondence between G-torsors and the fundamental group allows us to describe G-torsors via characters. Specifically, there is an isomorphism (Corollary 14):

$$\operatorname{Hom}_{\text{cont.}}(\pi_1^{\text{ét}}(U, \overline{x}), G) \xrightarrow{\cong} \operatorname{Tors}(U_{\text{ét}}, G)$$

In our local setting, where we consider the restriction to the complete local field $\widehat{K}_P$, the fundamental group identifies with the absolute Galois group $G_{\widehat{K}_P} := \mathrm{Gal}(\widehat{K}_P^{\mathrm{sep}}/\widehat{K}_P)$. Thus, the restricted G-torsor $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$ corresponds to a unique continuous homomorphism:

$$\rho : G_{\widehat{K}_P} \to G$$

Under this correspondence, the torsor $\mathcal{P}$ has ramification bounded by r at P if and only if the homomorphism ρ trivializes the r-th ramification group in the upper numbering:

$$\rho(G^r_{\widehat{K}_P}) = \{1\}$$

Basic Properties of Ramification of G-Torsors

We now prove some elementary lemmas regarding the ramification of G-torsors.

Lemma 42. *Let G be a finite abelian group and X be a locally Noetherian scheme over a field k. Let $U \subset X$ be a dense open subset and let (Z, ξ) be a prime divisor in the complement $X \setminus U$. Assume $\mathcal{P}_1$ and $\mathcal{P}_2$ are two G-torsors on $U_{\acute{e}t}$, such that $\mathcal{P}_1$ has ramification bounded by r_1 at (Z, ξ), and $\mathcal{P}_2$ has ramification bounded by r_2 at (Z, ξ). Then their product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ has ramification bounded by* $\max(r_1, r_2)$ *at (Z, ξ).*

Proof. Let $A = \widehat{\mathcal{O}_{X,\xi}}$ and let $K = \mathrm{Frac}(A)$. Let $\rho_1, \rho_2 : G_K \to G$ be the associated continuous homomorphisms corresponding to the G-torsors $\mathcal{P}_1|_{\mathrm{Spec}(K)}$, $\mathcal{P}_2|_{\mathrm{Spec}(K)}$. We finish by noticing that the associated character of $(\mathcal{P}_1 \otimes^G \mathcal{P}_2)|_{\mathrm{Spec}(K)}$ is $\rho = \rho_1 + \rho_2$. □

Lemma 43. *Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Let $U_X \subset X$ and $U_Y \subset Y$ be dense open subschemes such that $f^{-1}(U_Y) \subset U_X$.*

Let (Z_X, ξ_X) and (Z_Y, ξ_Y) be prime divisors of X and Y such that $Z_X \cap U_X = \emptyset$ and $Z_Y \cap U_Y = \emptyset$. Suppose $f(\xi_X) = \xi_Y$ and that f is étale at ξ_X. Let $\mathcal{P}$ be a G-torsor over U_Y, and let $f^{-1}\mathcal{P}$ be its pullback to $f^{-1}(U_Y) \subset U_X$. Then, the ramification of $f^{-1}\mathcal{P}$ at ξ_X is bounded by r if and only if the ramification of $\mathcal{P}$ at ξ_Y is bounded by r.

Proof. The boundedness of ramification is determined by the behavior of the torsor over the completion of the local rings at the generic points. Let $A = \mathcal{O}_{Y,\xi}$ and $B = \mathcal{O}_{X,\xi_X}$ be the discrete valuation rings at the generic points, with fraction fields K and L respectively. Since f is étale at ξ_X, the map $A \to B$ is a flat, unramified (hence weakly unramified) local homomorphism. Consequently, the extension of completions $\widehat{L}/\widehat{K}$ is a finite unramified extension of complete discretely valued fields. We finish by noticing that the upper numbering filtration on the absolute Galois group is compatible with unramified base change.[1] □

Lemma 44. *Let G_1, G_2 be finite abelian groups and let $X, U, (Z, \xi)$ be as in Lemma 42. Let $\mathcal{P}_1$ be a G_1-torsor and $\mathcal{P}_2$ a G_2-torsor on $U_{\acute{e}t}$, and consider the fiber product $\mathcal{P}_1 \times_U \mathcal{P}_2$, which is a $(G_1 \times G_2)$-torsor on $U_{\acute{e}t}$. If $\mathcal{P}_i$ has ramification bounded by r_i at (Z, ξ) for $i = 1, 2$, then $\mathcal{P}_1 \times_U \mathcal{P}_2$*

[1] Specifically, let $G_K = \mathrm{Gal}(K^{sep}/K)$ and $G_L = \mathrm{Gal}(L^{sep}/L)$. For an unramified extension, the Herbrand function is the identity, which implies that for any $r \geq 0$:

$$G^r_L = G^r_K \cap G_L$$

has ramification bounded by $\max(r_1, r_2)$ *at* (Z, ξ)*. Moreover,* $\mathcal{P}_1 \times_U \mathcal{P}_2$ *is tamely ramified (resp. unramified) at* (Z, ξ) *if and only if both* $\mathcal{P}_1$ *and* $\mathcal{P}_2$ *are.*

Proof. Let $A = \widehat{\mathcal{O}_{X,\xi}}$, $K = \mathrm{Frac}(A)$, and let $\rho_i : G_K \to G_i$ be the characters of $\mathcal{P}_i|_{\mathrm{Spec}(K)}$. The character of $(\mathcal{P}_1 \times_U \mathcal{P}_2)|_{\mathrm{Spec}(K)}$ is $\rho = (\rho_1, \rho_2) : G_K \to G_1 \times G_2$, since torsors under a product group correspond to pairs of characters, $H^1(K, G_1 \times G_2) = H^1(K, G_1) \times H^1(K, G_2)$, compatibly with the projections. Now for any closed subgroup $N \subset G_K$ (we take the upper-numbering group G_K^r, resp. the wild inertia subgroup, resp. the inertia subgroup) we have $\rho(N) = \{1\}$ if and only if $\rho_1(N) = \{1\}$ and $\rho_2(N) = \{1\}$. □

Lemma 45. *Let* $\iota : H \hookrightarrow G$ *be an injective homomorphism of finite abelian groups, let* $X, U, (Z, \xi)$ *be as above, let* $\mathcal{Q}$ *be an* H*-torsor on* $U_{\acute{e}t}$*, and let* $\iota_*\mathcal{Q} := (\mathcal{Q} \times G)/H$ *be the induced* G*-torsor (where* H *acts by* $h \cdot (q, g) = (qh,\ \iota(h)^{-1}g)$*). Then* $\iota_*\mathcal{Q}$ *has ramification bounded by* r *at* (Z, ξ) *if and only if* $\mathcal{Q}$ *does; likewise for tame ramification and for unramifiedness.*

Proof. Let $A = \widehat{\mathcal{O}_{X,\xi}}$, $K = \mathrm{Frac}(A)$, and let $\chi : G_K \to H$ be the character of $\mathcal{Q}|_{\mathrm{Spec}(K)}$; the character of $(\iota_*\mathcal{Q})|_{\mathrm{Spec}(K)}$ is $\iota \circ \chi$. Since ι is injective, $\iota \circ \chi$ and χ have the same kernel, so for any closed subgroup $N \subset G_K$ we have $(\iota \circ \chi)(N) = \{1\}$ if and only if $\chi(N) = \{1\}$. (Concretely: the connected components of the two torsors over $\mathrm{Spec}(K)$ are spectra of the *same* field extension F/K, the fixed field of $\ker \chi$; the induced torsor merely has $[G : H]$ times as many components.) □

2.3 Two Reductions Before the Cyclic Calculations

Before introducing explicit equations, we explain the strategy of the proof and isolate two reductions that simplify the situation. The theorem concerns a torsor under an arbitrary finite abelian group on an arbitrary smooth curve, whereas the calculation we ultimately wish to perform is local and cyclic. We show that both the structure group and the geometry may be simplified without changing the ramification problem relevant to the exceptional divisor.

The first reduction is algebraic: using the decomposition of a finite abelian group into cyclic factors, together with the compatibility of the symmetric-power construction and ramification bounds with products and quotients, we reduce the theorem to cyclic groups of order prime to p and cyclic groups of order p^r. This explains why the three cyclic theories recalled in the next section suffice. The second reduction is geometric and formal–local: the ramification of the induced torsor at the generic point of the exceptional divisor depends only on the restriction of the original torsor to the punctured formal neighbourhood of the point supporting the modulus. It therefore tells us exactly what the cyclic theories must accomplish: realize the resulting local cyclic torsor by an explicit equation on $\mathbb{G}_m \subset \mathbb{P}^1_k$.

Together, these reductions transform the original problem into three parallel explicit calculations, corresponding to Kummer, Artin–Schreier, and Artin–Schreier–Witt theory. The following section recalls precisely the conclusions from these theories that provide the required realizations.

2.3.1 Reduction from Finite Abelian Groups to Cyclic Groups

Fix a single-point submodulus $\mathfrak{n}$ and write $d = \deg(\mathfrak{n})$. The first reduction concerns the structure group and is independent of the geometry of the curve.

Proposition 46 (Reduction to cyclic groups). *Suppose that the conclusion of Theorem 33 is known whenever the structure group is cyclic of order prime to p, and whenever the structure group is a cyclic p-group. Then it holds for every finite abelian group G.*

Proof. By the structure theorem for finite abelian groups, choose a decomposition

$$G \cong \prod_{i=1}^{s} \mathbb{Z}/e_i\mathbb{Z} \times \prod_{j=1}^{t} \mathbb{Z}/p^{r_j}\mathbb{Z}, \qquad (e_i, p) = 1.$$

Let $\rho : \pi_1^{\text{ét}}(U, \bar{x}) \to G$ be the character corresponding to $\mathcal{P}$. Composing ρ with the projections to the displayed cyclic factors gives torsors $\mathcal{P}_i$ under $\mathbb{Z}/e_i\mathbb{Z}$ and torsors $\mathcal{Q}_j$ under $\mathbb{Z}/p^{r_j}\mathbb{Z}$. Since the character of each factor is a quotient of ρ, the ramification bound on $\mathcal{P}$ passes to every $\mathcal{P}_i$ and $\mathcal{Q}_j$.

The original torsor is recovered as their fiber product over U:

$$\mathcal{P} \cong \mathcal{P}_1 \times_U \cdots \times_U \mathcal{P}_s \times_U \mathcal{Q}_1 \times_U \cdots \times_U \mathcal{Q}_t.$$

The construction of the symmetric-power torsor commutes with this product. Indeed, after pullback to U^d, both sides are obtained by taking the contracted product of the d pullbacks along multiplication in the structure group. Hence

$$\mathcal{P}^{[d]} \cong \mathcal{P}_1^{[d]} \times_{U^{(d)}} \cdots \times_{U^{(d)}} \mathcal{P}_s^{[d]} \times_{U^{(d)}} \mathcal{Q}_1^{[d]} \times_{U^{(d)}} \cdots \times_{U^{(d)}} \mathcal{Q}_t^{[d]}.$$

By the assumed cyclic cases, every factor on the right is tamely ramified at $\eta_{\mathfrak{n}}$. The fiber product is therefore tamely ramified there by Lemma 44. This proves the assertion for G. ☐

Thus it remains to understand cyclic groups. Their prime-to-p part is described by Kummer theory, while cyclic groups of order p and p^r are described by Artin–Schreier and Artin–Schreier–Witt theory, respectively.

2.3.2 Formal–Local Reduction to $\mathbb{P}^1$

We argue that it is enough to prove Theorem 33 for $C = \mathbb{P}^1_k$, $\mathfrak{n} = d \cdot 0$, and $\mathcal{P} \to \mathbb{G}_m = \mathbb{P}^1_k \setminus \{0, \infty\}$ a G-torsor given by an explicit global equation. Informally, this is because *the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is a formal–local invariant of $\mathcal{P}$ at P*: it depends only on the completed local ring $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$ together with the restriction of $\mathcal{P}$ to the punctured formal disc $\operatorname{Spec}(k((x)))$, since all of the constructions involved — self products, quotients by finite groups, and blowups — commute with the flat base change to the completion; for blowups, this is Theorem 30. We emphasize that this is not a general principle that can simply be quoted: it is a statement about the particular constructions at hand, and it is proved by checking each construction separately. This is the content of Lemma 47 and Corollary 48 below. The second ingredient of the reduction is Lemma 53: every local torsor class appearing in Theorem 33 is realized by an explicit torsor over $\mathbb{G}_m \subset \mathbb{P}^1_k$.

Throughout this section we assume that k is algebraically closed; in particular every closed point of C is k-rational, and $\mathfrak{n} = dP$ with $P \in C(k)$ and $d = \deg \mathfrak{n}$.

Remark. The assumption that k is algebraically closed is harmless in our situation, for the following reasons.

1. **(Base change to $\bar{k}$.)** Assume k is perfect, so that $\bar{k} = k^{sep}$. The formation of $C^{(d)}$ (a quotient by a finite group), of the blowup $X_{\mathfrak{n}}$ along $Z_{\mathfrak{n}}$, and of $\mathcal{P}^{[d]}$ (Corollary 27) commutes with the flat base change $\mathrm{Spec}(\bar{k}) \to \mathrm{Spec}(k)$. Since $E_{\mathfrak{n}} \cong \mathbb{P}_k^{\deg \mathfrak{n}-1}$ is geometrically irreducible, $(E_{\mathfrak{n}})_{\bar{k}}$ is irreducible and its generic point $\bar{\eta}$ lies over $\eta_{\mathfrak{n}}$; moreover $\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}} \to \mathcal{O}_{(X_{\mathfrak{n}})_{\bar{k}},\bar{\eta}}$ is a weakly unramified extension of discrete valuation rings with separable residue field extension. The proof of Lemma 43 uses only these two properties of the extension (there the extension was moreover finite étale), so it applies verbatim: the ramification of $(\mathcal{P}_{\bar{k}})^{[d]} = (\mathcal{P}^{[d]})_{\bar{k}}$ at $\bar{\eta}$ is bounded by r if and only if the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is. Hence unramifiedness and tameness at $\eta_{\mathfrak{n}}$ may be checked after base change to $\bar{k}$.

2. **(Rationality of P — a caveat.)** If P is a closed point with $\kappa(P) \neq k$, then after base change to $\bar{k}$ the point $Z_{\mathfrak{n}}$ corresponds to the divisor $\sum_{\bar{P}\mapsto P} n\bar{P}$ on $C_{\bar{k}}$, whose support consists of $[\kappa(P):k]$ distinct points, whereas the computation below treats a modulus supported at a single rational point. We therefore prove Theorem 33 under the standing assumption that k is algebraically closed; this suffices for our purposes, since in the proof of Theorem 62 the field is assumed algebraically closed before the present theorem is invoked.

We now formulate the formal–local invariance precisely. Fix $P \in C(k)$ lying in the support of $\mathfrak{m}$, set $\mathfrak{n} = dP$, and fix a uniformizer x of $\mathcal{O}_{C,P}$; it induces an isomorphism $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$. The scheme $\mathrm{Spec}(k[[x]])$ has two points — the closed point, mapping to P, and the generic point, mapping to the generic point of C — so the punctured formal disc $\mathrm{Spec}(k((x)))$ maps to U, and we denote by

$$\mathcal{P}_x := \mathcal{P} \times_U \mathrm{Spec}(k((x)))$$

the corresponding restriction of $\mathcal{P}$; it is a G-torsor over $\mathrm{Spec}(k((x)))$. Let $e_1, \ldots, e_d \in k[[x_1, \ldots, x_d]]$ be the elementary symmetric polynomials in $x_1, \ldots, x_d$, and set

$$V = \mathrm{Spec}\Big(k[[e_1, \ldots, e_d]][e_d^{-1}]\Big), \qquad W = \mathrm{Spec}\Big(k[[x_1, \ldots, x_d]][(x_1 \cdots x_d)^{-1}]\Big).$$

The inclusion $k[[e_1, \ldots, e_d]] \subset k[[x_1, \ldots, x_d]]$ induces a finite flat morphism $\hat{r} : W \to V$ (note that inverting $e_d = x_1 \cdots x_d$ inverts each x_i), and for each i we let

$$q_i : W \to \mathrm{Spec}(k((x))), \qquad x \mapsto x_i.$$

The group S_d acts on W over V by permuting $x_1, \ldots, x_d$, and $q_{\sigma(i)} = q_i \circ \sigma$.

Lemma 47 (Formal–local invariance). *In the above notation:*

1. *The choice of x induces an isomorphism $\widehat{\mathcal{O}_{C^{(d)},Z_{\mathfrak{n}}}} \cong k[[e_1, \ldots, e_d]]$ carrying the maximal ideal to $(e_1, \ldots, e_d)$, such that the induced morphism $c : \mathrm{Spec}(k[[e_1, \ldots, e_d]]) \to C^{(d)}$ is flat and satisfies $c^{-1}(U^{(d)}) = V$; we write $c_V : V \to U^{(d)}$ for the restriction.*

2. *There is an induced isomorphism $\widehat{\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}}} \cong \kappa[[e_d]]$, where $\kappa = k(\frac{e_1}{e_d}, \ldots, \frac{e_{d-1}}{e_d})$. In particular $\widehat{K}_{\eta_{\mathfrak{n}}} := \mathrm{Frac}(\widehat{\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}}}) \cong \kappa((e_d))$, and the canonical morphism $\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}}) \to U^{(d)}$ factors as $\mathrm{Spec}(\kappa((e_d))) \to V \xrightarrow{c_V} U^{(d)}$, where the first map does not depend on $(C, \mathcal{P})$.*

3. *There is a canonical isomorphism of G-torsors over V*

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \;\cong\; \Xi \backslash \Big(q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x\Big).$$

Consequently, the G-torsor $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ — and hence whether the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is bounded by r (resp. tame, resp. unramified) — depends only on d and on the isomorphism class of the G-torsor $\mathcal{P}_x$ over $k((x))$, and not otherwise on the pair $(C,\mathcal{P})$.

Proof. **(1)** Since P is a k-rational point of the smooth curve C, the uniformizer x induces $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$. As completion at rational points takes products of k-varieties to completed tensor products, we get $\widehat{\mathcal{O}_{C^d,(P,\dots,P)}} \cong k[[x]]\widehat{\otimes}_k \cdots \widehat{\otimes}_k k[[x]] = k[[x_1,\dots,x_d]]$. Write $A = \mathcal{O}_{C^{(d)},Z_{\mathfrak{n}}}$ and $B = \mathcal{O}_{C^d,(P,\dots,P)}$. Since k is algebraically closed, a closed point $(Q_1,\dots,Q_d) \in C^d$ maps to $Z_{\mathfrak{n}} = dP$ under the finite projection $r : C^d \to C^{(d)}$ if and only if $Q_1 = \cdots = Q_d = P$; hence $(P,\dots,P)$ is the unique point of C^d over $Z_{\mathfrak{n}}$, $B = (r_*\mathcal{O}_{C^d})_{Z_{\mathfrak{n}}}$ is a finite A-module, and $A = B^{S_d}$ (formation of invariants commutes with localization at invariant primes). Since $\mathfrak{m}_B$ is the unique maximal ideal of B over $\mathfrak{m}_A$, we have $\mathfrak{m}_B^N \subset \mathfrak{m}_A B \subset \mathfrak{m}_B$ for some N, so the $\mathfrak{m}_A$-adic and $\mathfrak{m}_B$-adic topologies on B agree, and therefore $B \otimes_A \hat{A} \cong \hat{B} \cong k[[x_1,\dots,x_d]]$. Now $A = B^{S_d}$ is the kernel of $B \to \prod_{\sigma\in S_d} B$, $b \mapsto (\sigma(b)-b)_\sigma$; as $\hat{A}$ is flat over A, tensoring with $\hat{A}$ preserves this kernel, so

$$\hat{A} \cong (B \otimes_A \hat{A})^{S_d} \cong k[[x_1,\dots,x_d]]^{S_d}.$$

Finally, $k[[x_1,\dots,x_d]]^{S_d} = k[[e_1,\dots,e_d]]$: the ring $k[x_1,\dots,x_d]$ is a finite free module over $k[e_1,\dots,e_d]$, the same topological argument identifies $k[[x_1,\dots,x_d]]$ with $k[x_1,\dots,x_d] \otimes_{k[e_1,\dots,e_d]} k[[e_1,\dots,e_d]]$, and taking S_d-invariants — again a kernel, preserved by the flat base change $k[e_1,\dots,e_d] \to k[[e_1,\dots,e_d]]$ — yields $k[[e_1,\dots,e_d]]$. Under these identifications the maximal ideal of $\hat{A}$ is $(e_1,\dots,e_d)$, and c is flat as the composition of the flat morphisms $\mathrm{Spec}(\hat{A}) \to \mathrm{Spec}(A) \to C^{(d)}$.

It remains to check that $c^{-1}(C^{(d)} \setminus U^{(d)}) = V(e_d)$ as sets. We have $C^{(d)} \setminus U^{(d)} = \bigcup_{Q\in\mathrm{Supp}(\mathfrak{m})} Z_Q$ with $Z_Q = \{D : Q \in \mathrm{Supp}(D)\}$. For $Q \neq P$ the divisor Z_Q does not contain the point $Z_{\mathfrak{n}} = dP$, so its ideal is not contained in $\mathfrak{m}_A$, and $c^{-1}(Z_Q) = \emptyset$. For $Q = P$, pull back along the finite surjective morphism $\mathrm{Spec}(\hat{B}) \to \mathrm{Spec}(\hat{A})$: the preimage of Z_P in C^d is $\bigcup_i p_i^{-1}(P)$, whose trace on $\mathrm{Spec}(\hat{B}) = \mathrm{Spec}(k[[x_1,\dots,x_d]])$ is $V(x_1)\cup\cdots\cup V(x_d) = V(x_1\cdots x_d) = V(e_d)$; by surjectivity, $c^{-1}(Z_P) = V(e_d)$ as claimed. Hence $c^{-1}(U^{(d)}) = V$, and likewise $\mathrm{Spec}(\hat{B}) \times_{C^d} U^d = W$.

(2) Blowing up commutes with flat base change, and the center $Z_{\mathfrak{n}}$ pulls back under c to the closed point $V(e_1,\dots,e_d)$; hence

$$\hat{Y} := \mathrm{Bl}_{(e_1,\dots,e_d)}\big(\mathrm{Spec}\, k[[e_1,\dots,e_d]]\big) \cong X_{\mathfrak{n}} \times_{C^{(d)}} \mathrm{Spec}(\hat{A}),$$

and the exceptional divisor $\hat{E} \subset \hat{Y}$ is the base change of $E_{\mathfrak{n}}$ along $\mathrm{Spec}(\hat{A}/\hat{\mathfrak{m}}) = \mathrm{Spec}(\kappa(Z_{\mathfrak{n}}))$, i.e. the projection $\mathrm{pr} : \hat{Y} \to X_{\mathfrak{n}}$ restricts to an isomorphism $\hat{E} \cong E_{\mathfrak{n}} \cong \mathbb{P}^{d-1}_k$. Let $\hat{\eta}$ be the generic point of $\hat{E}$. Then $\mathrm{pr}(\hat{\eta}) = \eta_{\mathfrak{n}}$, and the induced local homomorphism of discrete valuation rings $\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}} \to \mathcal{O}_{\hat{Y},\hat{\eta}}$ carries a uniformizer to a uniformizer (the ideal of $E_{\mathfrak{n}}$ pulls back to the ideal of $\hat{E}$) and induces an isomorphism of residue fields (both are the function field of $\mathbb{P}^{d-1}_k$); hence it induces an isomorphism on completions. The completion of $\mathcal{O}_{\hat{Y},\hat{\eta}}$ is computed exactly as for the blowup of affine space at the origin, which we carry out in detail below: on the chart $u_d \neq 0$ we have $e_i = \frac{u_i}{u_d} e_d$, the local ring at $\hat{\eta}$ is $\left(\hat{A}[\frac{u_1}{u_d},\dots,\frac{u_{d-1}}{u_d}]\right)_{(e_d)}$, its residue field is $\kappa = k(\frac{u_1}{u_d},\dots,\frac{u_{d-1}}{u_d}) = k(\frac{e_1}{e_d},\dots,\frac{e_{d-1}}{e_d})$, and its completion is $\kappa[[e_d]]$. Finally, e_d is invertible in $\widehat{K}_{\eta_{\mathfrak{n}}} = \kappa((e_d))$, so the composite $\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}}) \to \mathrm{Spec}(\mathcal{O}_{\hat{Y},\hat{\eta}}) \to \hat{Y} \to \mathrm{Spec}(\hat{A})$ factors through V, giving the asserted factorization of $\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}}) \to U^{(d)}$ through c_V.

(3) Let $h : \mathcal{P}^{\times_k d} \to U^{(d)}$ denote the canonical morphism; it is finite, being the composition of the finite étale morphism $\mathcal{P}^{\times_k d} \to U^d$ with the finite morphism $U^d \to U^{(d)}$. By Corollary 27,

$\mathcal{P}^{[d]} = \Xi\backslash\mathcal{P}^{\times_k d} = \underline{\mathrm{Spec}}_{U^{(d)}}\Big((h_*\mathcal{O}_{\mathcal{P}^{\times_k d}})^{\Xi}\Big)$. Pushforward along the affine morphism h commutes with the flat base change $c_V : V \to U^{(d)}$, and the formation of Ξ-invariants is a kernel, hence also commutes with flat base change; therefore

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \cong \Xi\backslash\Big(\mathcal{P}^{\times_k d} \times_{U^{(d)}} V\Big).$$

By step (1), $U^d \times_{U^{(d)}} V \cong W$, so $\mathcal{P}^{\times_k d} \times_{U^{(d)}} V \cong \mathcal{P}^{\times_k d} \times_{U^d} W$. For each i, the composition $W \to \mathrm{Spec}(k[[x_1, \ldots, x_d]]) \to \mathrm{Spec}(k[[x]]) \to C$ — the middle map being the completion of the projection $p_i : C^d \to C$, i.e. $x \mapsto x_i$ — factors through $\mathrm{Spec}(k((x))) \to U$, since x_i is invertible on W; this composite is precisely q_i, and it commutes with $p_i : U^d \to U$. Since $\mathcal{P}^{\times_k d} = p_1^{-1}\mathcal{P} \times_{U^d} \cdots \times_{U^d} p_d^{-1}\mathcal{P}$, base changing to W gives an isomorphism

$$\mathcal{P}^{\times_k d} \times_{U^d} W \cong q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x,$$

which is equivariant for the actions of G^d (in particular of K_G) and of S_d — the latter acting on W by permuting the x_i, compatibly with $q_{\sigma(i)} = q_i \circ \sigma$. Passing to Ξ-quotients yields the displayed isomorphism.

For the final statement: the schemes V and W, the maps q_i and $\hat{r}$, the Ξ-action, and the point $\mathrm{Spec}(\kappa((e_d))) \to V$ of (2) are all constructed from d alone; by (3), the torsor $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ is the pullback along $\mathrm{Spec}(\kappa((e_d))) \to V$ of a torsor constructed from $\mathcal{P}_x$ and these data. Since the ramification at $\eta_{\mathfrak{n}}$ is by definition read off from $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ together with the discrete valuation ring $\kappa[[e_d]]$ (Definition 38, Definition 39), it depends only on $(\mathcal{P}_x, d)$. $\square$

Corollary 48 (Reduction to $\mathbb{P}^1$). *Let C, C' be smooth projective geometrically connected curves over the algebraically closed field k, let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ and $\mathcal{P}' \to U' = C' \setminus \mathfrak{m}'$ be G-torsors, let $P \in \mathrm{Supp}(\mathfrak{m})(k)$ and $P' \in \mathrm{Supp}(\mathfrak{m}')(k)$, and let x, x' be uniformizers at P, P' respectively. Assume that $\mathcal{P}_x \cong \mathcal{P}'_{x'}$ as G-torsors, compatibly with the isomorphism $k((x)) \cong k((x'))$, $x \mapsto x'$. Then for every $d \geq 1$ and every r, the ramification of $\mathcal{P}^{[d]}$ at η_{dP} is bounded by r if and only if the ramification of $\mathcal{P}'^{[d]}$ at $\eta_{dP'}$ is; in particular, the one is unramified (resp. tamely ramified) at η_{dP} if and only if the other is at $\eta_{dP'}$.*

Proof. Immediate from Lemma 47: both $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP}})}$ and $\mathcal{P}'^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP'}})}$ are identified, compatibly with the discrete valuation ring $\kappa[[e_d]]$, with the pullback along $\mathrm{Spec}(\kappa((e_d))) \to V$ of $\Xi\backslash(q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x)$. $\square$

The corollary reduces the geometry to $\mathbb{P}^1$ once the local cyclic torsor is realized by a global equation on $\mathbb{G}_m$. The next section introduces the three cyclic theories that provide precisely these equations and control their ramification.

2.4 Three Parallel Theories of Cyclic Torsors

The first reduction of the preceding section explains why cyclic theories suffice: it reduces the theorem from finite abelian groups to cyclic groups of order prime to p and cyclic groups of order p^r. The second reduction explains exactly what these theories must accomplish: they must realize a local cyclic torsor over $k((x))$ by an explicit equation on $\mathbb{G}_m \subset \mathbb{P}^1_k$, with the required control of its ramification. We therefore recall the three theories in the parallel form in which they will be used:

theory	H	$\varphi : H \to H$	$\ker(\varphi)$
Kummer	$\mathbb{G}_m$	$z \mapsto z^e$	$\mu_e \cong \mathbb{Z}/e\mathbb{Z}$
Artin–Schreier	$\mathbb{G}_a$	$z \mapsto z^p - z$	$\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$
Artin–Schreier–Witt	$\mathbb{W}_r$	$F - \mathrm{id}$	$W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}.$

Here the identification in the Kummer row is made after choosing a primitive e-th root of unity. In every row, pulling back φ along a section a produces the torsor given by the equation $\varphi(X) = a$, and changing a by an element of $\varphi(H)$ does not change its class.

Artin–Schreier–Witt theory contains Artin–Schreier theory as the case $r = 1$. We nevertheless treat the Artin–Schreier case separately: it displays, without the additional Witt-vector bookkeeping, the basic principle later used in the general case—choose a reduced representative of the local torsor, control its pole order by the ramification break, and use the same equation to realize the torsor on $\mathbb{G}_m$.

Only the conclusions used in Chapter 2 are stated here. The group-scheme constructions, classification arguments, normal-form results, and proofs are collected in Chapter A.

Throughout this section, let K be a complete discrete valuation field with perfect residue field κ of characteristic $p > 0$.

2.4.1 Kummer theory: order prime to p

Classes and equations. Assume that K contains the n-th roots of unity μ_n and $(n, p) = 1$. The Kummer sequence identifies $H^1(K, \mu_n)$ with $K^\times/(K^\times)^n$, the class of a being represented by $X^n = a$. Conversely, every cyclic extension of degree n is represented by such a Kummer equation.

Theorem 49 (Ramification in Kummer Extensions, [Koc97, Proposition 1.83]). *Assume K contains the n-th roots of unity μ_n and $(n, p) = 1$. Let L/K be the extension given by $X^n = a$, for $a \in K^\times$. Then:*

1. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in κ is an n-th power, the extension is trivial.*
2. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in κ is not an n-th power, the extension is unramified.*
3. *If $v_K(a) \notin n\mathbb{Z}$, the ramification index is $n/\gcd(n, v_K(a))$; in particular, if $\gcd(n, v_K(a)) = 1$, the extension is totally and tamely ramified of degree n.*

2.4.2 Artin–Schreier theory: order p

Classes and equations. Put $\wp(z) = z^p - z$. The Artin–Schreier sequence identifies $H^1(K, \mathbb{Z}/p\mathbb{Z})$ with $K/\wp(K)$, the class of a being represented by $X^p - X = a$. Conversely, every cyclic extension of degree p is represented by an Artin–Schreier equation. When $K = k((x))$ with k algebraically closed, every class has a representative in $x^{-1}k[x^{-1}]$ whose nonzero pole order is prime to p.

Theorem 50 (Ramification in Artin–Schreier Extensions, [Tho05]). *Let L/K be given by $X^p - X = a$. Then:*

1. *If $v_K(a) > 0$, or if $v_K(a) = 0$ and $a \in \wp(K)$, the extension is trivial.*

2. *If $v_K(a) = 0$ and $a \notin \wp(K)$, the extension is cyclic of degree p and unramified.*

3. *If $v_K(a) = -m < 0$ with $p \nmid m$, the extension is cyclic, totally ramified of degree p, and has unique upper ramification break m.*

2.4.3 Artin–Schreier–Witt theory: order p^r

Classes and equations. For an $\mathbb{F}_p$-algebra A, write $W_r(A) = A^r$ for the ring of p-typical Witt vectors of length r, with Witt addition and subtraction denoted by $\oplus$ and $\ominus$, and put $\wp = F - \mathrm{id}$. For a field K of characteristic p, Artin–Schreier–Witt theory identifies

$$H^1(K, \mathbb{Z}/p^r\mathbb{Z}) \cong W_r(K)/\wp W_r(K),$$

the class of $\vec{g}$ being represented by the equation

$$\wp\vec{X} = \vec{g}.$$

The construction of the Witt group scheme, the proof that this equation defines a $\mathbb{Z}/p^r\mathbb{Z}$-torsor, and the full classification theorem are given in Section A.6.

Definition 51. Let $K = k((x))$. For $g \in K$, put $m(g) := \max(0, -v_x(g))$. A vector $\vec{g} = (g_0, \ldots, g_{r-1}) \in W_r(K)$ is **reduced** if, for every j, either $m(g_j) = 0$ or $p \nmid m(g_j)$.

Theorem 52 (Ramification in Artin–Schreier–Witt Extensions (Schmid–Witt))**.** *Let $K = k((x))$, with k perfect of characteristic p, and let L/K be a cyclic totally ramified extension of degree p^r whose Artin–Schreier–Witt class is represented by a reduced vector $\vec{g} = (g_0, \ldots, g_{r-1})$, with $m_j := m(g_j)$. Then the largest upper-numbering ramification break of L/K is*

$$u = \max_{0 \le j \le r-1} p^{r-1-j} m_j.$$

The Schmid–Witt formula is an external ramification theorem: we use it as stated and do not reproduce its proof. References and the precise scope in which it is used are recorded in Section A.8.

2.4.4 The common realization conclusion

The three theories give the following single conclusion, which is the form needed after the formal–local reduction to $\mathbb{P}^1_k$.

Lemma 53 (Parallel realization over $\mathbb{G}_m$)**.** *Let k be algebraically closed of characteristic $p > 0$, and let $\mathcal{Q}$ be a G-torsor over* $\operatorname{Spec} k((x))$*.*

1. *If $\mathcal{Q}$ is tamely ramified, then there are an integer e prime to p, an injection $\iota : \mathbb{Z}/e\mathbb{Z} \hookrightarrow G$, and an integer a prime to e, such that $\mathcal{Q}$ is the restriction of the induced G-torsor on $\mathbb{G}_m$ obtained from*

$$T^e = x^a.$$

This torsor is tamely ramified at 0 and ∞.

2. *If $G = \mathbb{Z}/p\mathbb{Z}$ and the ramification of $\mathcal{Q}$ is bounded by d, then $\mathcal{Q}$ is represented by*

$$X^p - X = f(x), \qquad f \in x^{-1}k[x^{-1}],$$

where either $f = 0$, or $m := \deg_{x^{-1}} f < d$ and $p \nmid m$. The same equation defines a torsor on $\mathbb{G}_m$ extending étale across ∞.

3. *If $G = \mathbb{Z}/p^r\mathbb{Z}$, the torsor $\mathcal{Q}$ is connected, and its ramification is bounded by d, then $\mathcal{Q}$ is represented by*

$$\wp\vec{X} = \vec{f},$$

where $\vec{f} = (f_0, \dots, f_{r-1})$ is reduced and $f_j \in x^{-1}k[x^{-1}]$. Put $m_j := \deg_{x^{-1}} f_j$ when $f_j \neq 0$, and $m_j := 0$ when $f_j = 0$. Then it has the following properties:

   (a)
$$p^{r-1-j}m_j < d \qquad (0 \leq j < r). \tag{2.2}$$

   (b) The same equation defines a $\mathbb{Z}/p^r\mathbb{Z}$-torsor on $\mathbb{G}_m$ whose restriction to $\operatorname{Spec} k((x))$ is $\mathcal{Q}$.

   (c) This torsor extends étale across ∞.

Proof. For (1), let $\rho : \mathrm{Gal}(k((x))^{\mathrm{sep}}/k((x))) \to G$ classify $\mathcal{Q}$. Since k is algebraically closed, $k((x))$ has no nontrivial unramified extension and its tame quotient is procyclic. If $\mathcal{Q}$ is tame, the image of ρ is therefore a cyclic group of order e prime to p. Let $\iota : \mathbb{Z}/e\mathbb{Z} \hookrightarrow G$ identify this image with its subgroup of G. Kummer theory identifies the corresponding surjective character with the class of x^a for some a prime to e. Hence $\mathcal{Q}$ is induced from $T^e = x^a$. The same equation over $k[x, x^{-1}]$ is étale and defines the required torsor on $\mathbb{G}_m$; its ramification at both boundary points is tame.

For (2), Artin–Schreier theory identifies $\mathcal{Q}$ with a class in $k((x))/\wp(k((x)))$. The length-one construction in the proof of Proposition 89 produces a representative $f \in x^{-1}k[x^{-1}]$ which is zero or has pole order m prime to p. By Theorem 50, the ramification bound is equivalent to $m < d$. The equation $X^p - X = f$ is étale over $k[x, x^{-1}]$; since $f \in k[x^{-1}]$, it also extends over $\operatorname{Spec} k[x^{-1}] = \mathbb{P}^1_k \setminus \{0\}$.

For (3), let $[\vec{f_0}] \in W_r(k((x)))/\wp$ be the class of $\mathcal{Q}$ and choose a reduced polynomial representative $\vec{f}$ using Proposition 89. Connectedness and Corollary 86 imply $f_0 \neq 0$. We use the convention for m_j in the statement. The Schmid–Witt formula gives

$$\max_j p^{r-1-j}m_j < d,$$

which is (2.2). By Lemma 84, the equation $\wp\vec{X} = \vec{f}$ defines a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\mathbb{G}_m$ with the prescribed local class. Since every f_j lies in $k[x^{-1}]$, the same lemma applied over $k[x^{-1}]$ extends the torsor étale across ∞. □

2.4.5 Three common inputs for the cyclic calculations

The explicit equations above enter the symmetric-power construction through one contracted-product calculation. We record it once for all three cyclic theories.

Let S be a scheme, let H be a commutative S-group scheme, and let $\varphi : H \to H$ be a finite étale surjective homomorphism with constant kernel G. For a section $a : X \to H$, write

$$\mathcal{T}_a := X \times_{a,H,\varphi} H.$$

Thus $\mathcal{T}_a$ is a G-torsor on X. We write $\star$ for the group law of H: multiplication on $\mathbb{G}_m$, ordinary addition on $\mathbb{G}_a$, and Witt addition on $\mathbb{W}_r$.

Proposition 54 (Common cyclic product calculation). *Let $a_i : X \to H$, $1 \leq i \leq d$, be sections. There is a canonical isomorphism of G-torsors*

$$\mathcal{T}_{a_1} \otimes^G \cdots \otimes^G \mathcal{T}_{a_d} \cong \mathcal{T}_{a_1 \star \cdots \star a_d}.$$

On equation coordinates it is induced by

$$(X_1, \ldots, X_d) \longmapsto Y := X_1 \star \cdots \star X_d, \qquad \varphi(Y) = a_1 \star \cdots \star a_d.$$

If a group Γ permutes the indices, this isomorphism is Γ-equivariant and therefore compatible with descent to the quotient.

Proof. The product $\prod_i \mathcal{T}_{a_i}$ is a G^d-torsor. Contracting along the multiplication homomorphism $G^d \to G$ gives its quotient by K_G. The displayed map is K_G-invariant because the kernel translations cancel under $\star$, and it is equivariant for the residual G-action. Its target is a G-torsor because it is the pullback of φ. Hence it is an isomorphism by Proposition 5. Commutativity and associativity of $\star$ give the permutation-equivariance. □

We shall also pass from the connected inertia torsor to the original torsor. The following compatibility ensures that this passage commutes with the symmetric-power construction.

Lemma 55 (Symmetric powers and induced torsors). *Let $\iota : H \hookrightarrow G$ be an injection of finite abelian groups, and let $\mathcal{Q}$ be an H-torsor on U. There is a canonical isomorphism of G-torsors over $U^{(d)}$*

$$(\iota_* \mathcal{Q})^{[d]} \cong \iota_*(\mathcal{Q}^{[d]}).$$

Proof. The canonical morphism $j : \mathcal{Q} \to \iota_* \mathcal{Q} = (\mathcal{Q} \times G)/H$ is H-equivariant, where H acts on the target through ι. Its d-fold product commutes with the S_d-actions and is equivariant for $H^d \to G^d$. Writing $K_H := \ker(H^d \to H)$, it therefore descends to an H-equivariant morphism

$$\mathcal{Q}^{[d]} = (K_H \rtimes S_d) \backslash \mathcal{Q}^{\times d} \longrightarrow (K_G \rtimes S_d) \backslash (\iota_* \mathcal{Q})^{\times d} = (\iota_* \mathcal{Q})^{[d]}.$$

The universal property of induction extends this morphism uniquely to a G-equivariant morphism $\iota_*(\mathcal{Q}^{[d]}) \to (\iota_* \mathcal{Q})^{[d]}$. Both sides are G-torsors over $U^{(d)}$, so it is an isomorphism by Proposition 5. □

Finally, the two wild calculations use the same elementary fact about the exceptional valuation. Let $e_1, \ldots, e_d$ be the elementary symmetric polynomials in $x_1, \ldots, x_d$, put $y_i = x_i^{-1}$, and write

$$\kappa := k(e_1/e_d, \ldots, e_{d-1}/e_d).$$

Lemma 56 (Symmetric regularity). *If $\Phi \in k[y_1, \ldots, y_d]^{S_d}$ and every monomial of Φ has total degree strictly less than d, then*

$$\Phi \in k(e_1/e_d, \ldots, e_{d-1}/e_d) = \kappa.$$

In particular, Φ is regular at the generic point of the exceptional divisor.

Proof. The identity
$$e_j(y_1,\dots,y_d) = \frac{e_{d-j}(x_1,\dots,x_d)}{e_d(x_1,\dots,x_d)}$$
puts $e_1(y),\dots,e_{d-1}(y)$ in κ. By the fundamental theorem of symmetric polynomials, each homogeneous part of Φ is a polynomial in $e_1(y),\dots,e_d(y)$ with their natural degrees. Since every homogeneous part has degree $< d$, the generator $e_d(y)$ cannot occur. Hence $\Phi \in \kappa \subset \kappa[[e_d]]$. $\square$

2.5 Proof of the Torsor Formulation

The reductions and cyclic theories of the preceding sections leave one explicit calculation. We state its three cyclic forms together and then prove them in parallel. This is the only intermediate statement needed for the proof of Theorem 33.

Theorem 57 (The three cyclic cases)**.** *Let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ be a G-torsor with G cyclic, and let $\mathfrak{n} = dP \subset \mathfrak{m}$ be a single-point submodulus.*

1. *If $G = \mathbb{Z}/e\mathbb{Z}$ with $(e,p) = 1$, then $\mathcal{P}^{[d]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*
2. *If $G = \mathbb{Z}/p\mathbb{Z}$ and the ramification of $\mathcal{P}$ at P is bounded by d, then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*
3. *If $G = \mathbb{Z}/p^r\mathbb{Z}$, $r \geq 2$, and the ramification of $\mathcal{P}$ at P is bounded by d, then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*

Proof. By Section 2.3, we may work over an algebraically closed field and regard P as rational. Choose a uniformizer x at P, let
$$\rho : \mathrm{Gal}(k((x))^{\mathrm{sep}}/k((x))) \longrightarrow G$$
classify the punctured-formal-disc restriction $\mathcal{P}_x$, and put $H := \mathrm{im}(\rho)$. If H is trivial, then $\mathcal{P}_x$ and its symmetric power are trivial, so the conclusion follows from Lemma 47.

Assume $H \neq 0$. The surjective character with image H defines a connected H-torsor $\mathcal{Q}$ over $\operatorname{Spec} k((x))$, and $\mathcal{P}_x \cong \iota_*\mathcal{Q}$ for the inclusion $\iota : H \hookrightarrow G$. Whenever a ramification bound is assumed, it passes from $\mathcal{P}_x$ to $\mathcal{Q}$. By Lemma 53, the torsor $\mathcal{Q}$ is realized by an explicit connected H-torsor $\mathcal{Q}'$ on $\mathbb{G}_m$. Moreover,
$$(\iota_*\mathcal{Q}')^{[d]} \cong \iota_*(\mathcal{Q}'^{[d]})$$
by Lemma 55, and induction preserves the ramification bound by Lemma 45. Therefore Corollary 48 reduces every case to the connected explicit model
$$C = \mathbb{P}^1_k, \qquad U = \mathbb{G}_m, \qquad \mathfrak{n} = d \cdot 0.$$
We rename H as G for the calculation.

Write $x_1,\dots,x_d$ for the coordinates on $\mathbb{G}_m^d$ and $e_1,\dots,e_d$ for their elementary symmetric polynomials. By Lemma 47, the completed local ring and field at the generic point $\eta = \eta_{d\cdot 0}$ of the exceptional divisor are
$$\widehat{\mathcal{O}}_\eta = \kappa[[e_d]], \qquad \widehat{K}_\eta = \kappa((e_d)), \qquad \kappa = k(e_1/e_d,\dots,e_{d-1}/e_d). \tag{2.3}$$

The three calculations have the common shape

	$\mathbb{Z}/e\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p^r\mathbb{Z}$
normal form	$T^e = x^a$	$X^p - X = f$	$\wp\vec{X} = \vec{f}$
product coordinate	$\prod_i T_i$	$\sum_i X_i$	$\bigoplus_i \vec{X}_i$
new datum	e_d^a	$\sum_i f(x_i)$	$\bigoplus_i \vec{f}(x_i)$
exceptional behavior	$v(e_d^a) = a$	regular	componentwise regular
conclusion	tame	unramified	unramified.

Case (1): Kummer. The relevant group scheme is $\mathbb{G}_m$. More precisely, the connected torsor is the pullback of the finite étale homomorphism

$$[e] : \mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad T \longmapsto T^e,$$

whose kernel is $\mu_e \cong G$, along the section $x \mapsto x^a$. Thus it has equation

$$T^e = x^a, \qquad (a, e) = 1,$$

by Lemma 53(1). Put

$$R := k[x, x^{-1}], \qquad S := R[T]/(T^e - x^a).$$

Then $\operatorname{Spec} S \to \operatorname{Spec} R = \mathbb{G}_m$ is this G-torsor. After pulling it back along the d projections from $\mathbb{G}_m^d$, the product torsor has coordinate algebra $S^{\otimes_k d}$ over $R^{\otimes_k d}$. Contracting the product action G^d along $G^d \to G$ means taking invariants under $K_G = \ker(G^d \to G)$. By Proposition 54, multiplication in $\mathbb{G}_m$ gives

$$Y = \prod_{i=1}^{d} T_i, \qquad Y^e = \prod_{i=1}^{d} x_i^a = (x_1 \cdots x_d)^a = e_d^a,$$

and hence an isomorphism of G-torsor algebras

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[Y]/(Y^e - e_d^a).$$

The group S_d permutes the tensor factors and fixes both Y and e_d^a. The algebra on the right is free over $R^{\otimes_k d}$ with the S_d-invariant basis $1, Y, \ldots, Y^{e-1}$, so taking S_d-invariants acts only on its coefficients. Since

$$(R^{\otimes_k d})^{S_d} = k[x_1^{\pm 1}, \ldots, x_d^{\pm 1}]^{S_d} = k[e_1, \ldots, e_d, e_d^{-1}],$$

Corollary 27 identifies the descended torsor with

$$\mathcal{P}^{[d]} \cong \operatorname{Spec} \frac{k[e_1, \ldots, e_d, e_d^{-1}][Y]}{(Y^e - e_d^a)}.$$

Its generic fiber over $\widehat{K}_\eta$ is therefore the restriction of $\mathcal{P}^{[d]}$ and is given by the same Kummer equation. The defining element e_d^a has valuation a; thus Theorem 49, together with $(e, p) = 1$, shows that every field factor is tamely ramified. This proves part (1).

Case (2): Artin–Schreier. Here the relevant group scheme is $\mathbb{G}_a$. The torsor is the pullback of the finite étale homomorphism

$$\wp : \mathbb{G}_a \longrightarrow \mathbb{G}_a, \qquad X \longmapsto X^p - X,$$

whose kernel is $\mathbb{F}_p \cong G$, along the section $x \mapsto f(x)$ on $\mathbb{G}_m$. By Lemma 53(2), its equation is

$$X^p - X = f(x), \qquad f \in x^{-1}k[x^{-1}],$$

where either $f = 0$, or $m := \deg_{x^{-1}} f < d$ and $p \nmid m$. Put

$$R := k[x, x^{-1}], \qquad S := R[X]/(X^p - X - f(x)).$$

Thus $\operatorname{Spec} S \to \operatorname{Spec} R = \mathbb{G}_m$ is the realized G-torsor. After pullback along the d projections, $S^{\otimes_k d}$ is the coordinate algebra of the product torsor over $R^{\otimes_k d}$. Contracting along $G^d \to G$ again amounts to taking K_G-invariants. This time the group law on $\mathbb{G}_a$ is addition, so Proposition 54 gives

$$Y = \sum_{i=1}^{d} X_i, \qquad Y^p - Y = \alpha, \qquad \alpha := \sum_{i=1}^{d} f(x_i),$$

and hence

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[Y]/(Y^p - Y - \alpha).$$

The group S_d fixes Y and α. The displayed algebra is free over $R^{\otimes_k d}$ with the S_d-invariant basis $1, Y, \dots, Y^{p-1}$, so taking S_d-invariants acts only on the coefficients. Using $(R^{\otimes_k d})^{S_d} = k[e_1, \dots, e_d, e_d^{-1}]$ and Corollary 27, we obtain

$$\mathcal{P}^{[d]} \cong \operatorname{Spec} \frac{k[e_1, \dots, e_d, e_d^{-1}][Y]}{(Y^p - Y - \alpha)}.$$

Put $y_i = x_i^{-1}$. If $f \neq 0$, then α is a symmetric polynomial in $y_1, \dots, y_d$ whose monomials have total degree at most $m < d$; if $f = 0$, then $\alpha = 0$. Hence Lemma 56 gives

$$\alpha \in \kappa \subset \widehat{\mathcal{O}}_\eta.$$

Set $A := \widehat{\mathcal{O}}_\eta$ and

$$B := A[Y]/(Y^p - Y - \alpha).$$

Because the defining polynomial is monic of degree p, the algebra B is finite free of rank p over A. Its derivative with respect to Y is -1, a unit in B; hence B/A is finite étale. The preceding descended presentation identifies its generic fiber

$$B \otimes_A \widehat{K}_\eta = \widehat{K}_\eta[Y]/(Y^p - Y - \alpha)$$

with the restriction of $\mathcal{P}^{[d]}$ to $\widehat{K}_\eta$.

Since A is a complete discrete valuation ring, the finite étale A-algebra B is a product of finite étale local A-algebras. Each factor is a discrete valuation ring whose maximal ideal is generated by the uniformizer of A, and whose residue field is finite separable over κ. Consequently, every field factor of the generic fiber has ramification index 1 and separable residue extension, so it is unramified in the sense of Definition 35. This proves part (2).

Case (3): Artin–Schreier–Witt. If the connected inertia group H produced by the common reduction has order p, the Artin–Schreier calculation in Case (2) already gives the required unramified conclusion. We may therefore assume that H has order p^r with $r \geq 2$, and continue to denote it by G. By Lemma 53(3), the connected torsor has an equation

$$\wp \vec{X} = \vec{f} = (f_0, \dots, f_{r-1}), \qquad f_j \in x^{-1}k[x^{-1}],$$

where, putting $m_j = \deg_{x^{-1}} f_j$ for $f_j \neq 0$ and $m_j = 0$ for $f_j = 0$, the pole bounds are

$$p^{r-1-j} m_j < d \qquad (0 \le j < r). \tag{2.4}$$

Put $R = k[x, x^{-1}]$ and

$$S = R[\vec{X}]/(\wp\vec{X} \ominus \vec{f}).$$

After pulling back to $\mathbb{G}_m^d$, define the single Witt vectors

$$\vec{Y} := \bigoplus_{i=1}^{d} \vec{X}_i, \qquad \vec{\alpha} := \bigoplus_{i=1}^{d} \vec{f}(x_i).$$

Here and below $\bigoplus$ denotes iterated Witt addition, not a direct sum.

Proposition 58 (Identification of the Witt product torsor)**.** *Every component of $\vec{Y}$ is K_G-invariant, every component of $\vec{\alpha}$ is a symmetric Laurent polynomial, and*

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[\vec{Y}]/(\wp\vec{Y} \ominus \vec{\alpha})$$

as G-torsors over $R^{\otimes_k d}$. Taking S_d-invariants gives

$$\big((S^{\otimes_k d})^{K_G}\big)^{S_d} \cong k[e_1, \ldots, e_d, e_d^{-1}][\vec{Y}]/(\wp\vec{Y} \ominus \vec{\alpha}).$$

Consequently,

$$\mathcal{P}^{[d]}\big|_{\operatorname{Spec} \widehat{K}_\eta} \cong \operatorname{Spec} \widehat{K}_\eta[\vec{Y}]/(\wp\vec{Y} \ominus \vec{\alpha}).$$

Proof. Write an element of $K_G \cong G^{d-1}$ as $(g_1, \ldots, g_{d-1})$. Under the usual embedding in G^d, it acts in Witt coordinates by

$$\begin{aligned}
\vec{X}_1 &\longmapsto \vec{X}_1 \oplus \underline{g}_1, \\
\vec{X}_i &\longmapsto \vec{X}_i \oplus \underline{g}_i \ominus \underline{g}_{i-1} \quad (1 < i < d), \\
\vec{X}_d &\longmapsto \vec{X}_d \ominus \underline{g}_{d-1},
\end{aligned}$$

where $\underline{g}$ is the image of g in $W_r(\mathbb{F}_p)$. Functoriality of W_r means that this action is componentwise and commutes with $\oplus$ and $\ominus$. The translations telescope in $\vec{Y}$, so each component of $\vec{Y}$ is invariant. Additivity of $\wp$ gives

$$\wp\vec{Y} = \bigoplus_{i=1}^{d} \wp\vec{X}_i = \bigoplus_{i=1}^{d} \vec{f}(x_i) = \vec{\alpha}.$$

The resulting coordinate map is equivariant for the residual G-action. Both the displayed Artin–Schreier–Witt cover and the contracted product are G-torsors over $R^{\otimes_k d}$, by Lemma 84 and proposition 54; hence the map is an isomorphism by Proposition 5.

The group S_d permutes the tensor factors. Commutativity and associativity of Witt addition show that it fixes $\vec{Y}$ and $\vec{\alpha}$ componentwise. Thus every component of $\vec{\alpha}$ lies in

$$(R^{\otimes_k d})^{S_d} = k[e_1, \ldots, e_d, e_d^{-1}].$$

By Lemma 84, the ASW algebra is free over $R^{\otimes_k d}$ with basis $\prod_n Y_n^{a_n}$, $0 \le a_n < p$. Every basis element is S_d-invariant, so an element is invariant exactly when all its coefficients are invariant. This proves the second displayed isomorphism. Finally, Corollary 27 identifies the quotient by $K_G \rtimes S_d$ with $\mathcal{P}^{[d]}$, and base change to $\widehat{K}_\eta$ gives the last assertion. ∎

It remains to show that every component of $\vec{\alpha}$ is regular at the exceptional valuation. The Witt components are not ordinary componentwise sums, so this is the weighted analogue of the Artin–Schreier degree count. We claim that every component α_n, $0 \le n < r$, belongs to κ.

Give $f_j(x_i)$ weight p^j. By Lemma 81, every nonzero monomial

$$\prod_{i,j} f_j(x_i)^{a_{ij}}$$

appearing in α_n satisfies

$$\sum_{i,j} a_{ij} p^j = p^n.$$

Put $A_j := \sum_i a_{ij}$. Since $f_j(x_i)$ is a polynomial in $y_i = x_i^{-1}$ of degree m_j, every monomial obtained after expansion in the y_i has total degree at most

$$\sum_j A_j m_j < d \sum_j A_j p^{j-(r-1)} = d\, p^{n-(r-1)} \le d.$$

The inequality is strict: a nonzero monomial has a positive exponent on at least one nonzero f_j, and every pole bound in (2.4) is strict. By Proposition 58, the whole component α_n is symmetric. Therefore Lemma 56 gives $\alpha_n \in \kappa$. Thus

$$\vec{\alpha} \in W_r(\kappa) \subset W_r(\widehat{\mathcal{O}}_\eta).$$

It follows that the equation

$$\wp \vec{Y} = \vec{\alpha}$$

is defined over $\widehat{\mathcal{O}}_\eta$. By Lemma 84, it defines a finite étale $\widehat{\mathcal{O}}_\eta$-algebra. Its generic fiber is the restriction of $\mathcal{P}^{[d]}$ to $\widehat{K}_\eta$ by Proposition 58. Hence every field factor is unramified. This proves part (3). $\square$

Proof of Theorem 33. Fix a single-point submodulus $\mathfrak{n} = dP$. By Proposition 46, it is enough to consider cyclic groups of order prime to p and cyclic p-groups. The prime-to-p case is Theorem 57(1). For $G = \mathbb{Z}/p^r\mathbb{Z}$, the ramification bound on $\mathcal{P}$ gives the unramified conclusion by part (2) when $r = 1$ and by part (3) when $r \ge 2$. The reduction proposition reassembles the cyclic factors and proves the theorem for every finite abelian group. $\square$

Chapter 3

Geometric Class Field Theory

The purpose of this chapter is to deduce the reduced form of geometric class field theory, Theorem 2, from the ramification calculation established in Chapter 2.

Let C be a smooth, projective, geometrically connected curve over the perfect field k, let $\mathfrak{m}$ be a modulus on C, and put $U = C \setminus \mathfrak{m}$. Given a rank-one local system $\mathcal{F}$ on U whose ramification is bounded by $\mathfrak{m}$, our goal is to descend its symmetric power $\mathcal{F}^{[d]}$ along the Abel–Jacobi morphism

$$\Phi_d \colon U^{(d)} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}$$

for all sufficiently large d.

When $\mathfrak{m} = 0$, the fibers of the Abel–Jacobi morphism are projective spaces. Their trivial étale fundamental groups allow the desired descent directly. When $\mathfrak{m} > 0$, however, the fibers of Φ_d are affine spaces, so this argument cannot be applied to Φ_d itself. Following Takeuchi, we resolve this difficulty by extending Φ_d to a projective space bundle

$$\widetilde{\Phi}_d \colon \widetilde{C}^{(d)}_{\mathfrak{m}} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}},$$

where $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is an open subscheme of the blowup of $C^{(d)}$ along the locus obtained by adding the modulus.

The proof then has three remaining steps. First, we use the single-point ramification calculation of Chapter 2, together with compatibility under products and addition of divisors, to prove that $\mathcal{F}^{[d]}$ is tamely ramified along the boundary

$$H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}.$$

Second, on the geometric generic Abel–Jacobi fiber, the boundary is the hyperplane at infinity. A Tendler-style prime-to-p power argument, together with the structure of the abelian fundamental group of affine space, forces the corresponding boundary inertia representation to be trivial. Purity then extends $\mathcal{F}^{[d]}$ uniquely across the boundary. Finally, since $\widetilde{\Phi}_d$ is a projective space bundle, it induces an isomorphism on étale fundamental groups. The extended local system therefore descends uniquely to $\mathrm{Pic}^d_{C,\mathfrak{m}}$, proving Theorem 2.

We begin by recalling the generalized Picard scheme and the geometry of the Abel–Jacobi morphism. We then introduce Takeuchi's compactification, prove the required ramification and extension results, and conclude with the descent argument.

3.1 Generalized Picard Schemes and the Abel–Jacobi Map

3.1.1 The generalized Picard scheme

We recall the notion of generalized Jacobian varieties and study their fundamental properties. The material presented here is primarily adapted from [Gui19] and [Tak19]. For further background on the general theory of abelian varieties and Jacobians, the reader may also consult [Mil08]. Let S be a scheme and let C be a projective smooth S-scheme whose geometric fibers are connected and of dimension 1. Let $\mathfrak{m}$ be a modulus on C, defined as an effective Cartier divisor of C/S (i.e., a closed subscheme of C which is finite flat of finite presentation over S). We denote the projection $C \times_S T \to T$ by pr for any S-scheme T.

The Functor of Points

Let d be an integer. For an S-scheme T, we consider the set of data $(\mathcal{L}, \psi)$ where:

- $\mathcal{L}$ is an invertible sheaf of degree d on C_T.
- $\psi : \mathcal{O}_{\mathfrak{m}_T} \xrightarrow{\sim} \mathcal{L}|_{\mathfrak{m}_T}$ is a trivialization of $\mathcal{L}$ along the modulus.

Two such pairs $(\mathcal{L}, \psi)$ and $(\mathcal{L}', \psi')$ are said to be isomorphic if there exists an isomorphism of invertible sheaves $f : \mathcal{L} \to \mathcal{L}'$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathcal{O}_{\mathfrak{m}_T} \arrow[dl, "\psi"'] \arrow[dr, "\psi'"] & \\ \mathcal{L}|_{\mathfrak{m}_T} \arrow[rr, "f|_{\mathfrak{m}_T}"'] & & \mathcal{L}'|_{\mathfrak{m}_T} \end{tikzcd}$$

We define the presheaf $\mathrm{Pic}^{d,\mathrm{pre}}_{C,\mathfrak{m}}$ on Sch/S by assigning to T the set of isomorphism classes of such pairs. Let $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ denote the étale sheafification of this presheaf.

Representability and Structure

The fundamental properties of this functor are as follows:

1. $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ is represented by an S-scheme. (Note: If $\mathfrak{m}$ is faithfully flat over S, the presheaf is already a étale sheaf).
2. $\mathrm{Pic}^{0}_{C,\mathfrak{m}}$ is a smooth commutative group S-scheme with geometrically connected fibers, referred to as the *generalized Jacobian variety* of C with modulus $\mathfrak{m}$.
3. For any d, $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ is a $\mathrm{Pic}^{0}_{C,\mathfrak{m}}$-torsor.

In the case where $\mathfrak{m} = 0$, we recover the standard Jacobian variety, denoted simply as Pic^{d}_{C}.

Relation to the Standard Jacobian

We now examine the behavior of the generalized Picard scheme under the variation of the modulus. By viewing the structure along the modulus as an additional rigidification, we obtain natural transition maps corresponding to the inclusion of moduli.

Let $\mathfrak{m}_1$ and $\mathfrak{m}_2$ be moduli such that $\mathfrak{m}_1 \subset \mathfrak{m}_2$. There exists a natural map

$$\mathrm{Pic}^d_{C,\mathfrak{m}_2} \to \mathrm{Pic}^d_{C,\mathfrak{m}_1}$$

obtained by restricting the rigidification from $\mathfrak{m}_2$ to $\mathfrak{m}_1$. More precisely, it is an epimorphism for the étale topology: if $(\mathcal{L}, \psi_1)$ is an object over an S-scheme T, then, after an étale cover $T' \to T$, the rigidification ψ_1 of $\mathcal{L}|_{(\mathfrak{m}_1)_{T'}}$ extends to a rigidification of $\mathcal{L}|_{(\mathfrak{m}_2)_{T'}}$. Thus every local section of $\mathrm{Pic}^d_{C,\mathfrak{m}_1}$ lifts étale-locally to a section of $\mathrm{Pic}^d_{C,\mathfrak{m}_2}$. In particular, for any modulus $\mathfrak{m}$, there is a natural surjective morphism of étale sheaves:

$$\mathrm{Pic}^d_{C,\mathfrak{m}} \to \mathrm{Pic}^d_C.$$

Local Freeness and Base Change

Let $\mathfrak{m}$ be a modulus which is everywhere strictly positive. Let g denote the genus of C, which is a locally constant function on S. We restrict our attention to degrees d satisfying the condition:

$$d \geq \max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}. \tag{3.1}$$

Assuming S is quasi-compact, such a d always exists.

Fix an integer d satisfying the condition above. Let T be an S-scheme and let $\mathcal{L}$ be an invertible sheaf of degree d on C_T. One can show that the pushforwards $\mathrm{pr}_*\mathcal{L}$ and $\mathrm{pr}_*\mathcal{L}(-\mathfrak{m})$ are locally free sheaves and their formations commute with any base change. Explicitly, for any morphism of S-schemes $f : T' \to T$, the base change morphisms are isomorphisms:

$$f^*\mathrm{pr}_*\mathcal{L} \xrightarrow{\sim} \mathrm{pr}_* f^*\mathcal{L}$$

and

$$f^*\mathrm{pr}_*(\mathcal{L}(-\mathfrak{m})) \xrightarrow{\sim} \mathrm{pr}_* f^*(\mathcal{L}(-\mathfrak{m})).$$

In particular, following [Gui19], if $\mathcal{L}$ is invertible $\mathcal{O}_C$-module with degree d satisfying 3.1 on each fiber of f then, $\mathrm{pr}_*\mathcal{L}$ is a locally free $\mathcal{O}_S$-module of rank $d - g + 1$.

For further background and verification of these constructions, we refer the reader to Milne's notes on abelian Varieties ([Mil08]).

3.1.2 The Abel–Jacobi Morphism and Its Fibers

Let $U = C \backslash \mathfrak{m}$ be the complement of the modulus in C. The effective cartier divisors of degree d which are prime to $\mathfrak{m}$ are parameterized by the symmetric power $\mathrm{Sym}^d_S(U) = U^{(d)}$ over S (See [Gui19] Proposition 4.12, [Mil08] Theorem 3.13). For any such divisor $D \in U^{(d)}$, the associated line bundle $\mathcal{O}_C(D)$ admits a canonical trivialization along $\mathfrak{m}$. Specifically, the canonical section 1_D is regular and non-vanishing on $\mathfrak{m}$ because $\mathrm{supp}(D) \cap \mathrm{supp}(\mathfrak{m}) = \emptyset$. This section restricts to a nowhere-vanishing section on the subscheme $\mathfrak{m}$, thereby determining a trivialization $\psi_D^{-1} : \mathcal{O}_C(D)|_{\mathfrak{m}} \xrightarrow{\sim} \mathcal{O}_{\mathfrak{m}}$. This is done functorially in families, yielding a morphism from the symmetric power to the generalized Picard scheme (over S):

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}, \quad D \mapsto [(\mathcal{O}_C(D), \psi_D)], \tag{3.2}$$

When $\mathfrak{m} = 0$, $d \geq \max\{2g-1, 0\}$ and C admits a section over S, $C^{(d)}$ is a projective space bundle over Pic^d_C, It is proper, surjective with geometrically connected fibers.

Guignard ([Gui19] Theorem 4.14) proves that for $\mathfrak{m} > 0$ and d satisfying (3.1), the Abel-Jacobi morphism Φ_d is surjective smooth of relative dimension $d - \deg \mathfrak{m} - g + 1$, with geometrically connected fibers.

When $S = \mathrm{Spec}(k)$, the geometric fibers of Φ_d are well understood:

Theorem 59. *Assume that $S = \mathrm{Spec}(k)$ and $d \geq \max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}$. Let $\bar{s} = \mathrm{Spec}\,\Omega \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ be a geometric point, where Ω is separably closed. Then the geometric fiber of the Abel–Jacobi morphism*

$$\Phi_d : U^{(d)} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}$$

at $\bar{s}$ is isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg \mathfrak{m}-g+1}_{\Omega} & \text{if } \mathfrak{m} > 0, \\ \mathbb{P}^{d-g}_{\Omega} & \text{if } \mathfrak{m} = 0. \end{cases}$$

Thus every geometric fiber is an affine space when $\mathfrak{m} > 0$ and a projective space when $\mathfrak{m} = 0$. This statement does not, by itself, assert that Φ_d is Zariski-locally trivial.

Proof. See [Ten15] Propositions 3.13-3.14, or [Tót11] Proposition 2.1.4. □

3.2 Takeuchi's Compactification of the Abel–Jacobi Morphism

We recall that our objective is to descend the local system $\mathcal{F}^{[d]}$ from $U^{(d)}$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$ along the Abel-Jacobi map Φ_d:

$$\begin{array}{ccc} \mathcal{F}^{[d]} & & \\ \downarrow & & \\ U^{(d)} & \xrightarrow{\Phi_d} & \mathrm{Pic}^d_{C,\mathfrak{m}} \end{array}$$

(Here, the purple arrow emphasizes that the morphism is of sheaves on the étale site).

However, we encounter an obstruction: in the case we are considering ($\mathfrak{m} > 0$), the fibers of Φ_d are affine spaces (of the same degree) rather than the better-behaved projective spaces. This suggests that a solution to this problem is to compactify the morphism to yield projective fibers.

This section describes the result of the compactification constructed by [Tak19] by blowing up the symmetric power.

Let $\mathfrak{m} = \sum_{i=1}^{n} k_i P_i$, where the P_i are distinct closed points of C, be a modulus on C, and let d satisfy (3.1). Takeuchi ([Tak19]) defines $Z_0 = Z_0(\mathfrak{m}, d)$ as the closed subscheme of $C^{(d)}$ defined by the map $C^{(d-\deg \mathfrak{m})} \to C^{(d)}$ adding $\mathfrak{m}$. He also defines $X_{\mathfrak{m},d}$ as the blowup of $C^{(d)}$ along Z_0. Let $E_0 = E_{\mathfrak{m},d} = Z_0(\mathfrak{m}, d) \times_{C^{(d)}} X_{\mathfrak{m},d}$ be the exceptional divisor of the blowup. It is irreducible of codimension 1, and we let $\eta_0 = \eta_{\mathfrak{m},d}$ be its generic point.

Diagrammatically:

$$\begin{tikzcd}
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
Z_0 \arrow[r, "c.i"] & C^{(d)}
\end{tikzcd}$$

Incorporating $U^{(d)}$, the local system $\mathcal{F}^{[d]}$ and the Abel-Jacobi map, we have:

$$\begin{tikzcd}
 & & \mathcal{F}^{[d]} \arrow[d] & \\
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] & U^{(d)} \arrow[l, hook'] \arrow[r, "\Phi_d"] \arrow[dl, hook'] & \mathrm{Pic}^d_{C,\mathfrak{m}} \\
Z_0 \arrow[r, "c.i"] & C^{(d)} & &
\end{tikzcd}$$

In Section 3 of [Tak19] Takeuchi constructs, for d satisfying (3.1) a compactification denoted by $\tilde{C}^{(d)}_{\mathfrak{m}}$ and proves the following:

Theorem 60 (Takeuchi)**.** *The scheme $\tilde{C}^{(d)}_{\mathfrak{m}}$ is an open subscheme of $X_{\mathfrak{m},d}$ containing $U^{(d)}$. The morphism $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ extends to a morphism $\tilde{\Phi}_d : \tilde{C}^{(d)}_{\mathfrak{m}} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ which makes $\tilde{C}^{(d)}_{\mathfrak{m}}$ a projective space bundle over $\mathrm{Pic}^d_{C,\mathfrak{m}}$. Furthermore, the complement of $U^{(d)}$ in $\tilde{C}^{(d)}_{\mathfrak{m}}$ is the scheme-theoretic intersection*

$$E_0 \times_{X_{\mathfrak{m},d}} \tilde{C}^{(d)}_{\mathfrak{m}} \cong Z_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}.$$

Diagrammatically we have:

$$\begin{tikzcd}
 & & \mathcal{F}^{[d]} \arrow[d] \\
E_0 \times_{X_{\mathfrak{m},d}} \tilde{C}^{(d)}_{\mathfrak{m}} \arrow[r] \arrow[d] & \tilde{C}^{(d)}_{\mathfrak{m}} \arrow[d, hook] \arrow[dr, "\tilde{\Phi}_d"] & U^{(d)} \arrow[l, hook'] \arrow[d, "\Phi_d"] \\
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] & \mathrm{Pic}^d_{C,\mathfrak{m}} \\
Z_0 \arrow[r, "c.i"] & C^{(d)} &
\end{tikzcd}$$

Here the boundary may equivalently be written as $Z_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}}$, since $E_0 = Z_0 \times_{C^{(d)}} X_{\mathfrak{m},d}$.

The construction gives more than the description of the open complement in Theorem 60: it identifies the boundary as a relative hyperplane.

Corollary 61 (The boundary is a relative hyperplane)**.** *The boundary*

$$H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$$

is a smooth relative hyperplane divisor in the projective-space bundle $\widetilde{\Phi}_d : \tilde{C}^{(d)}_{\mathfrak{m}} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$*. Consequently, if* $\bar{s} = \operatorname{Spec}\Omega \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ *is any geometric point and* $r = d - \deg\mathfrak{m} - g + 1$*, then there is an isomorphism of pairs*

$$\Big((\tilde{C}^{(d)}_{\mathfrak{m}})_{\bar{s}}, H_{\bar{s}}\Big) \cong \Big(\mathbb{P}^r_\Omega, \mathbb{P}^{r-1}_\Omega\Big),$$

and hence $U^{(d)}_{\bar{s}} \cong \mathbb{A}^r_\Omega$*.*

Proof. In Takeuchi's construction, the projective-space bundle is obtained from a locally free sheaf $E_{\mathfrak{m}}$ fitting into an exact sequence whose kernel is $\mathrm{pr}_* \mathcal{L}(-\mathfrak{m})$ and whose quotient is invertible. The resulting closed immersion

$$\mathbb{P}\big(\mathrm{pr}_* \mathcal{L}(-\mathfrak{m})\big) \hookrightarrow \mathbb{P}(E_{\mathfrak{m}})$$

is therefore a relative hyperplane. Takeuchi identifies this hyperplane scheme-theoretically with the inverse image of Z_0, and identifies $\mathbb{P}(E_{\mathfrak{m}})$ with the blowup [Tak19, Lemmas 3.2 and 3.3(1), Theorem 3.4]. Pulling these identifications back along the open immersion defining $\tilde{C}^{(d)}_{\mathfrak{m}}$ gives the asserted description of H. The statement on geometric fibers follows by base change. □

3.3 Tame Ramification Along the Full Boundary

The ramification calculation of Chapter 2 concerns a modulus supported at a single point and a symmetric power of the corresponding degree. We now combine those local calculations over the different points of $\mathfrak{m}$ and then pass from degree $\deg\mathfrak{m}$ to every sufficiently large degree d.

We retain the blowup notation of Chapter 2. For every nonzero effective submodulus $\mathfrak{n} \le \mathfrak{m}$, let $z_{\mathfrak{n}} \in C^{(\deg\mathfrak{n})}$ be the point representing the divisor $\mathfrak{n}$, let $Z_{\mathfrak{n}}$ be the corresponding closed point, and put

$$X_{\mathfrak{n}} = \mathrm{Bl}_{Z_{\mathfrak{n}}} C^{(\deg\mathfrak{n})}, \qquad E_{\mathfrak{n}} = X_{\mathfrak{n}} \times_{C^{(\deg\mathfrak{n})}} Z_{\mathfrak{n}}.$$

We write $\eta_{\mathfrak{n}}$ for the generic point of $E_{\mathfrak{n}}$. For the full modulus in degree d, the corresponding center, blowup, exceptional divisor, and generic point are denoted by $Z_0 = Z_0(\mathfrak{m}, d)$, $X_{\mathfrak{m},d}$, $E_0 = E_{\mathfrak{m},d}$, and $\eta_0 = \eta_{\mathfrak{m},d}$, as in Section 3.2.

Theorem 62. *Let* Λ *be a finite commutative ring of cardinality invertible in* k*, and let* $\mathcal{F}$ *be an étale sheaf of* Λ*-modules, locally free of rank* 1 *on* U*, with ramification bounded by* $\mathfrak{m}$*. For every sufficiently large integer* d*, the local system* $\mathcal{F}^{[d]}$ *is tamely ramified along*

$$H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)} = E_0 \times_{X_{\mathfrak{m},d}} \widetilde{C}^{(d)}_{\mathfrak{m}} \cong Z_0 \times_{C^{(d)}} \widetilde{C}^{(d)}_{\mathfrak{m}}.$$

3.3.1 From Point-Supported Moduli to the Full Modulus

The first step is to combine the single-point calculations. The following product-compatibility result is placed here because it is used only in the coprime-modulus reduction that immediately follows it.

Compatibility under products

Let X and Y be smooth integral schemes over a field k, and let $x \in X(k)$ and $y \in Y(k)$. We denote the blowups of these schemes at the given points by $\pi_X : \mathrm{Bl}_x(X) \to X$ and $\pi_Y : \mathrm{Bl}_y(Y) \to Y$. Furthermore, let $\pi_{X\times Y} : \mathrm{Bl}_{(x,y)}(X \times_k Y) \to X \times_k Y$ be the blowup of the product scheme at the point (x, y). We denote by E_X, E_Y, and $E_{X\times Y}$ the respective exceptional divisors, and let η_X, η_Y, and $\eta_{X\times Y}$ be their generic points.

In this section, we establish the following result concerning the stability of ramification bounds under the external product of torsors.

Proposition 63. *Let G be a finite abelian group. Suppose $\mathcal{G}_X$ and $\mathcal{G}_Y$ are G-torsors defined on open subsets $U_X \subset \mathrm{Bl}_x(X)$ and $U_Y \subset \mathrm{Bl}_y(Y)$ that are disjoint from the exceptional divisors. If the ramification of $\mathcal{G}_X$ at η_X and $\mathcal{G}_Y$ at η_Y is bounded by r, then the external product torsors*

$$\mathcal{G}_{X\times Y} := pr_1^{-1}\mathcal{G}_X \otimes pr_2^{-1}\mathcal{G}_Y$$

have ramification bounded by r at the generic point $\eta_{X\times Y}$ of the exceptional divisor in the product blowup.

Proof. The proposition is purely local in nature, so it suffices to consider the case where X and Y are affine. More precisely, by the smoothness of X and Y, we may restrict our attention to open neighborhoods of x and y that are etale over affine spaces. The rest of this section treats that case.

The Affine Case

Let $X = \mathbb{A}^n_k$ be the affine n-space over a field k, and let $0 \in X$ be the origin. Let $\tilde{X} = \mathrm{Bl}_0(X)$ be the blowup of X at the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}^{n-1}_k$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}^{n-1}_k$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin, $E = \{(0, [u_1 : \cdots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E, and let $R = \mathcal{O}_{\tilde{X},\eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $K = K(X) = k(x_1, \ldots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1}x_1$. The coordinate ring is:

$$\mathcal{O}_{\tilde{X}}(U_1) = k\left[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}\right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R.

For any monomial $M = x_1^{a_1} \ldots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i}\left(\frac{u_2}{u_1}\right)^{a_2} \cdots \left(\frac{u_n}{u_1}\right)^{a_n}$$

Since the term in parentheses is a unit in R, the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \cdots + f_l$, where f_i is the homogeneous part of degree i, we have $\nu_E(f) = d$ (the order of vanishing at the origin).

The Product Case

Now, let $X = \mathbb{A}^n$ and $Y = \mathbb{A}^m$ with origins $x = 0$ and $y = 0$. As before, $\mathrm{Bl}_0(X) \subset X \times \mathbb{P}^{n-1}$ and $\mathrm{Bl}_0(Y) \subset Y \times \mathbb{P}^{m-1}$ have exceptional divisors E_X and E_Y respectively. Consider the product $X \times_k Y \cong \mathbb{A}_k^{n+m}$. The blowup of the product at the origin $(0,0)$, denoted $\mathrm{Bl}_{(0,0)}(X \times_k Y)$, is a subscheme of $(X \times Y) \times \mathbb{P}^{n+m-1}$ defined by:

$$\begin{cases} x_i w_j = x_j w_i & 1 \leq i,j \leq n \\ y_k w_{n+l} = y_l w_{n+k} & 1 \leq k,l \leq m \\ x_i w_{n+j} = y_j w_i & 1 \leq i \leq n,\ 1 \leq j \leq m \end{cases}$$

where $[w_1 : \cdots : w_{n+m}]$ are the homogeneous coordinates of $\mathbb{P}^{n+m-1}$. The exceptional divisor $E_{X\times Y}$ is isomorphic to $\mathbb{P}^{n+m-1}$.

Comparison of Blowups

Both $\mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$ and $\mathrm{Bl}_{(0,0)}(X \times_k Y)$ are birational to $X \times_k Y$; their common dense open is

$$(X \setminus \{0\}) \times_k (Y \setminus \{0\}).$$

Put $B = \mathrm{Bl}_{(0,0)}(X \times_k Y)$, and let

$$q \colon B \longrightarrow \mathbb{P}_k^{n+m-1}$$

be the second projection in the above realization of B. Define

$$\Omega_X = \bigcup_{i=1}^{n} D_+(w_i), \qquad \Omega_Y = \bigcup_{j=1}^{m} D_+(w_{n+j}),$$

and

$$\widetilde{U} := q^{-1}(\Omega_X \cap \Omega_Y) = \bigcup_{\substack{1\leq i\leq n \\ 1\leq j\leq m}} q^{-1}\big(D_+(w_i w_{n+j})\big).$$

There is a morphism

$$\begin{aligned} f \colon \widetilde{U} &\longrightarrow \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y), \\ ((x,y),[w_1 : \cdots : w_{n+m}]) &\longmapsto \big((x,[w_1 : \cdots : w_n]),(y,[w_{n+1} : \cdots : w_{n+m}])\big). \end{aligned}$$

Thus we have a diagram

$$\begin{tikzcd} & \widetilde{U} \arrow[dl, "f"'] \arrow[dr, hook, "j"] & \\ \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y) & & B, \end{tikzcd}$$

where j is the open immersion. The restriction of q induces an isomorphism $E_{X\times Y} \xrightarrow{\sim} \mathbb{P}_k^{n+m-1}$. Since neither $V_+(w_1,\ldots,w_n)$ nor $V_+(w_{n+1},\ldots,w_{n+m})$ contains the generic point of $\mathbb{P}_k^{n+m-1}$, we have $\eta_{X\times Y} \in \widetilde{U}$.

Extensions of DVRs

Let

$$S = \mathcal{O}_{B,\eta_{X\times Y}}, \qquad R_X = \mathcal{O}_{\mathrm{Bl}_0(X),\eta_X}, \qquad R_Y = \mathcal{O}_{\mathrm{Bl}_0(Y),\eta_Y}.$$

The morphisms $\mathrm{pr}_X \circ f$ and $\mathrm{pr}_Y \circ f$ send $\eta_{X\times Y}$ to η_X and η_Y, respectively, and therefore induce local homomorphisms of DVRs

$$R_X \longrightarrow S, \qquad R_Y \longrightarrow S.$$

The point $\eta_{X\times Y}$ belongs to $q^{-1}(D_+(w_1 w_{n+1}))$. On this chart,

$$\begin{aligned} S &= k\left[x_1, \frac{x_2}{x_1}, \dots, \frac{x_n}{x_1}, \frac{y_1}{x_1}, \dots, \frac{y_m}{x_1}\right]_{(x_1)} \\ &= k\left[y_1, \frac{x_1}{y_1}, \dots, \frac{x_n}{y_1}, \frac{y_2}{y_1}, \dots, \frac{y_m}{y_1}\right]_{(y_1)}. \end{aligned}$$

Consequently, x_1 is a uniformizer in both R_X and S, while y_1 is a uniformizer in both R_Y and S. Both extensions have ramification index one. Their residue-field extensions are

$$k\left(\frac{x_2}{x_1}, \dots, \frac{x_n}{x_1}\right) \subset k\left(\frac{x_2}{x_1}, \dots, \frac{x_n}{x_1}, \frac{y_1}{x_1}, \dots, \frac{y_m}{x_1}\right)$$

and

$$k\left(\frac{y_2}{y_1}, \dots, \frac{y_m}{y_1}\right) \subset k\left(\frac{x_1}{y_1}, \dots, \frac{x_n}{y_1}, \frac{y_2}{y_1}, \dots, \frac{y_m}{y_1}\right),$$

respectively. In particular, both are purely transcendental and hence separable.

Ramification of G-Torsors

The extensions $R_X \to S$ and $R_Y \to S$ are weakly unramified and have separable residue-field extensions. Hence the pullbacks of $\mathcal{G}_X$ and $\mathcal{G}_Y$ to $\mathrm{Frac}(S)$ both have ramification bounded by r. Their contracted product is the restriction of $\mathcal{G}_{X\times Y}$ to $\mathrm{Frac}(S)$, so Lemma 42 gives the same bound for $\mathcal{G}_{X\times Y}$ at $\eta_{X\times Y}$. □

Lemma 64. *Let $\mathfrak{n}_1, \mathfrak{n}_2 \subset \mathfrak{m}$ be two coprime nonzero submoduli. Assume that $\mathcal{P}^{[\deg \mathfrak{n}_1]}$ and $\mathcal{P}^{[\deg \mathfrak{n}_2]}$ are tamely ramified at $\eta_{\mathfrak{n}_1}$ and $\eta_{\mathfrak{n}_2}$, respectively. Then $\mathcal{P}^{[\deg \mathfrak{n}_1 + \deg \mathfrak{n}_2]}$ is tamely ramified at $\eta_{\mathfrak{n}_1+\mathfrak{n}_2}$.*

Proof. Apply Proposition 63 to the two torsors and then use the compatibility of symmetric powers with addition of divisors from Proposition 28. □

3.3.2 From the Minimal Degree to Large Degrees

We now propagate the result in degree $\deg \mathfrak{m}$ to degree d by adding an arbitrary effective divisor of degree $d - \deg \mathfrak{m}$.

The following addition lemma is adapted from [Tak19, Lemma 4.1].

Lemma 65. *Let C be a projective, smooth, and geometrically connected curve over a perfect field k. Let $\mathfrak{m} = \sum_{i=1}^{r} k_i P_i$ be an effective divisor where $P_1, \dots, P_r$ are distinct closed points. Let $U = C \setminus \mathfrak{m}$ and let $d \geq \deg \mathfrak{m}$.*

Suppose $\mathfrak{n}_1, \ldots, \mathfrak{n}_l$ are pairwise coprime submoduli of $\mathfrak{m}$ such that $\mathfrak{m} = \sum_{j=1}^{l} \mathfrak{n}_j$. Consider the summation morphism:

$$\pi : C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow C^{(d)}$$

defined by $(D_1, \ldots, D_l, D_{extra}) \mapsto \sum_{j=1}^{l} D_j + D_{extra}$.

Then π is étale at the generic point of the closed subvariety

$$V = \{\mathfrak{n}_1\} \times_k \cdots \times_k \{\mathfrak{n}_l\} \times_k C^{(d-\deg \mathfrak{m})}$$

inside the domain $C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$.

Proof. We may assume that k is algebraically closed (hence $\deg P_i = 1$ for all i). The morphism π is finite: it is induced by the finite quotient morphism

$$C^d / \big(\mathfrak{S}_{\deg \mathfrak{n}_1} \times \cdots \times \mathfrak{S}_{\deg \mathfrak{n}_l} \times \mathfrak{S}_{d-\deg \mathfrak{m}}\big) \longrightarrow C^d / \mathfrak{S}_d.$$

Its source and target are smooth k-schemes of pure dimension d. Hence π is flat by miracle flatness, and therefore finite flat. It is enough to show that there exists a closed point Q of $\mathfrak{n}_1 + \ldots \mathfrak{n}_l + C^{(d-\deg \mathfrak{m})} \subset C^{(d)}$ over which there are $\deg \pi$ points on $C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$. (Because it will be unramified at this point and thus also at the generic point of V.) Choose Q as a point corresponding to a divisor $\mathfrak{n}_1 + \cdots + \mathfrak{n}_l + P_{r+1} + \cdots + P_{r+d-\deg \mathfrak{m}}$, where $P_1, \ldots, P_{r+d-\deg \mathfrak{m}}$ are distinct points of $U(k)$. □

Corollary 66. *The morphism $C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \xrightarrow{\pi} C^{(d)}$ is finite flat everywhere, and étale at the generic point of the closed subvariety $Z_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$.*

Proof. This is the case $l = 1$ and $\mathfrak{n}_1 = \mathfrak{m}$ of Lemma 65. □

Following from this, we look at the following diagram, coming from the flat base change $C^{(d-\deg \mathfrak{m})} \to \mathrm{Spec}(k)$ (Proposition 20) of (2.1):

$$\begin{tikzcd}
E_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})} \arrow[r] \arrow[d] & X_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})} \arrow[d] & \mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})} \arrow[d] \\
Z_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})} \arrow[r] & C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} & U^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \arrow[ul]
\end{tikzcd}$$

Note that $U^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$ is dense open subscheme of $X_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})}$, And $E_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})}$ is a prime divisor of $X_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})}$. Hence it is a well-defined question according to Definition 38 to ask whether $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})}$ is tamely ramified at the generic point θ of $E_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})}$.

Lemma 67. *If $\mathcal{F}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_\mathfrak{m}$, then $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})}$ is tamely ramified at the generic point θ of $E_\mathfrak{m} \times_k C^{(d-\deg \mathfrak{m})}$.*

Proof. This follows from [Stacks, Tag 0EYD] □

By Lemma 67, replacing $C^{(d-\deg \mathfrak{m})}$ with the dense open subscheme $U^{(d-\deg \mathfrak{m})} \subset C^{(d-\deg \mathfrak{m})}$, we get that the G-torsor $p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]}$ ($\mathcal{P}$ corresponds to $\mathcal{F}$ under Proposition 9) is tamely ramified at θ the generic point of $E_\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$, viewed as a prime divisor of $X_\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$; the torsor itself is defined on the dense open $U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$, where $p_1 : U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to U^{(\deg \mathfrak{m})}$ is the projection to the first factor.

Looking at the second projection $p_2 : X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \to U^{(d-\deg \mathfrak{m})}$, and the fact that $\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $U^{(d-\deg \mathfrak{m})}$ we get that $p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]} = X_{\mathfrak{m}} \times_k \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Hence, its restriction to $U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ is unramified at θ.

Thus, by Lemma 68, we conclude that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]} = p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]} \otimes^G p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$.

Lemma 68. *Let $X \to \operatorname{Spec} k$ be a scheme over a field k, and let $\mathcal{P}_1, \mathcal{P}_2$ be two G-torsors on U_{et}. Let ξ be the generic point of a prime divisor $D \subset X$. If $\mathcal{P}_1$ is tamely ramified at ξ, and $\mathcal{P}_2$ is unramified at ξ, then the product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ is tamely ramified at ξ.*

Proof. This follows from Lemma 42. ☐

Combining this with Corollary 66 we get

Corollary 69. *Let $\mathfrak{m}$ be a modulus as above, and let $\eta_{\mathfrak{m}}$ be the generic point of $E_{\mathfrak{m}}$. Let $\mathcal{P}$ be a G-torsor on U_{et} with ramification bounded by $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Then $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$, and $\mathcal{P}^{[d]}$ is tamely ramified at the generic point $\eta_0 = \eta_{\mathfrak{m},d}$ of $E_0 = E_{\mathfrak{m},d}$*

Proof. The first assertion, that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ, follows from the preceding discussion. Thus, it remains to show that $\mathcal{P}^{[d]}$ is tamely ramified at η_0.

Consider the blowup diagram defining $X_{\mathfrak{m},d}$:

$$\begin{tikzcd}
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
Z_0 \arrow[r, "c.i"] & C^{(d)}
\end{tikzcd}$$

By performing a base change along the flat addition map $+ : C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(d)}$, we obtain the following commutative diagram:

$$\begin{tikzcd}
 & \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} E_0 \arrow[r] \arrow[d] & \overline{\{\eta_0\}} = E_0 \arrow[d] \\
 & \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d} \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "c.i"] & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "+"] & C^{(d)}
\end{tikzcd}$$

By Theorem 30, blowups commute with flat base change. Since the inverse image of the center Z_0 under the map $+$ is $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$, the scheme $\left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d}$ is the blowup of $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ along $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. This, in turn, is isomorphic to the base change of the blowup $X_{\mathfrak{m}}$ (of $C^{(\deg \mathfrak{m})}$ along $\mathfrak{m}$) via the (flat) projection $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(\deg \mathfrak{m})}$.

Assembling these facts, we obtain the following Cartesian square:

$$
\begin{tikzcd}
 & E_{\mathfrak{m}} \times U^{(d-\deg\mathfrak{m})} \arrow[r, "\tilde{+}"] \arrow[d] & \overline{\{\eta_0\}} = E_0 \arrow[d] \\
 & X_{\mathfrak{m}} \times_k U^{(d-\deg\mathfrak{m})} \arrow[r, "\tilde{+}"] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
\mathfrak{m} \times_k U^{(d-\deg\mathfrak{m})} \arrow[r, "c.i"] & C^{(\deg\mathfrak{m})} \times_k U^{(d-\deg\mathfrak{m})} \arrow[r, "+"] & C^{(d)}
\end{tikzcd}
$$

The point θ defined in the Corollary is the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg\mathfrak{m})}$. Let η be the generic point of $\mathfrak{m} \times_k U^{(d-\deg\mathfrak{m})}$. By Corollary 66, the map $+$ is étale at η. Consequently, the lifted map $\tilde{+}$ is étale at θ. Given the isomorphism $(+^{-1})(\mathcal{P}^{[d]}) \cong \mathcal{P}^{[\deg\mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg\mathfrak{m}]}$ from Proposition 28, the tame ramification of the box product at θ descends to the tame ramification of $\mathcal{P}^{[d]}$ at η_0 by applying Lemma 43.

□

3.3.3 Conclusion of the Tameness Argument

Proof of Theorem 62. Let $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$ be the $\Lambda^\times$-torsor of frames. For each single-point submodulus $\mathfrak{n} = k_P P$ of $\mathfrak{m}$, Theorem 3 shows that $\mathcal{P}^{[\deg\mathfrak{n}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$. Repeated application of Lemma 64 shows that $\mathcal{P}^{[\deg\mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Therefore Corollary 69 implies that $\mathcal{P}^{[d]}$ is tamely ramified at the generic point η_0 of E_0 for every sufficiently large d. By Proposition 9 and definition 26, the same conclusion holds for $\mathcal{F}^{[d]}$. Since H is the nonempty open part of E_0 described in Corollary 61, its generic point is η_0. Thus $\mathcal{F}^{[d]}$ is tamely ramified along H. □

3.4 Extension Across the Boundary

We now use the geometry of the Abel–Jacobi fibers to upgrade the tameness proved above to extension across the boundary. On the geometric generic fiber, a Tendler-style prime-to-p power argument forces the boundary inertia representation to be trivial.

Lemma 70. *Let Λ be a finite commutative ring whose cardinality is invertible in k, and let $\mathcal{F}^{[d]}$ be a locally constant sheaf of free rank-one Λ-modules on $U^{(d)}$. If $\mathcal{F}^{[d]}$ is tamely ramified at the generic point of*

$$H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)},$$

then it extends uniquely to a locally constant sheaf $\widetilde{\mathcal{F}}^{[d]}$ on $\widetilde{C}^{(d)}_{\mathfrak{m}}$.

Proof. Put $r = d - \deg\mathfrak{m} - g + 1$, let ξ be the generic point of $\mathrm{Pic}^d_{C,\mathfrak{m}}$, let $\Omega = \kappa(\xi)^{\mathrm{sep}}$, and write $\bar{\xi} = \operatorname{Spec}\Omega$. By Corollary 61, the compactified geometric generic Abel–Jacobi fiber and its boundary are

$$\left(\widetilde{C}^{(d)}_{\mathfrak{m},\bar{\xi}}, H_{\bar{\xi}}\right) \cong \left(\mathbb{P}^r_\Omega, \mathbb{P}^{r-1}_\Omega\right),$$

and hence the geometric generic open fiber is

$$U^{(d)}_{\bar{\xi}} \cong \mathbb{A}^r_\Omega.$$

Let

$$\chi\colon \pi_1\big(\mathbb{A}^r_\Omega\big) \longrightarrow \Lambda^\times$$

be the character corresponding to the restriction of $\mathcal{F}^{[d]}$ to this fiber.

Let η be the generic point of H, and let $\bar{\eta}$ be the generic point of $H_{\bar{\xi}}$. Because H is a relative hyperplane divisor, the passage from η to $\bar{\eta}$ preserves a local equation of the boundary and induces a separable extension of residue fields. The corresponding extension of henselian discrete valuation rings is therefore unramified, and it identifies the inertia group at η with the inertia group

$$I_\infty := I_{\bar{\eta}}$$

at the hyperplane at infinity. In particular, tameness and triviality of the finite inertia representation may both be checked after this base change.

If $\mathrm{char}(k) = 0$, then

$$\pi_1\big(\mathbb{A}^r_\Omega\big) = 1,$$

so χ is trivial [Ten15, Example 4.14].

Suppose now that $\mathrm{char}(k) = p > 0$. Since $\mathcal{F}^{[d]}$ is tame along H, the finite group $\chi(I_\infty)$ has order prime to p. Let N be its exponent. Then $(N, p) = 1$, and the character χ^N is trivial on I_∞.

Consequently, the local system corresponding to χ^N is unramified at the generic point of $\mathbb{P}^{r-1}_\Omega$. By purity, it extends to $\mathbb{P}^r_\Omega$. Since

$$\pi_1\big(\mathbb{P}^r_\Omega\big) = 1,$$

this extension is constant, and therefore

$$\chi^N = 1.$$

On the other hand, χ factors through

$$\pi_1\big(\mathbb{A}^r_\Omega\big)^{\mathrm{ab}},$$

which is a pro-p group [Ten15, Example 4.15]. Hence the finite group $\mathrm{im}(\chi)$ is a p-group. Since its exponent divides N, and N is prime to p, it follows that

$$\mathrm{im}(\chi) = 1.$$

Thus χ is trivial.

In either characteristic, the restriction of χ to I_∞ is trivial. By the inertia comparison above, the inertia group at η acts trivially on $\mathcal{F}^{[d]}$.

Therefore $\mathcal{F}^{[d]}$ is unramified at the generic point of H. Since $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is regular and its complement of $U^{(d)}$ is the single smooth relative hyperplane divisor H, purity of the branch locus ([Stacks, Tag 0BMB]) gives an extension of $\mathcal{F}^{[d]}$ to $\widetilde{C}^{(d)}_{\mathfrak{m}}$. The extension is unique because

$$\pi_1\big(U^{(d)}\big) \longrightarrow \pi_1\big(\widetilde{C}^{(d)}_{\mathfrak{m}}\big)$$

is surjective. □

Remark. The argument does not assert that tame ramification is generally unramified. Tameness is used only to choose N prime to p. The conclusion that the inertia character is trivial then follows from the global geometry of the geometric generic fiber and from the fact that the abelian fundamental group of affine space is pro-p.

3.5 Descent and Reduced Geometric Class Field Theory

We finish by recording the fundamental-group input and applying it to the ordinary and generalized Abel–Jacobi morphisms.

3.5.1 The Fundamental-Group Input

Proposition 71 ([Stacks, Tag 0C0J]). *Let $f\colon X \to S$ be a flat proper morphism of finite presentation whose geometric fibers are connected and reduced. Assume that S is connected, and let $\bar{s}$ be a geometric point of S. Then there is an exact sequence*

$$\pi_1(X_{\bar{s}}) \longrightarrow \pi_1(X) \longrightarrow \pi_1(S) \longrightarrow 1.$$

Corollary 72. *Let $f\colon X \to S$ be a proper smooth morphism of finite presentation whose geometric fibers are connected. Then the sequence of Proposition 71 is exact.*

Proof. A smooth geometric fiber is reduced, so this is an immediate application of Proposition 71. ☐

Corollary 73. *Let $f\colon X \to S$ be a projective space bundle over a connected scheme. Then pullback induces an isomorphism*

$$\pi_1^{\mathrm{et}}(X) \xrightarrow{\sim} \pi_1^{\mathrm{et}}(S).$$

Proof. The geometric fibers are projective spaces and therefore have trivial étale fundamental group. The assertion follows from Corollary 72. ☐

3.5.2 The Descent Argument

We work over $S = \operatorname{Spec} k$, where k is perfect. By Proposition 9, it is equivalent first to prove the following torsor formulation.

Theorem 74 (Torsor formulation). *Let $G = \Lambda^\times$, and let $\mathcal{P}$ be a G-torsor on U whose ramification is bounded by $\mathfrak{m}$. For every sufficiently large integer d, there exists a unique, up to isomorphism, G-torsor $\mathcal{Q}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$ such that*

$$\Phi_d^*\mathcal{Q}_d \cong \mathcal{P}^{[d]}.$$

Proof. We divide the proof into two cases.

Case 1: $\mathfrak{m} = 0$. For d sufficiently large, the usual Abel–Jacobi morphism

$$\Phi_d\colon C^{(d)} \longrightarrow \mathrm{Pic}^d_C$$

is proper and smooth with geometrically connected fibers, and Theorem 59 identifies every geometric fiber with $\mathbb{P}^{d-g}$. Applying Proposition 71, and using $\pi_1^{\mathrm{et}}(\mathbb{P}^{d-g}) = 1$, gives

$$\pi_1^{\mathrm{et}}\big(C^{(d)}\big) \xrightarrow{\sim} \pi_1^{\mathrm{et}}\big(\mathrm{Pic}^d_C\big).$$

The G-torsor $\mathcal{P}^{[d]}$ therefore descends uniquely to a G-torsor on Pic^d_C.

Case 2: $\mathfrak{m} > 0$. Let $\mathcal{F} = \mathcal{P} \times^G \Lambda$ be the rank-one local system associated with $\mathcal{P}$. By Definition 26, the frame torsor of $\mathcal{F}^{[d]}$ is $\mathcal{P}^{[d]}$. For d sufficiently large, Theorem 60 extends Φ_d to a projective space bundle

$$\widetilde{\Phi}_d \colon \widetilde{C}^{(d)}_{\mathfrak{m}} \longrightarrow \mathrm{Pic}^d_{C,\mathfrak{m}}.$$

By Theorem 62, $\mathcal{F}^{[d]}$ is tamely ramified along $H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$. Hence Lemma 70 gives a unique extension $\widetilde{\mathcal{F}}^{[d]}$ to $\widetilde{C}^{(d)}_{\mathfrak{m}}$.

By Corollary 73,

$$\pi_1^{\mathrm{et}}\big(\widetilde{C}^{(d)}_{\mathfrak{m}}\big) \xrightarrow{\sim} \pi_1^{\mathrm{et}}\big(\mathrm{Pic}^d_{C,\mathfrak{m}}\big).$$

Thus $\widetilde{\mathcal{F}}^{[d]}$ descends uniquely to a rank-one local system $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$. Restricting to $U^{(d)}$ gives $\Phi_d^* \mathcal{G}_d \cong \mathcal{F}^{[d]}$. Taking frame torsors yields

$$\Phi_d^* \,\mathrm{Fr}(\mathcal{G}_d) \cong \mathcal{P}^{[d]},$$

so $\mathcal{Q}_d = \mathrm{Fr}(\mathcal{G}_d)$ has the required property. Uniqueness follows from the fundamental-group isomorphism and from the surjection

$$\pi_1^{\mathrm{et}}(U^{(d)}) \twoheadrightarrow \pi_1^{\mathrm{et}}(\widetilde{C}^{(d)}_{\mathfrak{m}})$$

induced by the dense open immersion into the normal scheme $\widetilde{C}^{(d)}_{\mathfrak{m}}$. $\square$

Proof of Theorem 2. Let $\mathcal{F}$ be as in Theorem 2, and set $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$. Apply Theorem 74 to obtain $\mathcal{Q}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, and let

$$\mathcal{G}_d = \mathcal{Q}_d \times^{\Lambda^\times} \Lambda.$$

The equivalence of Proposition 9, together with the compatibility of frame torsors and symmetric powers from Definition 26, gives $\Phi_d^* \mathcal{G}_d \cong \mathcal{F}^{[d]}$. The same equivalence carries uniqueness of $\mathcal{Q}_d$ to uniqueness of $\mathcal{G}_d$. $\square$

Appendix A

Cyclic Torsor Theories and Witt Vectors

This appendix contains the supporting theory and proofs for the three cyclic torsor descriptions stated in Chapter 2. We first record the general algebra of finite-length p-typical Witt vectors, and then give the group-scheme, classification, normal-form, and realization arguments used there. The Schmid–Witt ramification-break formula remains an externally cited theorem.

A.1 The ring of Witt vectors

For $n \geq 0$, the n-th **Witt polynomial** in the variables $X_0, \dots, X_n$ is

$$w_n(X_0, \dots, X_n) \;=\; \sum_{j=0}^{n} p^j X_j^{p^{n-j}} \;=\; X_0^{p^n} + pX_1^{p^{n-1}} + \cdots + p^n X_n.$$

Theorem 75 (Witt; [Ser79, II, §6, Theorem 6], [Rab14, §1])**.** *There exists a unique family of polynomials $S_n, P_n \in \mathbb{Z}[X_0, \dots, X_n; Y_0, \dots, Y_n]$, $n \geq 0$, such that*

$$w_n(S_0, \dots, S_n) = w_n(X_0, \dots, X_n) + w_n(Y_0, \dots, Y_n), \qquad w_n(P_0, \dots, P_n) = w_n(X_0, \dots, X_n) \cdot w_n(Y_0, \dots, Y_n)$$

for all n. For any commutative ring A, the operations

$$\vec{a} \oplus \vec{b} := \big(S_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}, \qquad \vec{a} \odot \vec{b} := \big(P_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}$$

*make the set A^r into a commutative ring $W_r(A)$, the ring of **p-typical Witt vectors of length** r **over** A, with zero $(0, \dots, 0)$ and unit $(1, 0, \dots, 0)$. Additive inverses are likewise given by universal integral polynomials; we write $\ominus$ for Witt subtraction. The **ghost map***

$$w \colon W_r(A) \longrightarrow A^r, \qquad \vec{a} \longmapsto \big(w_0(\vec{a}), w_1(\vec{a}), \dots, w_{r-1}(\vec{a})\big)$$

is a ring homomorphism to the product ring A^r; it is injective when p is a non-zero-divisor in A, and bijective when $p \in A^\times$.

We index the components of a Witt vector $\vec{a} = (a_0, \dots, a_{r-1})$ by $0, \dots, r-1$. For $r = 1$ the polynomials reduce to $S_0 = X_0 + Y_0$ and $P_0 = X_0 Y_0$, so $W_1(A) = A$ as a ring.

Since the ring operations are given by universal polynomials with integer coefficients, W_r is a functor: a ring homomorphism $\sigma \colon A \to B$ induces the ring homomorphism

$$W_r(\sigma) \colon W_r(A) \to W_r(B), \qquad W_r(\sigma)(a_0, \dots, a_{r-1}) = (\sigma a_0, \dots, \sigma a_{r-1}),$$

acting *componentwise*. Two consequences of this trivial-looking observation are used repeatedly and deserve emphasis: a ring automorphism σ of A commutes with $\oplus$ and $\ominus$ on $W_r(A)$, and a Witt vector $\vec{a} \in W_r(A)$ satisfies $W_r(\sigma)(\vec{a}) = \vec{a}$ if and only if *each component* a_n is σ-invariant. In particular, for a group Γ acting on A by ring automorphisms,

$$W_r(A)^\Gamma \;=\; W_r\big(A^\Gamma\big) \qquad \text{inside } W_r(A). \tag{A.1}$$

The uniqueness statement in Theorem 75 is an instance of a general principle that we invoke several times, so we record it explicitly.

Lemma 76 (ghost principle)**.** *Let Φ and Ψ be two operations on Witt vectors given by universal polynomials with integer coefficients in the components of their arguments. If $w \circ \Phi = w \circ \Psi$ holds over the polynomial ring $\mathbb{Z}[X_0, X_1, \dots]$ on the components of the arguments, then $\Phi = \Psi$ over every commutative ring.*

Proof. Over $\mathbb{Z}[X_0, X_1, \dots]$ the prime p is a non-zero-divisor, so the ghost map is injective by Theorem 75 and $\Phi = \Psi$ there; this is an identity of integral polynomials, which persists under every base change $\mathbb{Z}[X_0, X_1, \dots] \to A$. □

The only structural feature of the addition polynomials S_n that we shall need is their "triangular" shape:

Proposition 77 (shape of Witt addition)**.** *For every $n \geq 0$ there is a polynomial $C_n \in \mathbb{Z}[X_0, \dots, X_{n-1}; Y_0, \dots, Y_{n-}$ involving only the variables of index* strictly less *than n, such that*

$$(\vec{a} \oplus \vec{b})_n \;=\; a_n + b_n + C_n(a_0, \dots, a_{n-1}; b_0, \dots, b_{n-1}). \tag{A.2}$$

We refer to C_n as the ***carry polynomial****. The analogous statement holds for $\ominus$.*

Proof. By induction on n, using the recursion defining S_n:

$$p^n S_n \;=\; \sum_{j \leq n} p^j \big(X_j^{p^{n-j}} + Y_j^{p^{n-j}}\big) \;-\; \sum_{j<n} p^j S_j^{p^{n-j}}.$$

The terms with $j = n$ contribute $p^n(X_n + Y_n)$, and all remaining terms involve only variables and polynomials S_j of index $j < n$, which by induction lie in $\mathbb{Z}[X_{<n}; Y_{<n}]$. Since S_n has integer coefficients (Theorem 75), $C_n := S_n - X_n - Y_n$ is an integral polynomial in the variables of index $< n$. The statement for $\ominus$ follows by the same argument applied to the inversion polynomials. □

A.2 Frobenius, Verschiebung and truncation

From now on, A denotes an $\mathbb{F}_p$-algebra; this is the only case we use. Three operators act on the rings $W_r(A)$:

Definition 78. Let A be an $\mathbb{F}_p$-algebra.

1. The **Frobenius** $F \colon W_r(A) \to W_r(A)$ is $F := W_r(\varphi_A)$, where $\varphi_A \colon a \mapsto a^p$ is the absolute Frobenius of A. Explicitly, F acts componentwise: $F(a_0, \dots, a_{r-1}) = (a_0^p, \dots, a_{r-1}^p)$. By functoriality, F is a ring endomorphism of $W_r(A)$.

2. The **Verschiebung** is the shift

$$V\colon W_{r-1}(A) \to W_r(A), \qquad V(a_0,\dots,a_{r-2}) = (0,a_0,\dots,a_{r-2}).$$

Iterating, $V^j\colon W_{r-j}(A) \to W_r(A)$; in particular $V^{r-1}\colon A = W_1(A) \to W_r(A)$ sends $c \mapsto (0,\dots,0,c)$.

3. The **truncation** is

$$t\colon W_r(A) \to W_{r-1}(A), \qquad t(a_0,\dots,a_{r-1}) = (a_0,\dots,a_{r-2}).$$

Remark. For a general (not necessarily $\mathbb{F}_p$-) algebra A, the Frobenius is defined by universal polynomials characterized by $w_n \circ F = w_{n+1}$, and the componentwise formula above holds precisely in characteristic p; see [Ser79, II, §6] or [Rab14, §2]. We will not need the general case.

We record from the general theory ([Ser79, II, §6], [Rab14, §2]) how multiplication by p interacts with these operators.

Proposition 79. *Let A be an $\mathbb{F}_p$-algebra. Then:*

1. *t is a natural ring homomorphism commuting with F, and its kernel is*

$$\ker\bigl(t\colon W_r(A) \to W_{r-1}(A)\bigr) \;=\; V^{r-1}A \;=\; \{(0,\dots,0,c) : c \in A\}.$$

2. *V is additive, and $FV = VF$.*

3. *Multiplication by p on $W_r(A)$ equals $V \circ F = F \circ V$ (composed with the appropriate truncation); explicitly,*

$$p \cdot (a_0,\dots,a_{r-1}) \;=\; (0,\, a_0^p,\, \dots,\, a_{r-2}^p).$$

Iterating, $p^j \cdot \vec{a} = (0,\dots,0,a_0^{p^j},\dots,a_{r-1-j}^{p^j})$; in particular $p^r = 0$ on $W_r(A)$, and

$$p^{r-1} \cdot (a_0,\dots,a_{r-1}) \;=\; \bigl(0,\dots,0,\, a_0^{p^{r-1}}\bigr). \tag{A.3}$$

Proof. (1) That t is a ring homomorphism follows from the recursive definition of the S_n, P_n: the first $r-1$ components of a sum or product depend only on the first $r-1$ components of the arguments (Proposition 77). Naturality and the commutation with the (componentwise) F are clear, and the kernel is computed componentwise. (2) Additivity of V follows from Lemma 76 and the ghost identity $w_n(V\vec{a}) = p\,w_{n-1}(\vec{a})$ (with $w_{-1} := 0$); the commutation $FV = VF$ is immediate since F is componentwise. (3) The identity $FV = p$ holds over arbitrary rings for the general Frobenius, by Lemma 76 applied to $w_n(FV\vec{a}) = w_{n+1}(V\vec{a}) = p\,w_n(\vec{a}) = w_n(p\vec{a})$; in characteristic p the general Frobenius is componentwise, giving the displayed formula. See [Ser79, II, §6, Proposition 8 ff.] for details. ☐

Example 80. $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ canonically. Indeed, by (A.3) the unit $\underline{1} = (1,0,\dots,0)$ satisfies $p^j \cdot \underline{1} = (0,\dots,0,1,0,\dots,0)$ (1 in position j), so $\underline{1}$ has additive order p^r; since $|W_r(\mathbb{F}_p)| = p^r$, the additive group is cyclic of order p^r generated by the unit, and $m \mapsto m \cdot \underline{1}$ is a ring isomorphism $\mathbb{Z}/p^r\mathbb{Z} \xrightarrow{\cong} W_r(\mathbb{F}_p)$. We write $\underline{m} \in W_r(\mathbb{F}_p)$ for the image of $m \in \mathbb{Z}/p^r\mathbb{Z}$. (Similarly, the full ring of Witt vectors satisfies $W(\mathbb{F}_p) \cong \mathbb{Z}_p$; we shall not need this.)

A.3 Isobaricity of the Witt addition polynomials

The following homogeneity property of the universal addition polynomials is what makes degree estimates on Witt sums possible: even though the carry polynomials C_n of (A.2) are complicated, they are *isobaric* for the weighting in which the j-th component has weight p^j.

Lemma 81 (isobaricity of Witt addition)**.** *Assign to the variables X_j and Y_j the weight p^j. Then the n-th addition polynomial S_n is isobaric of weight p^n: every monomial of S_n has total weight exactly p^n. The same holds for each component of a d-fold iterated Witt sum $\bigoplus_{i=1}^{d} \vec{a}_i$, viewed as a universal polynomial in the components $a_{i,j}$ of weight p^j.*

Proof. The n-th ghost polynomial $w_n = \sum_{j \le n} p^j X_j^{p^{n-j}}$ is isobaric of weight p^n for this weighting. The recursion

$$p^n S_n = \sum_{j \le n} p^j \big(X_j^{p^{n-j}} + Y_j^{p^{n-j}}\big) - \sum_{j<n} p^j S_j^{p^{n-j}}$$

preserves isobaricity of weight p^n by induction on n: each S_j is isobaric of weight p^j, so $S_j^{p^{n-j}}$ is isobaric of weight p^n. Isobaricity persists under iterated substitution of Witt sums into Witt sums (substituting an isobaric polynomial of weight p^j for a variable of weight p^j preserves weights), giving the statement for d-fold sums. □

A.4 The common group-scheme construction

The three cyclic theories used in Chapter 2 share the same formal construction. Let H be a commutative group scheme over a scheme S, and let $\varphi : H \to H$ be a finite étale surjective homomorphism with constant kernel G. If X is an S-scheme and $a : X \to H$ is a section, then

$$T_a := X \times_{a,H,\varphi} H$$

is a G-torsor over X. In coordinates it is the torsor $\varphi(Y) = a$. If $a' = a + \varphi(b)$ for a section $b : X \to H$, translation by b identifies T_a with $T_{a'}$.

For Kummer theory one takes $H = \mathbb{G}_m$ and $\varphi(z) = z^e$; for Artin–Schreier theory, $H = \mathbb{G}_a$ and $\varphi(z) = z^p - z$; and for Artin–Schreier–Witt theory, $H = \mathbb{W}_r$ and $\varphi = F - \mathrm{id}$. This is why the three equations in Chapter 2 behave in parallel.

A.5 Kummer and Artin–Schreier theory

Let K be a complete discretely valued field with residue characteristic $p > 0$. If $(e, p) = 1$ and K contains μ_e, the Kummer sequence

$$1 \longrightarrow \mu_e \longrightarrow \mathbb{G}_m \xrightarrow{(-)^e} \mathbb{G}_m \longrightarrow 1$$

and Hilbert 90 give

$$H^1(K, \mu_e) \cong K^\times/(K^\times)^e.$$

The class of $a \in K^\times$ is represented by $T^e = a$, with $\zeta \in \mu_e$ acting by $T \mapsto \zeta T$.

Proof of Theorem 49. Write $a = \pi^m u$ with $u \in \mathcal{O}_K^\times$. Replacing a by an element of the same class in $K^\times/(K^\times)^n$ changes m by a multiple of n. If $n \mid m$, we may therefore assume that $a = u$ is a unit. Since n is invertible in the residue field, Hensel's lemma shows that u is an n-th power in K precisely when its residue is an n-th power; this is the trivial case. Otherwise the extension is obtained by lifting the corresponding separable residue-field extension and is unramified.

In general, if $y^n = a$, then $v_L(y) = m/n$ with respect to the extension of v_K. Hence the subgroup of the value group generated by $\mathbb{Z}$ and m/n has index $n/\gcd(n,m)$ over $\mathbb{Z}$, which is the ramification index. When $\gcd(n,m) = 1$ this index is n, so the extension is totally ramified; it is tame because $(n,p) = 1$. The converse is the Kummer classification supplied by the displayed cohomological isomorphism. See [Koc97, Proposition 1.83] for the general statement. $\square$

In characteristic p, the Artin–Schreier sequence

$$0 \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{G}_a \xrightarrow{\wp} \mathbb{G}_a \longrightarrow 0, \qquad \wp(z) = z^p - z,$$

and additive Hilbert 90 give

$$H^1(K, \mathbb{Z}/p\mathbb{Z}) \cong K/\wp(K).$$

The class of $a \in K$ is represented by $X^p - X = a$, with $c \in \mathbb{F}_p$ acting by $X \mapsto X + c$.

Proof of Theorem 50. If $v_K(a) > 0$, the convergent series $u = -\sum_{i \geq 0} a^{p^i}$ satisfies $u^p - u = a$, so the class is trivial. If $v_K(a) = 0$, the same argument removes every term of positive valuation. The extension is therefore determined by the Artin–Schreier class of the residue of a: it is trivial when this residue class vanishes and otherwise is the unramified cyclic extension of degree p.

Suppose now that $v_K(a) = -m < 0$ and $p \nmid m$, and let $y^p - y = a$. Since the residue class cannot be removed by an Artin–Schreier change of variable, the extension has degree p. Normalize v_L so that $v_L|_K = p v_K$. The equation gives $v_L(y) = -m$, so the extension is totally ramified. Choose integers $1 \leq c \leq p-1$ and b with $-cm + bp = 1$. Then $\varpi_L = y^c \pi^b$ is a uniformizer. For a nontrivial $\sigma \in \mathrm{Gal}(L/K)$, after rescaling the generator we have $\sigma(y) = y + 1$, and

$$v_L\big(\sigma(\varpi_L) - \varpi_L\big) = v_L\big(\pi^b((y+1)^c - y^c)\big) = bp - (c-1)m = m + 1.$$

Thus the unique lower ramification break is m; for a cyclic group of order p this is also the unique upper break. Finally, the Artin–Schreier cohomological classification gives every cyclic extension of degree p. The reduced polynomial representative used in the last assertion is constructed in the proof of Lemma 53(2) below. $\square$

A.6 Artin–Schreier–Witt bookkeeping

We retain the notation for Witt vectors from the beginning of this appendix. For an $\mathbb{F}_p$-algebra A, Frobenius acts componentwise on $W_r(A)$.

Definition 82. The **Artin–Schreier–Witt operator** is

$$\wp := F - \mathrm{id}, \qquad \wp\vec{u} = F\vec{u} \ominus \vec{u}.$$

It is an endomorphism of the additive group of $W_r(A)$ and commutes with truncation and Verschiebung.

Lemma 83 (Witt bookkeeping)**.** *Let A be an $\mathbb{F}_p$-algebra.*

(i) The truncation $t : W_r(A) \to W_{r-1}(A)$ is a ring homomorphism commuting with F, hence with $\wp$, and $\ker(t) = V^{r-1}A$.

(ii) The map $V^{r-1} : A \to W_r(A)$ is additive and $\wp \circ V^{r-1} = V^{r-1} \circ \wp$.

(iii) For every $\vec{a} \in W_r(A)$ and $c \in A$,

$$\vec{a} \oplus V^{r-1}(c) = (a_0, \ldots, a_{r-2}, a_{r-1} + c).$$

Proof. The first two assertions follow from Proposition 79. For (iii), apply the ghost principle over $\mathbb{Z}[a_0, \ldots, a_{r-1}, c]$. If $\vec{b} = (0, \ldots, 0, c)$ and $\vec{s} = \vec{a} \oplus \vec{b}$, then $w_n(\vec{b}) = 0$ for $n < r-1$, so induction gives $s_n = a_n$ for $n \leq r-2$. At level $r-1$, the identity $w_{r-1}(\vec{b}) = p^{r-1}c$ gives $p^{r-1}s_{r-1} = p^{r-1}(a_{r-1} + c)$, hence $s_{r-1} = a_{r-1} + c$. The resulting identity of universal integral polynomials holds over every A. ☐

The last assertion is special to the top Verschiebung slice. Witt addition is not componentwise in lower positions, and the carry polynomials in (A.2) cannot in general be discarded.

A.7 The ASW group scheme and classification

The Witt addition polynomials make $\mathbb{A}^r_{\mathbb{F}_p}$ into the additive Witt vector group scheme $\mathbb{W}_r$. The operator $\wp = F - \mathrm{id}$ is an endomorphism of $\mathbb{W}_r$ and fits into an exact sequence for the étale topology

$$0 \longrightarrow W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z} \longrightarrow \mathbb{W}_r \xrightarrow{\wp} \mathbb{W}_r \longrightarrow 0. \tag{A.4}$$

Lemma 84 (ASW covers are étale torsors)**.** *Let A be an $\mathbb{F}_p$-algebra and $\vec{g} \in W_r(A)$. Then*

$$B := A[\vec{X}]/(\wp\vec{X} \ominus \vec{g})$$

is finite free over A, with basis $\{\prod_n X_n^{a_n} : 0 \leq a_n \leq p-1\}$, and is étale over A. Translation by $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ makes Spec B *a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over* Spec A.

Proof. By the triangular shape of Witt addition ((A.2)), the n-th equation in $\wp\vec{X} \ominus \vec{g} = 0$ has the form

$$X_n^p - X_n - g_n + C'_n(X_0, \ldots, X_{n-1}; g_0, \ldots, g_{n-1}) = 0.$$

Successively adjoining $X_0, \ldots, X_{r-1}$ therefore makes B finite free over A with the stated basis. The Jacobian matrix is lower triangular with diagonal entries -1, so B/A is étale. Moreover, this cover is the pullback of $\wp : \mathbb{W}_r \to \mathbb{W}_r$ along the section $\vec{g}$. Its kernel is $\{\vec{u} : F\vec{u} = \vec{u}\} = W_r(\mathbb{F}_p)$, and translation by this kernel makes the pullback a torsor. ☐

Theorem 85 (Artin–Schreier–Witt theory; [Tho05, §3–4])**.** *Let $M \supseteq \mathbb{F}_p$ be a field. There is a canonical isomorphism*

$$W_r(M)/\wp W_r(M) \xrightarrow{\cong} H^1(M, \mathbb{Z}/p^r\mathbb{Z}).$$

The class of $\vec{g}$ corresponds to the torsor

$$\operatorname{Spec} M[\vec{X}]/(\wp\vec{X} \ominus \vec{g}).$$

If the class has order p^s, the corresponding cyclic extension has degree p^s; in particular the torsor is connected precisely when its class has order p^r. Every cyclic extension of degree dividing p^r arises in this way.

Proof. Let M^s be a separable closure of M and $\Gamma = \mathrm{Gal}(M^s/M)$. The triangular equations in the preceding proof show that $\wp$ is surjective on $W_r(M^s)$ and that its kernel is $W_r(\mathbb{F}_p)$. The connecting map sends the class of $\vec{g}$ to

$$\sigma \longmapsto \sigma(\vec{u}) \ominus \vec{u}, \qquad \wp\vec{u} = \vec{g}.$$

This is a continuous homomorphism $\Gamma \to W_r(\mathbb{F}_p)$, independent of the choices. It is injective on classes because a Γ-invariant solution lies in $W_r(M)$. It is surjective by additive Hilbert 90, applied inductively to

$$0 \longrightarrow M^s \xrightarrow{V^{r-1}} W_r(M^s) \xrightarrow{t} W_{r-1}(M^s) \longrightarrow 0.$$

Thus $W_r(M)/\wp W_r(M) \cong H^1(M, \mathbb{Z}/p^r\mathbb{Z})$. The order and connectedness assertions follow from the correspondence between finite constant-group torsors and continuous characters: the degree of the connected field extension is the order of the image of the associated character. □

Corollary 86. *Let $M \supseteq \mathbb{F}_p$ and let $\vec{g} \in W_r(M)$ have $g_0 = 0$. Then $p^{r-1}[\vec{g}] = 0$ in $W_r(M)/\wp W_r(M)$. Consequently, a class of order p^r has nonzero zeroth component in every representative.*

Proof. By (A.3), $p^{r-1}\vec{g} = (0, \ldots, 0, g_0^{p^{r-1}}) = 0$ already in $W_r(M)$. □

A.8 The Schmid–Witt ramification formula

The break formula stated in Theorem 52 is due, for finite residue fields, to Kanesaka–Sekiguchi [KS79] and Thomas [Tho05], building on Schmid [Sch37] for $r = 1$, and for arbitrary perfect residue fields to Elder–Keating [EK25]. We use it in one direction only: evaluating the formula on one reduced representative computes the largest upper break; no minimization over representatives is needed.

Reducedness prevents cancellation between Witt levels. Indeed, if two nonzero pole orders m_j and $m_{j'}$, with $j < j'$, contributed the same weighted value, then $m_{j'} = p^{j'-j}m_j$ would be divisible by p, contradicting reducedness. For $r = 1$ the formula is exactly the Artin–Schreier break computation.

A.9 Reduced polynomial representatives

Definition 87. A vector $\vec{g} = (g_0, \ldots, g_{r-1}) \in W_r(k((x)))$ is a **reduced polynomial vector** if it is reduced and every $g_j \in x^{-1}k[x^{-1}]$.

Lemma 88 (Surjectivity on integral Witt vectors)**.** *If k is algebraically closed, then $\wp : W_r(k[[x]]) \to W_r(k[[x]])$ is surjective.*

Proof. Proceed by induction on r. For $r = 1$, given $h = \sum_{i \geq 0} b_i x^i$, choose u_0 with $u_0^p - u_0 = b_0$ and define the remaining coefficients recursively by $u_i = u_{i/p}^p - b_i$, with $u_{i/p} = 0$ when $p \nmid i$. Then $u^p - u = h$.

For $r > 1$, choose by induction $\vec{u}' \in W_{r-1}(k[[x]])$ with $\wp\vec{u}' = t(\vec{h})$ and lift it to $\widetilde{u} = (\vec{u}', 0)$. By Lemma 83(i), $\wp\widetilde{u} \ominus \vec{h} = V^{r-1}(g)$ for some $g \in k[[x]]$. Choose $w \in k[[x]]$ with $\wp w = -g$. Then Lemma 83(ii) gives $\wp(\widetilde{u} \oplus V^{r-1}w) = \vec{h}$. □

Proposition 89 (Reduced polynomial representatives)**.** *Every class in $W_r(k((x)))/\wp W_r(k((x)))$ has a reduced polynomial representative.*

Proof. Proceed by induction on r. For $r = 1$, write a Laurent series as the sum of its principal part and an element of $k[[x]]$. The latter can be removed by Lemma 88. If the remaining principal part has leading term cx^{-m} with $p \mid m$, subtract

$$\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}.$$

This strictly decreases the pole order; iteration produces zero or a pole order prime to p.

For the inductive step, first reduce the truncation of the given vector and lift the change of representative to length r. The first $r-1$ components are then reduced polynomials. By Lemma 83(iii), subtracting an element of the form $\wp(V^{r-1}z) = V^{r-1}(\wp z)$ changes only the last component. Apply the length-one procedure to that component: remove its integral part and then successively remove leading terms whose pole orders are divisible by p. This leaves all earlier components unchanged and produces a reduced polynomial vector representing the original class. □

Remark. The Witt pole bound forces $f_j = 0$ whenever $p^{r-1-j} \geq d$. Thus the restriction imposed by a break smaller than d is visible in every Witt component, not only in the last one.

Appendix B

A Wrong Path

In this appendix we explain why part (3) of Theorem 57 is difficult to prove by induction on r from the degree-p case, part (2) of the same theorem.

B.1 Quotient Towers of Symmetric-Power Torsors

We now study how the symmetric power construction of Proposition 25 interacts with the canonical decomposition of Proposition 15. Throughout this section, G denotes a finite abelian group, $H \subseteq G$ a subgroup, and $\mathcal{P} \to X$ a G-torsor over an S-scheme X that is Zariski-locally quasi-projective over S. Recall that $\mathcal{P} \to X$ factors canonically as

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}/H \xrightarrow{\psi} X,$$

where $\mathcal{P}/H = (G/H) \times^G \mathcal{P}$ is a G/H-torsor over X and ϕ is an H-torsor.

In addition to the kernel $K_G = \ker(G^d \to G)$ of the summation morphism appearing in Proposition 25, we shall need the analogous *summation kernels*

$$K_H := \ker(H^d \to H), \qquad K_{G/H} := \ker((G/H)^d \to G/H).$$

Applying the snake lemma to the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & H^d \arrow[r] \arrow[d, two heads] & G^d \arrow[r] \arrow[d, two heads] & (G/H)^d \arrow[r] \arrow[d, two heads] & 0 \\
0 \arrow[r] & H \arrow[r] & G \arrow[r] & G/H \arrow[r] & 0,
\end{tikzcd}$$

in which the vertical maps are the surjective summation morphisms, yields a short exact sequence

$$0 \to K_H \to K_G \to K_{G/H} \to 0. \tag{B.1}$$

The induced isomorphism $K_G/K_H \cong K_{G/H}$ will play a central role below.

We define four quotient schemes

$$\begin{aligned}
\mathcal{P}^{[d]_G} &:= (K_G \rtimes S_d)\backslash \mathcal{P}^{\times_S d},\\
\mathcal{P}^{[d]_H} &:= (K_H \rtimes S_d)\backslash \mathcal{P}^{\times_S d},\\
(\mathcal{P}/H)^{(d)} &:= (H^d \rtimes S_d)\backslash \mathcal{P}^{\times_S d},\\
(\mathcal{P}/H)^{[d]} &:= ((H^d + K_G) \rtimes S_d)\backslash \mathcal{P}^{\times_S d},
\end{aligned}$$

where S_d acts on $\mathcal{P}^{\times_S d}$ as in Corollary 27 and the subgroups $K_H \subseteq K_G \subseteq H^d + K_G$ and $H^d \subseteq H^d + K_G$ of G^d act diagonally on $\mathcal{P}^{\times_S d}$ via the G-action on $\mathcal{P}$. The first space is the symmetric G-power of $\mathcal{P}$ as in Proposition 25; the remaining three are, respectively, the symmetric H-power of $\mathcal{P}$ relative to ϕ, the symmetric power of the scheme $\mathcal{P}/H$, and the symmetric G/H-power of $\mathcal{P}/H$. **The latter two identifications follow from $(\mathcal{P}/H)^{\times_S d} \cong H^d\backslash\mathcal{P}^{\times_S d}$ together with the fact that the preimage of $K_{G/H} \subseteq (G/H)^d$ under $G^d \twoheadrightarrow (G/H)^d$ equals $H^d + K_G$.** the last isomorphism like that, is from the correspondence theorem, claude generated at the end of the article, add it somewhere

Proposition 90. *The four quotient schemes above exist and fit into a canonical commutative diagram*

$$\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}/H)^{(d)} \arrow[d, "q"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}/H)^{[d]} \arrow[d, "G/H"] \\
& X^{(d)},
\end{tikzcd}$$

in which each labelled arrow is a torsor under the indicated constant group. The right-hand vertical q is the canonical projection of Corollary 27, applied to the G/H-torsor $\mathcal{P}/H \to X$; in general it is not *a torsor under any constant group scheme (see the remark below). The unlabelled left-hand vertical $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ is the canonical projection induced by the inclusion of S_d-stable subgroups $K_H \rtimes S_d \subseteq K_G \rtimes S_d$ in $G^d \rtimes S_d$; it is the base change of q along the structural map $\mathcal{P}^{[d]_G} \to (\mathcal{P}/H)^{[d]}$ via the Cartesian property below, and likewise fails to be a torsor under a constant group scheme in general.*

Moreover, the upper square is Cartesian:

$$\mathcal{P}^{[d]_H} \cong \mathcal{P}^{[d]_G} \times_{(\mathcal{P}/H)^{[d]}} (\mathcal{P}/H)^{(d)}.$$

Proof. The four labelled arrows are torsors under the indicated constant groups by direct application of Proposition 25 and Proposition 15:

- Applied to the G-torsor $\mathcal{P} \to X$, Proposition 25 gives the G-torsor $\mathcal{P}^{[d]_G} \to X^{(d)}$ (the diagonal arrow).
- Applied to the H-torsor $\phi : \mathcal{P} \to \mathcal{P}/H$ (taking $\mathcal{P}/H$ in place of the base X), Proposition 25 gives the H-torsor $\mathcal{P}^{[d]_H} \to (\mathcal{P}/H)^{(d)}$ (the top arrow).

- Applied to the G/H-torsor $\psi : \mathcal{P}/H \to X$, Proposition 25 gives the G/H-torsor $(\mathcal{P}/H)^{[d]} \to X^{(d)}$ (the bottom-right vertical), and Corollary 27 produces the canonical morphism $q : (\mathcal{P}/H)^{(d)} \to (\mathcal{P}/H)^{[d]}$.

The factorization $\mathcal{P}^{[d]_G} \xrightarrow{H} (\mathcal{P}/H)^{[d]} \xrightarrow{G/H} X^{(d)}$ of the diagonal arrow into an H-torsor followed by a G/H-torsor is the canonical decomposition of Proposition 15, applied to the G-torsor $\mathcal{P}^{[d]_G} \to X^{(d)}$ and the subgroup $H \subseteq G$. Indeed, both $\mathcal{P}^{[d]_G} \times^G (G/H)$ and $(\mathcal{P}/H)^{[d]}$ are realized as the admissible quotient $((H^d + K_G) \rtimes S_d) \backslash \mathcal{P}^{\times_S d}$. here maybe add more info, like isomorphism theorem for torsors and qoutients, or something like that, to make it more clear, from which it follows

To prove the upper square is Cartesian, set $F := \mathcal{P}^{[d]_G} \times_{(\mathcal{P}/H)^{[d]}} (\mathcal{P}/H)^{(d)}$. Base-changing the H-torsor $\mathcal{P}^{[d]_G} \to (\mathcal{P}/H)^{[d]}$ along q exhibits F as an H-torsor over $(\mathcal{P}/H)^{(d)}$. The universal property of the fiber product furnishes a canonical morphism $\mathcal{P}^{[d]_H} \to F$ over $(\mathcal{P}/H)^{(d)}$, which is H-equivariant since the H-actions on both source and target are induced by the diagonal H-action on $\mathcal{P}^{\times_S d}$ add earlier in the section of torsors that in fact base changing an H-torsor along a morphism gives another H-torsor is a torsor. Any morphism of H-torsors is an isomorphism, so $\mathcal{P}^{[d]_H} \cong F$ and the upper square is Cartesian. The left vertical $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ is then identified, via this Cartesian square, with the base change of q along $\mathcal{P}^{[d]_G} \to (\mathcal{P}/H)^{[d]}$. □

Remark. The two vertical morphisms $q : (\mathcal{P}/H)^{(d)} \to (\mathcal{P}/H)^{[d]}$ and $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ are not, in general, torsors under any constant group scheme. The obstruction is purely group-theoretic: in $K_G \rtimes S_d$ the subgroup $K_H \rtimes S_d$ is not normal, because the conjugation action of S_d on K_G by coordinate permutation is non-trivial — for $k \in K_G$ and $\sigma \in S_d$, the components of $k - \sigma \cdot k$ are $k_i - k_{\sigma^{-1}(i)} \in G$, which need not lie in H. Equivalently, $H^d \rtimes S_d$ is not normal in $(H^d + K_G) \rtimes S_d$. So the naive set-theoretic quotients $(K_G \rtimes S_d)/(K_H \rtimes S_d)$ and $((H^d + K_G) \rtimes S_d)/(H^d \rtimes S_d)$ are not groups, even though $K_G/K_H \cong K_{G/H}$ and $(H^d + K_G)/H^d \cong K_{G/H}$ as abstract groups.

B.2 Setup and notation

Let $\mathcal{P} \to U$ be a G-torsor over $U = C \setminus \mathfrak{m}$, where G is a finite abelian group, $H \trianglelefteq G$, and C is a smooth projective curve. Fix $p_\infty \in \mathfrak{m}$ and an integer d. Assume the G torsor $\mathcal{P} \to U$ has ramification strictly bounded by d at p_∞. (i.e. $< d$) according to the definition in Definition 39 (i.e. the higher d'th-ramification group of the extension of DVRs at p_∞ is trivial).

Recall the four quotient schemes, defined in: Section B.1

$$\mathcal{P}^{[d]_H}, \qquad \mathcal{P}^{[d]_G}, \qquad (\mathcal{P}/H)^{(d)}, \qquad (\mathcal{P}/H)^{[d]},$$

together with the canonical Cartesian square

$$\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}/H)^{(d)} \arrow[d, "q"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}/H)^{[d]} \arrow[d, "G/H"] \\
 & U^{(d)},
\end{tikzcd}$$

in which the arrows labelled H, G/H and G are torsors under the indicated finite constant groups (in particular, *étale*), and the upper square is Cartesian (Proposition 90).

The G/H-torsor $\mathcal{P}/H \to U$ compactifies to a finite map of smooth projective curves $f\colon C' \to C$. Pick $p'_\infty \in f^{-1}(p_\infty)$

$$f^{(d)}\colon C'^{(d)} \to C^{(d)}$$

for the induced map of d-fold symmetric powers; it sends $d \cdot p'_\infty \mapsto d \cdot p_\infty$. Let

$$\pi_C\colon X_C \to C^{(d)}, \qquad \pi_{C'}\colon X_{C'} \to C'^{(d)}$$

be the blowups at $d \cdot p_\infty$ and $d \cdot p'_\infty$ respectively, with exceptional divisors $E_C, E_{C'}$ and generic points $\eta_C, \eta_{C'}$. The DVRs of these generic points are

$$V_C := \mathcal{O}_{X_C,\eta_C} \subset K(C^{(d)}), \qquad V_{C'} := \mathcal{O}_{X_{C'},\eta_{C'}} \subset K(C'^{(d)}).$$

A base-change lemma

We will use the following standard lemma when comparing a Galois extension with the function field of a base change.

Lemma 91. *Let E/F be a finite Galois extension with group H, and let K/F be an arbitrary field extension. Fix a common overfield $\Omega \supseteq E, K$ and form inside it the intersection $M := E \cap K$, the compositum $E \cdot K$, and the subgroup $N := \mathrm{Gal}(E/M) \leq H$. Then:*

1. *$E \cdot K/K$ is Galois, and restriction induces an isomorphism*

$$\mathrm{Gal}(E \cdot K/K) \xrightarrow{\sim} \mathrm{Gal}(E/M) = N;$$

2. *as K-algebras,*

$$E \otimes_F K \cong \prod_{i=1}^{[M:F]} E \cdot K.$$

In particular, $E \otimes_F K$ is a field if and only if $M = F$, equivalently, if and only if E and K are linearly disjoint over F.

Proof. Write $E = F[\alpha]$ and let $g \in F[x]$ be the minimal polynomial of α. Then $\deg(g) = |H|$, and g splits completely in E because E/F is Galois. The classical translation theorem shows that $E \cdot K/K$ is Galois and that restriction identifies its Galois group with the subgroup of H fixing $E \cap K = M$ pointwise, namely $\mathrm{Gal}(E/M) = N$.

For the second assertion, factor $g = g_1 \cdots g_m$ into monic irreducible polynomials over K. Adjoining any root of g to K produces $E \cdot K$, so each g_i has degree

$$[E \cdot K : K] = [E : M] = |N|.$$

It follows that $m = |H|/|N| = [M : F]$, and hence

$$E \otimes_F K \cong K[x]/(g) \cong \prod_{i=1}^{m} K[x]/(g_i) \cong \prod_{i=1}^{[M:F]} E \cdot K.$$

This product is a field exactly when $m = 1$, or equivalently when $M = F$. $\square$

B.3 The master diagram

The torsor tower over the open base $U^{(d)}$ embeds into the projective geometry of $C^{(d)}$ and $C'^{(d)}$, which in turn admit canonical blowups at the points at infinity. We assemble all of this into a single commutative diagram.

$$\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}/H)^{(d)} \arrow[r, hook, "\iota'"] \arrow[d, "q"] & C'^{(d)} \arrow[dd, "f^{(d)}"] & X_{C'} \arrow[l, "\pi_{C'}"'] \arrow[dd, dashed, "f_X"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}/H)^{[d]} \arrow[d, "G/H"] & & \\
& U^{(d)} \arrow[r, hook, "\iota"'] & C^{(d)} & X_C \arrow[l, "\pi_C"]
\end{tikzcd}$$

Reading the diagram:

- The **left block** (columns 1–2) is the torsor tower of Proposition 90: H-torsors horizontally, the G/H-torsor on the lower right, and the composite G-torsor $\mathcal{P}^{[d]_G} \to U^{(d)}$.
- The **middle block** (column 3) records the compactifications. The map ι is the canonical open immersion $U^{(d)} \hookrightarrow C^{(d)}$. The map ι' is the open immersion $(\mathcal{P}/H)^{(d)} \hookrightarrow C'^{(d)}$ obtained from compactifying $\mathcal{P}/H \subset C'$. The vertical $f^{(d)}$ is the finite morphism of symmetric powers; restricting along ι' recovers the canonical map $(\mathcal{P}/H)^{(d)} \to U^{(d)}$, which is the composite of q with the G/H-torsor.
- The **right block** (column 4) records the blowups at the diagonal points at infinity. The dashed arrow $f_X \colon X_{C'} \dashrightarrow X_C$ is the rational lift of $f^{(d)}$, defined on the strict transform; on generic points of the exceptional divisors it sends $\eta_{C'} \mapsto \eta_C$ so we have induced extension of DVRs $V_C \hookrightarrow V_{C'}$

B.4 Unramifiedness of $\mathcal{P}^{[d]_G}$ at η_C

So, returning to the beginning of this appendix, a natural attempt to prove part (3) of Theorem 57 from part (2) by induction on r comes down to proving the following theorem.

Theorem 92. *Suppose*

(H1) $(\mathcal{P}/H)^{[d]} \to U^{(d)}$ *is unramified at* η_C*; and*

(H2) $\mathcal{P}^{[d]_H} \to (\mathcal{P}/H)^{(d)}$ *is unramified at* $\eta_{C'}$*.*

(H3) $G = \mathbb{Z}/p^r\mathbb{Z}$ *is cyclic of prime power order.*

Then $\mathcal{P}^{[d]_G} \to U^{(d)}$ *is unramified at* η_C*.*

And the next section is devoted to explaining why this is not an easy task, and in fact, we were not able to prove it.

B.5 Difficulties in Proving the Theorem

First, note that we can assume $\mathcal{P} \to U$ is connected, and that H is non-trivial.

(R1) *$\mathcal{P} \to U$ is connected.* Let $\rho\colon \pi_1(U) \to G$ be the monodromy of $\mathcal{P}$ and put $G_0 := \operatorname{Im}(\rho)$, $H_0 := H \cap G_0$. Pick a connected component $\mathcal{P}_0 \subseteq \mathcal{P}$; it is a connected G_0-torsor, and $\mathcal{P}$ is the disjoint union of $[G : G_0]$ copies of $\mathcal{P}_0$ permuted by G. Since the symmetric-power quotients of Section B.1 commute with such disjoint unions, each of the four schemes in Proposition 90 for $(\mathcal{P}, G, H)$ decomposes as a disjoint union of copies of the corresponding scheme for $(\mathcal{P}_0, G_0, H_0)$. Unramifiedness at η_C (resp. $\eta_{C'}$) is detected componentwise, so (H1) and (H2) are inherited by $(\mathcal{P}_0, G_0, H_0)$; (H3) is inherited because subgroups of cyclic p-groups are cyclic p-groups. Replacing $(\mathcal{P}, G, H)$ by $(\mathcal{P}_0, G_0, H_0)$ we may assume $\mathcal{P}$ is connected and ρ is surjective.

(R2) *$H \neq 0$.* If $H = 0$ then $\mathcal{P}/H = \mathcal{P}$ and $(\mathcal{P}/H)^{[d]} = \mathcal{P}^{[d]_G}$, so (H1) is literally the conclusion.

We can read a decomposition of fields directly from the canonical Cartesian square of Proposition 90. After the reduction (R1) all four corners are integral: the three corners $\mathcal{P}^{[d]_G}$, $(\mathcal{P}/H)^{[d]}$ and $(\mathcal{P}/H)^{(d)}$ are integral, and the fourth, $\mathcal{P}^{[d]_H} = (K_H \rtimes S_d)\backslash\mathcal{P}^{\times_S d}$, is integral as a quotient of the integral scheme $\mathcal{P}^{\times_S d}$ (Theorem 21). So we may take function fields; using the open immersions ι, ι' of the master diagram together with $K_C = \operatorname{Frac} V_C = K(C^{(d)}) = K(U^{(d)})$ and $K_{C'} = \operatorname{Frac} V_{C'} = K(C'^{(d)}) = K\big((\mathcal{P}/H)^{(d)}\big)$, we identify

$$K(U^{(d)}) = K_C, \qquad K\big((\mathcal{P}/H)^{[d]}\big) = F, \qquad K\big((\mathcal{P}/H)^{(d)}\big) = K_{C'}, \qquad K\big(\mathcal{P}^{[d]_G}\big) = E,$$

with E/F Galois of group H, F/K_C Galois of group G/H, and the composite E/K_C Galois of group G. The fourth corner $\mathcal{P}^{[d]_H}$ is the fibre product $\mathcal{P}^{[d]_G} \times_{(\mathcal{P}/H)^{[d]}} (\mathcal{P}/H)^{(d)}$; being integral, its single function field is the compositum $E \cdot K_{C'}$. In particular its generic fibre $E \otimes_F K_{C'}$ over $(\mathcal{P}/H)^{(d)}$ is a field, so by Lemma 91 the extensions E and $K_{C'}$ are linearly disjoint over F; that is,

$$E \cap K_{C'} = F,$$

and $E \cdot K_{C'}/K_{C'}$ is Galois with group H. We denote:

1. $K_C := \operatorname{Frac} V_C$, $K_{C'} := \operatorname{Frac} V_{C'}$;
2. $F := K\big((\mathcal{P}/H)^{[d]}\big)$, $E := K(\mathcal{P}^{[d]_G})$;
3. $V_F := V_{C'}|_F$;
4. a place v_E of E above V_F, and a place $v_\star$ of $E \cdot K_{C'}$ above both v_E and $V_{C'}$.

Thus we have:

$$\begin{tikzcd}[column sep=small]
 & E \cdot K_{C'} \arrow[dl, dash] \arrow[dr, dash, "H"] & \\
E \arrow[dr, dash, "H"'] & & K_{C'} \arrow[dl, dash] \\
 & F \arrow[d, dash, "G/H"] & \\
 & K_C &
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}[column sep=small]
 & v_\star \arrow[dl, dash] \arrow[dr, dash] & \\
v_E \arrow[dr, dash] & & V_{C'} \arrow[dl, dash] \\
 & V_F \arrow[d, dash] & \\
 & V_C &
\end{tikzcd}$$

(lines denote field inclusion / place lying-over, read upwards; in the left diagram a label is the Galois group of the extension where it is Galois). In addition E/K_C is Galois with group G.

Read through this dictionary, the hypotheses of the theorem become statements about the ramification of the places above:

(H1) $(\mathcal{P}/H)^{[d]} \to U^{(d)}$ is unramified at η_C means that the G/H-extension F/K_C is unramified at V_C. In particular V_F is unramified over V_C, with $e(V_F \mid V_C) = 1$.

(H2) $\mathcal{P}^{[d]_H} \to (\mathcal{P}/H)^{(d)}$ is unramified at $\eta_{C'}$ means that the H-extension $E \cdot K_{C'}/K_{C'}$ is unramified at $V_{C'}$; equivalently the inertia group $I(v_\star \mid V_{C'}) \leq H$ is trivial.

(H3) $G = \mathbb{Z}/p^r\mathbb{Z}$ makes G, hence H and every inertia group, a cyclic p-group; as the residue characteristic is p, the ramification of E/F is potentially *wild*.

The conclusion we are after — that $\mathcal{P}^{[d]_G} \to U^{(d)}$ is unramified at η_C — is the statement that the G-extension E/K_C is unramified at V_C, i.e. that the inertia $I(v_E \mid V_C) \leq G$ is trivial. By (H1) the lower layer F/K_C is already unramified at V_C, so this inertia lies in $H = \mathrm{Gal}(E/F)$ and coincides with the inertia of the top layer,

$$I(v_E \mid V_C) \;=\; I(v_E \mid V_F) \;\leq\; H.$$

The theorem therefore reduces to the single assertion

$$E/F \text{ is unramified at } V_F, \qquad \text{i.e.} \qquad I(v_E \mid V_F) = 1.$$

Now $E \cdot K_{C'}/K_{C'}$ is the base change of E/F along $F \hookrightarrow K_{C'}$, and under the restriction isomorphism $\mathrm{Gal}(E \cdot K_{C'}/K_{C'}) \cong \mathrm{Gal}(E/F) = H$ the inertia injects

$$I(v_\star \mid V_{C'}) \;\hookrightarrow\; I(v_E \mid V_F).$$

Hypothesis (H2) says the source is trivial; the goal is that the target is trivial. These need *not* coincide: the two differ precisely by the ramification of $K_{C'}/F$ at $V_{C'} \mid V_F$ — the ramification that the symmetric-power base change q introduces at the point at infinity — which can be wild and can absorb the ramification of E/F. It is exactly this discrepancy that obstructs deducing the goal from (H1)–(H3).

Put in other words: The base change $K_{C'}/F$ is wildly ramified, so the inertia comparison is not onto .

The inertia injection, and why (H2) is not enough

The injection used above is an instance of the following general fact.

Lemma 93. *Let E/F be a finite Galois extension, linearly disjoint over F from a field extension K/F, with compositum $E \cdot K$; thus restriction is an isomorphism $r\colon \mathrm{Gal}(E \cdot K/K) \xrightarrow{\sim} \mathrm{Gal}(E/F)$, $\sigma \mapsto \sigma|_E$. Let w be a place of $E \cdot K$, and write $w_K := w|_K$, $w_E := w|_E$, $w_F := w|_F$ for its restrictions. Then r carries the inertia group of w over w_K into that of w_E over w_F:*

$$r\big(I(w \mid w_K)\big) \;\subseteq\; I(w_E \mid w_F),$$

so that $I(w \mid w_K) \hookrightarrow I(w_E \mid w_F)$ is injective. Moreover, completing at w identifies $I(w \mid w_K)$ with the inertia group of the local compositum $\widehat{E} \cdot \widehat{K}$ over $\widehat{K}$; in particular, if $\widehat{E} \subseteq \widehat{K}$ then $I(w \mid w_K) = 1$, no matter how ramified E/F is at w_F.

Proof. That r is an isomorphism is the translation theorem for the Galois extension E/F and the linearly disjoint K/F (Lemma 91(1) with $M = F$). For the inclusion, let $\sigma \in I(w \mid w_K)$. Since E/F is Galois and σ fixes $F \subseteq K$ pointwise, $\sigma(E) = E$ and $\sigma|_E \in \mathrm{Gal}(E/F)$ fixes w_F. As σ lies in the decomposition group of w we have $w \circ \sigma = w$; restricting to E gives $w_E \circ \sigma|_E = w_E$, so $\sigma|_E \in D(w_E \mid w_F)$. Finally the residue extension $\kappa(w_E) \hookrightarrow \kappa(w)$ is equivariant and σ acts trivially on $\kappa(w)$, hence $\sigma|_E$ acts trivially on $\kappa(w_E)$, i.e. $\sigma|_E \in I(w_E \mid w_F)$. Injectivity is inherited from that of r. For the last assertion, the completion of $E \cdot K$ at w is the local compositum $\widehat{E}\widehat{K}$, with $\mathrm{Gal}(\widehat{E}\widehat{K}/\widehat{K}) = D(w \mid w_K)$ and inertia subgroup $I(w \mid w_K)$; if $\widehat{E} \subseteq \widehat{K}$ then $\widehat{E}\widehat{K} = \widehat{K}$ and this group is trivial. □

Taking $w = v_\star$ (so $w_K = V_{C'}$, $w_E = v_E$, $w_F = V_F$) recovers the injection $I(v_\star \mid V_{C'}) \hookrightarrow I(v_E \mid V_F)$. The implication "(H2) $\Rightarrow$ goal" would follow if this were always onto; the next example shows it need not be, already for $G = \mathbb{Z}/p^2\mathbb{Z}$ over $\mathbb{F}_p$.

Example 94. Take $k = \mathbb{F}_p$, $G = \mathbb{Z}/p^2\mathbb{Z}$, and $H = p\,\mathbb{Z}/p^2\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z}$ the unique subgroup of order p, so $G/H \cong \mathbb{Z}/p\mathbb{Z}$. We describe the local fields at the place $V_{C'} \mid V_F$ — the only place where the obstruction lives — writing $\widehat{(\cdot)}$ for completions.

The base and the G-tower. Let $\widehat{K_C} = \mathbb{F}_p((t))$, and present the local G-extension $\widehat{E}/\widehat{K_C}$ (an Artin–Schreier–Witt extension) by its two layers:

- lower layer $\widehat{F}/\widehat{K_C}$ (group G/H): *split*, so $\widehat{F} = \mathbb{F}_p((t))$. Then F/K_C is unramified at V_C — this is (H1) — and $\widehat{V_F} = \widehat{V_C}$;
- upper layer $\widehat{E}/\widehat{F}$ (group H): the Artin–Schreier extension
$$\widehat{E} = \widehat{F}(y), \qquad y^p - y = t^{-1}.$$
By Theorem 50(3) (with $m = 1$, prime to p) this is totally, wildly ramified of degree p, with $I(v_E \mid V_F) = H \cong \mathbb{Z}/p\mathbb{Z} \neq 1$.

Hence $I(v_E \mid V_C) = I(v_E \mid V_F) = H \neq 1$: the extension E/K_C is ramified at V_C, so *the conclusion of the theorem fails.*

The base change. The places of $K_{C'}$ over V_F are governed by the symmetric-power cover q. Suppose at $V_{C'}$ this cover reproduces the *same* Artin–Schreier extension,
$$\widehat{K_{C'}} = \widehat{F}(y), \quad y^p - y = t^{-1}, \qquad \text{i.e.} \qquad \widehat{K_{C'}} = \widehat{E}.$$
This is compatible with the global linear disjointness $E \cap K_{C'} = F$: two *distinct* degree-p extensions of the global field F may have isomorphic completions at a place. Now the local compositum is $\widehat{E \cdot K_{C'}} = \widehat{E} \cdot \widehat{K_{C'}} = \widehat{E} = \widehat{K_{C'}}$, so $E \cdot K_{C'}/K_{C'}$ is split, hence unramified, at $V_{C'}$:
$$I(v_\star \mid V_{C'}) = \mathrm{Gal}\big(\widehat{E}/\widehat{K_{C'}}\big) = 1.$$
Thus (H2) holds, and (H3) holds by construction.

Outcome. All of (H1)–(H3) hold, yet E/K_C is wildly ramified at V_C with inertia $H \cong \mathbb{Z}/p\mathbb{Z}$. The inertia injection of Lemma 93 is here
$$\underbrace{I(v_\star \mid V_{C'})}_{=1} \hookrightarrow \underbrace{I(v_E \mid V_F)}_{=\mathbb{Z}/p\mathbb{Z}},$$

the inclusion of the trivial group: (H2) carries no information about the ramification of E/F, which has been entirely *absorbed* by that of $K_{C'}/F$ — the wild ramification introduced by q at the point at infinity.

Remark. This is globally consistent with everything above. Since $\widehat{E} \otimes_{\widehat{F}} \widehat{K_{C'}} = \widehat{E} \otimes_{\widehat{F}} \widehat{E}$ is not a field, $\mathcal{P}^{[d]_H}$ (function field $E \cdot K_{C'}$) *splits* at $V_{C'}$ into several places and is therefore unramified at $\eta_{C'}$, as (H2) demands; meanwhile $\mathcal{P}^{[d]_G}$ stays ramified at η_C. The hypotheses simply do not see the local coincidence $\widehat{K_{C'}} = \widehat{E}$, and ruling out such coincidences for a given $(\mathcal{P}, d)$ is exactly the global input the inductive argument lacks — which is why this is a wrong path.

Absorption made explicit: two local fields

Example 94 posited the local coincidence $\widehat{K_{C'}} = \widehat{E}$ abstractly. For completeness we exhibit it in the smallest possible setting — two honest local fields and a one-line Artin–Schreier reduction — to make the absorption phenomenon fully visible.

Example 95 (absorption of a wild break, explicit fields)**.** Let $k = \bar{\mathbb{F}}_p$ and let $F = k((t))$. Define a second local field $K = k((\pi))$, and embed $F \hookrightarrow K$ by

$$t \;=\; \frac{\pi^p}{1-\pi^{p-1}} \;=\; \pi^p\bigl(1 + \pi^{p-1} + \pi^{2(p-1)} + \cdots\bigr), \qquad \text{so} \qquad v_\pi(t) = p.$$

Since $t = \pi^p \cdot (\text{unit})$ and $dt/d\pi = -\pi^{2p-2}(1-\pi^{p-1})^{-2} \neq 0$, the extension K/F is separable, *totally (wildly) ramified of degree* p; equivalently it is the Artin–Schreier extension $\sigma^p - \sigma = t^{-1}$ with $\sigma = \pi^{-1}$, of break 1 (Theorem 50(3)).

Now let E/F be the Artin–Schreier extension

$$E = F(y), \qquad y^p - y = t^{-1}.$$

Over F this is ramified: $v_t(t^{-1}) = -1$ is prime to p, so by Theorem 50(3) it is totally wildly ramified of degree p, break 1, with $I(v_E \mid v_F) = \mathbb{Z}/p\mathbb{Z} \neq 1$.

Yet E is absorbed by K. Computing t^{-1} inside K,

$$t^{-1} \;=\; \pi^{-p}\bigl(1-\pi^{p-1}\bigr) \;=\; \pi^{-p} - \pi^{-1} \;=\; \wp\bigl(\pi^{-1}\bigr), \qquad \wp(x) := x^p - x,$$

so $y = \pi^{-1} \in K$ already solves $y^p - y = t^{-1}$. Hence $E = F(\pi^{-1}) \subseteq K$ (indeed $E = K$, both being degree p over F), the compositum is $E \cdot K = K$, and

$$E \cdot K/K \text{ is trivial, hence unramified:} \qquad I(v_\star \mid v_K) = 1,$$

while E/F is ramified. The ramification of E/F has been *entirely absorbed* by the wild base change to K — exactly the failure of surjectivity in the inertia injection $I(v_\star \mid v_K) \hookrightarrow I(v_E \mid v_F)$, here $1 \hookrightarrow \mathbb{Z}/p\mathbb{Z}$.

Remark. The mechanism, and why F cannot do it. Read over K, the datum t^{-1} has pole order $v_\pi(t^{-1}) = p$, *divisible by* p; this is what licenses the Artin–Schreier reduction $y \mapsto y - \pi^{-1}$, which removes the pole completely because $\wp(\pi^{-1})$ matches t^{-1} to all orders. The substitution is available over K but *not* over F: the reducing element π^{-1} has valuation -1, whereas $v_\pi(F^\times) = p\,\mathbb{Z}$ contains no element of valuation -1 at all. Thus the wild base change does not "fight" the ramification of E/F; it *supplies the one element of intermediate valuation* that the Artin–Schreier reduction needs. The ramification index $e(K/F) = p$ is the gate that turns the prime-to-p break 1 into the divisible-by-p pole order p over K.

General r. For $G = \mathbb{Z}/p^r$ the faithful local model has $\widehat{K_{C'}}/\widehat{F}$ totally ramified of degree p^{r-1} (the residue field k being algebraically closed, there is no residual part), and absorption is the proper inclusion $\widehat{E} \subsetneq \widehat{K_{C'}}$ of the top-layer $\mathbb{Z}/p$-extension. The explicit fields are obtained by iterating the recipe above: with $E = k((\pi))$, $\pi = y^{-1}$ as here, set $\pi = \rho^p(1-\rho^{p-1})^{-1}$ to get a degree-p overfield $k((\rho)) \supsetneq E$, and so on $r-1$ times. The pole order of the top datum is multiplied to $p^{r-1} \cdot 1 \equiv 0 \pmod{p}$ and reduced away by the same one-line substitution. Over an algebraically closed residue field, "$E \cdot K/K$ unramified with E/F ramified" is *equivalent* to $\widehat{E} \subseteq \widehat{K_{C'}}$ — there are no nontrivial unramified extensions to hide in — so absorption is precisely a containment of local fields, never a softer phenomenon.

Bibliography

[Sch37] Hermann Ludwig Schmid. “Zur Arithmetik der zyklischen p-Körper”. In: *Journal für die reine und angewandte Mathematik* 176 (1937), pp. 161–167.

[SGA1] Alexander Grothendieck. *Revêtements étales et groupe fondamental (SGA 1)*. Vol. 224. Lecture notes in mathematics. Springer-Verlag, 1971.

[KS79] Kimiaki Kanesaka and Koji Sekiguchi. “Representation of Witt vectors by formal power series and its applications”. In: *Tokyo Journal of Mathematics* 2.2 (1979), pp. 349–370. DOI: 10.3836/tjm/1270473027.

[Ser79] Jean-Pierre Serre. *Local Fields*. Springer, 1979.

[Koc97] Helmut Koch. *Algebraic Number Theory*. Vol. 62. Encyclopaedia of Mathematical Sciences. Berlin, Heidelberg: Springer-Verlag, 1997. ISBN: 978-3-540-63003-6.

[Tho05] Lara Thomas. “Ramification groups in Artin-Schreier-Witt extensions”. In: *Journal de théorie des nombres de Bordeaux* 17.2 (2005), pp. 689–720. DOI: 10.5802/jtnb.514. URL: https://jtnb.centre-mersenne.org/articles/10.5802/jtnb.514/.

[Mil08] James S. Milne. *Abelian Varieties (v2.00)*. Available at www.jmilne.org/math/. 2008.

[Tót11] Péter Tóth. “Geometric Abelian Class Field Theory”. In: (May 2011). Master of Science thesis. URL: https://math.bu.edu/people/rmagner/Seminar/GCFTthesis.pdf.

[Rab14] Joseph Rabinoff. *The Theory of Witt Vectors*. 2014. arXiv: 1409.7445 [math.RA]. URL: https://arxiv.org/abs/1409.7445.

[Ten15] Avichai Tendler. *Geometric Class Field Theory*. 2015. arXiv: 1507.00104 [math.AG]. URL: https://arxiv.org/abs/1507.00104.

[Stacks] The Stacks Project Authors. *Stacks Project*. https://stacks.math.columbia.edu. 2018.

[Gui19] Quentin Guignard. “On the ramified class field theory of relative curves”. In: *Algebra & Number Theory* 13 (May 2019). Revised version submitted on 29 May 2019, pp. 1299–1326. DOI: 10.2140/ant.2019.13.1299. arXiv: 1804.02243 [math.AG].

[Tak19] Daichi Takeuchi. “Blow-ups and the class field theory for curves”. In: *Algebraic Number Theory* 13.6 (2019), pp. 1327–1351. DOI: 10.2140/ant.2019.13.1327. URL: https://doi.org/10.2140/ant.2019.13.1327.

[You20] Alex Youcis. “Notes on Torsors”. Available at https://alex-youcis.github.io/torsors.pdf. 2020. URL: https://alex-youcis.github.io/torsors.pdf.

[EK25] G. Griffith Elder and Kevin Keating. *Artin–Schreier–Witt extensions and ramification breaks*. 2025. arXiv: 2503.16830. URL: https://arxiv.org/abs/2503.16830.